

# WJEC GCE Biology

AS UNIT 1 PAST PAPER BY TOPIC

IGC HK Exam

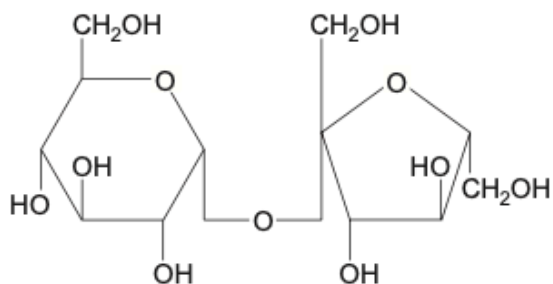


1.1

WJEC 2016

4. Sucrose is a disaccharide of glucose and fructose. The enzyme sucrase, catalyses the hydrolysis of sucrose into its monosaccharides. A colorimeter can be used to record absorbance values, which can then be used to determine the rate of hydrolysis.

(a) (i) Complete the diagram to show the hydrolysis of sucrose including the products formed. [2]





(ii) State the name of the bond broken during the reaction. [1]

(iii) Explain why glucose and fructose are referred to as structural isomers. [1]

(iv) A student was provided with two beakers, one containing sucrose and the other containing sucrase. Describe **one** biochemical test that the student could have carried out to distinguish between the two solutions. [2]

(b) A student wanted to investigate the hypothesis that sucrase catalyses the hydrolysis of sucrose fastest at a neutral pH. She was provided with the following method:

- Add 5 cm<sup>3</sup> of buffer solution to a test tube
- Add 5 cm<sup>3</sup> of sucrose solution to the test tube
- Add 1 cm<sup>3</sup> of sucrase solution to the test tube and mix
- After 20 minutes add 1 cm<sup>3</sup> of dinitrosalicylic acid (DNS)
- Pipette 5 cm<sup>3</sup> of the mixture into a cuvette and place into a colorimeter and record the absorbance of light passing through the solution.

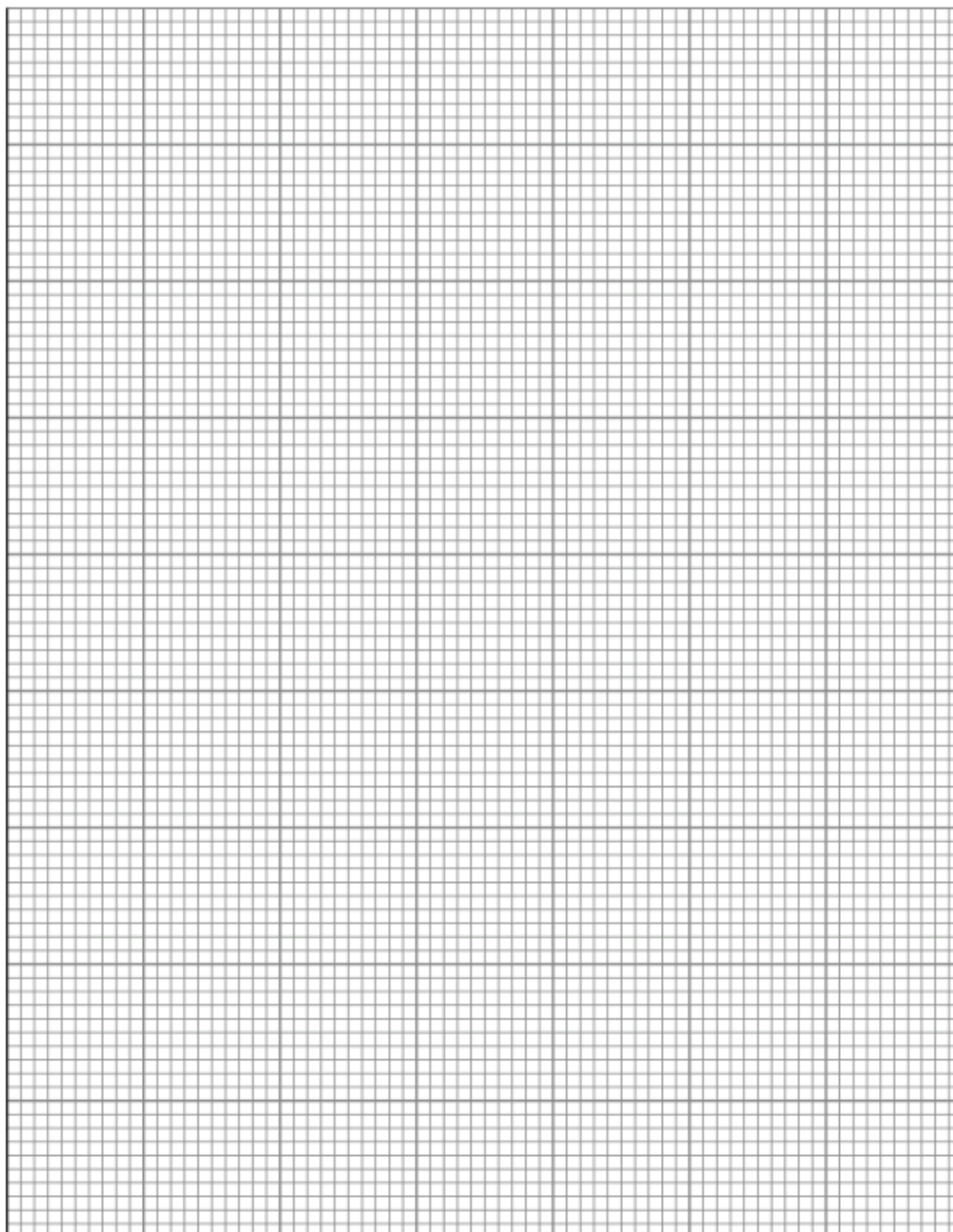
Dinitrosalicylic acid (DNS) will react with monosaccharides to produce amino-nitrosalicylic acid (ANS). ANS causes a colour change to occur which can be detected by a colorimeter. The greater the concentration of ANS the greater the absorbance of light.

(i) State **two** additional variables that the student should control to ensure that the results recorded would be repeatable. [2]

(ii) The results are shown in the table below.

| pH of sucrose and<br>sucrase solution | Absorbance of light (au) |         |         |      |
|---------------------------------------|--------------------------|---------|---------|------|
|                                       | trial 1                  | trial 2 | trial 3 | mean |
| 3                                     | 0.84                     | 0.63    | 0.76    | 0.74 |
| 4                                     | 0.89                     | 0.95    | 0.76    | 0.87 |
| 5                                     | 0.85                     | 0.91    | 0.72    | 0.83 |
| 7                                     | 0.27                     | 0.34    | 0.29    | 0.30 |
| 9                                     | 0.13                     | 0.11    | 0.12    | 0.12 |
| 10                                    | 0.05                     | 0.04    | 0.06    | 0.05 |

Plot the mean data on the graph below.



- (iii) What conclusions can be drawn from the data regarding the hypothesis being tested? [3]

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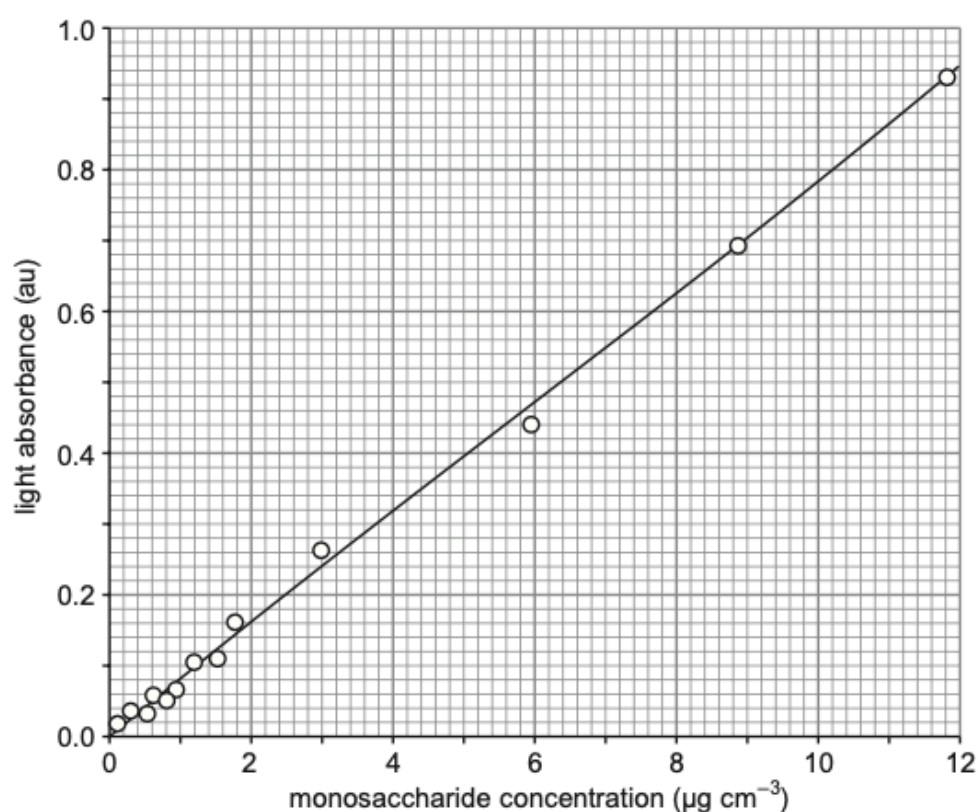
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- (iv) The student was also provided with the calibration curve below in order to calculate the concentration of glucose produced during the investigation.

The colour change, caused by ANS, is dependent on the concentration of monosaccharides, the greater the concentration the greater the absorbance value recorded.



Determine the maximum concentration of glucose produced during the investigation at pH 7. [2]

- (c) With reference to the data explain how, and why, you would modify the method to determine the optimum pH more accurately. [2]

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|   |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |  |   |  |   |
|---|-----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--|---|--|---|
| 4 | (a) | (i)   | Diagram shows the addition of water (1)<br>Glucose and fructose drawn with –OH groups on C1 and C2 in down positions (1)                                                                                                                                                                                                                                                                                       | 2 |   |  | 2 |  |   |
|   |     | (ii)  | Glycosidic                                                                                                                                                                                                                                                                                                                                                                                                     | 1 |   |  | 1 |  |   |
|   |     | (iii) | They have the same {chemical formula/ molecular formula/ number of atoms of each element} but different {structural formulae/ structures}/ both C <sub>6</sub> H <sub>12</sub> O <sub>6</sub> but different {structural formulae/ structures}                                                                                                                                                                  | 1 |   |  | 1 |  |   |
|   |     | (iv)  | Add Biuret solution to a sample of both solutions (1)<br>Reject reference to heat<br>The sucrose will cause a colour change from blue to lilac (and the sucrose will remain blue) (1)<br>OR<br>Heat both solutions with acid, <u>then</u> neutralise with alkali, <u>then</u> (heat) with Benedicts (1)<br>The sucrose will cause a colour change from blue to brick red (and the sucrose will remain blue)(1) | 1 | 1 |  | 2 |  | 2 |

| Question |       |  | Marking details                                                                                                                                                                                                                                                                                     | Marks available |     |     |       |       |      |
|----------|-------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |       |  |                                                                                                                                                                                                                                                                                                     | AO1             | AO2 | AO3 | Total | Maths | Prac |
| (b)      | (i)   |  | Any two (x 1) from:<br>concentration of {sucrose / sucrose / DNS} (1)<br>temperature (1)<br>filter/ wavelength of light in the colorimeter (1)<br>method of mixing (1)                                                                                                                              |                 | 2   |     | 2     |       | 2    |
|          | (ii)  |  | Linear axes with values at origin (1)<br>X-axis labelled pH and Y-axis labelled <u>mean</u> absorbance + au (1)<br>Correct plots (1) tolerance $\pm \frac{1}{2}$ small square<br>Plots joined (1) no extrapolation                                                                                  |                 | 4   |     | 4     | 4     | 4    |
|          | (iii) |  | Hypothesis is incorrect (1)<br>The {fastest rate of reaction/ most product produced} is {pH 4/ acidic/ between pH3-5}/ the optimum pH is pH 4 (1)<br>Use of data e.g. mean light absorption was 0.87 au at pH 4 and only 0.30 au at pH 7/ rate is approx three times faster at pH4 than at pH 7 (1) |                 | 3   |     | 3     |       | 3    |

# WJEC 2017

6. The cell surface membranes of root cells are responsible for differences in the concentration of ions in the roots when compared to those found in the surrounding soil solution. Inorganic ions are essential for cell metabolism in plants.

(a) Describe the role of magnesium, calcium, and phosphate ions in plant cells and tissues.

[3]

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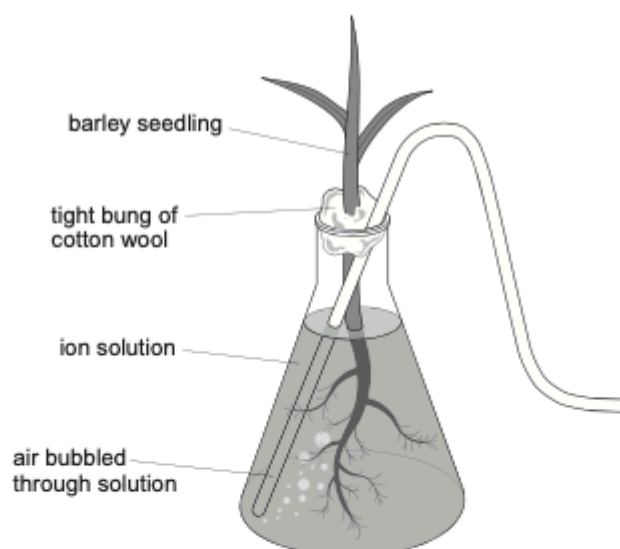
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To investigate the uptake of mineral ions by plants, barley seedlings were grown with their roots in a solution of mineral ions for four days. Air was bubbled through the solution. The concentration of the ions in the solution and the root tissue were determined.



| ion                                        | ion concentration / mmol dm <sup>-3</sup> |             |
|--------------------------------------------|-------------------------------------------|-------------|
|                                            | surrounding solution                      | root tissue |
| calcium (Ca <sup>2+</sup> )                | 1.0                                       | 57.0        |
| magnesium (Mg <sup>2+</sup> )              | 0.3                                       | 33.0        |
| phosphate (PO <sub>4</sub> <sup>3-</sup> ) | 0.9                                       | 1.6         |

- (b) (i) With reference to the data, explain the relative concentrations in the surrounding solution and the root tissue after four days. [3]

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- (ii) Explain why the growth of a barley seedling may be reduced if the solution did not have air bubbled through it. [4]

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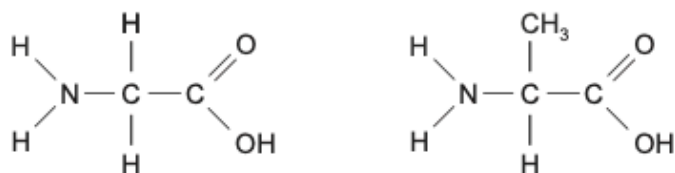
|   |     |      |                                                                                                                                                                                                                                                                                                                                                                |          |          |          |           |          |          |
|---|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
| 6 | (a) |      | Magnesium - chlorophyll (1) NOT chloroplast<br>Calcium – (structure of) cell <u>walls</u> in plants (1)<br>Phosphate - nucleic acids/ nucleotides/ phospholipids/ ATP/<br>NADP/ NAD/ FAD(1)                                                                                                                                                                    | 3        |          |          | 3         |          |          |
|   | (b) | (i)  | Active transport (1)<br>Against concentration gradient (1)<br>Correct use of data (1)                                                                                                                                                                                                                                                                          | 1        | 2        |          | 3         |          |          |
|   |     | (ii) | A. anaerobic conditions / lack of oxygen/ oxygen is required(1)<br>B. {Less/ no} ATP produced (1)<br>C. active transport {cannot occur/ reduced}, fewer ions can be<br>transported (against concentration gradient) (1)<br>D. Stunted growth because of a lack of named ion {e.g.<br>phosphate, required to make DNA and proteins essential for<br>growth} (1) |          | 3        | 1        | 4         |          |          |
|   |     |      | <b>Question 6 total</b>                                                                                                                                                                                                                                                                                                                                        | <b>4</b> | <b>5</b> | <b>1</b> | <b>10</b> | <b>0</b> | <b>0</b> |

WJEC 2018

4.  $\beta$ -Bungarotoxin is a neurotoxin found in the venom of Krait snakes. The neurotoxin is a protein that causes muscle paralysis and eventual death.

The diagram below shows the structure of two amino acids found within the protein.

- (a) (i) **Complete** the diagram, to show the products formed when these amino acid molecules are joined by a condensation reaction. [2]



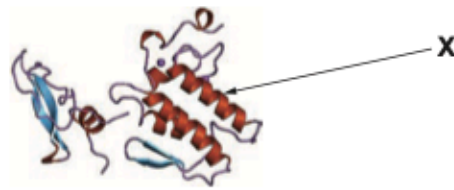
- (ii) State the bond formed.

[1]

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- (b) The diagram below shows the structure of the  $\beta$ -bungarotoxin. The protein is composed of two polypeptide chains.



- (i) State the highest level of protein structure exhibited by the  $\beta$ -bungarotoxin. [1]

- (ii) Name the structure labelled X and state how this structure is maintained. [2]

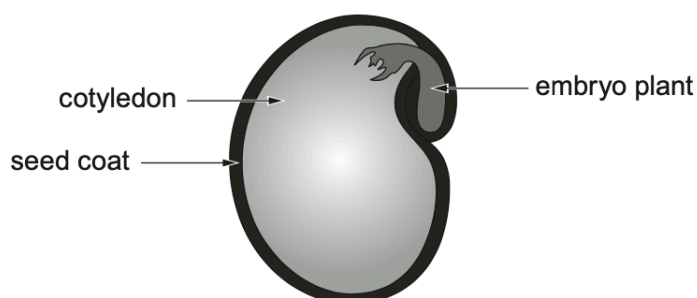
- (iii) State the minimum number of genes required to code for the  $\beta$ -bungarotoxin. Explain how you reached this conclusion. [1]

- (c) (i) One polypeptide chain of  $\beta$ -bungarotoxin contains 120 amino acids. State the minimum number of DNA nucleotides that would be required to code for this polypeptide. [1]

- (ii) Scientists have isolated, and sequenced, the gene for this polypeptide chain; it was found to contain over 2000 nucleotides. Scientists have also isolated molecules of mRNA, which code for this polypeptide, and found that some mRNA molecules had a higher molecular mass than the others. What conclusions can be drawn about the production of this polypeptide? [5]

|   |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |          |          |           |          |  |
|---|-----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|--|
| 4 | (a) | (i)   | Molecules drawn with a peptide bond correct (1)<br>Molecule of <u>water/H<sub>2</sub>O</u> also produced (1)                                                                                                                                                                                                                                                                                                                                                                                         | 2        |          |          | 2         |          |  |
|   |     | (ii)  | peptide bond                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1        |          |          | 1         |          |  |
|   | (b) | (i)   | Quaternary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1        |          |          | 1         |          |  |
|   |     | (ii)  | α helix (1)<br>By hydrogen bonding (1) Ignore reference to peptide bonds                                                                                                                                                                                                                                                                                                                                                                                                                             | 2        |          |          | 2         |          |  |
|   |     | (iii) | Two - one for each polypeptide chain (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |          | 1        | 1         |          |  |
|   | (c) | (i)   | 360 / 363 / 366 (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          | 1        |          | 1         | 1        |  |
|   |     | (ii)  | <b>Any five (×1) from:</b><br>1. DNA/ (pre)mRNA contains both introns and exons (1)<br>2. The entire gene is transcribed (1)<br>3. Introns are non-coding/ only exons are coding (1)<br>4. The (pre)mRNA is then edited to remove the introns / introns are removed from (pre)mRNA/ only leave exons(1)<br>5. 'pre-edited' (pre) mRNA molecules and 'post-edited' mRNA molecules have different masses/ OWTTE (1)<br>6. only the 'edited' mRNA is used {in translation / to produce the protein} (1) |          | 2        | 3        | 5         |          |  |
|   |     |       | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>6</b> | <b>3</b> | <b>4</b> | <b>13</b> | <b>1</b> |  |

7. The diagram below shows the generalised structure of a seed.



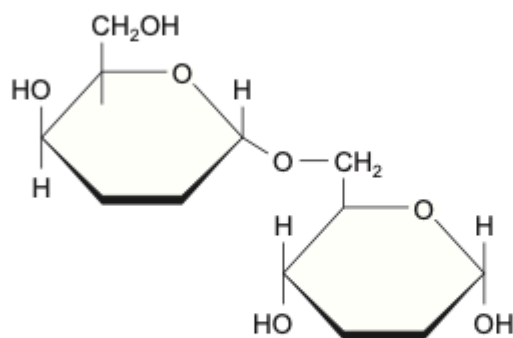
The cotyledon contains many organic molecules including starch and triglycerides. Describe the structure of starch and triglycerides and suggest how their structure and properties relate to their function as energy storage molecules. [9 QER]

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |   |  |  |  |  |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--|--|--|--|
| 7 | <p><b>Starch</b></p> <ul style="list-style-type: none"> <li>• Polymer of α-glucose</li> <li>• Composed of amylose and amylopectin</li> <li>• Amylose contains only 1,4 glycosidic bonds</li> <li>• Forms a helical structure</li> <li>• Amylopectin contains 1,4 and 1,6 glycosidic bonds</li> <li>• Forms a branched structure</li> </ul> <p><b>Triglycerides/lipids</b></p> <ul style="list-style-type: none"> <li>• Composed of glycerol and three fatty acids</li> <li>• Joined together by ester bonds</li> <li>• Saturated fatty acids contain only C-C single bonds</li> <li>• Unsaturated fatty acids contain at least one C=C double bond</li> <li>• Properties of triglyceride dependent upon the fatty acids they contain</li> </ul> <p><b>Functions in the seed</b></p> <ul style="list-style-type: none"> <li>• Starch and triglycerides are insoluble so are osmotically inert</li> <li>• Starch's helical/branched structure makes the molecule compact</li> <li>• Hydrolysis provides glucose readily</li> <li>• Required for respiration/produce ATP</li> <li>• Triglycerides also have a compact structure</li> <li>• Triglycerides have many high energy bonds/ provide approximately twice the quantity of energy than starch</li> </ul> | 6 | 3 |  |  |  |  |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--|--|--|--|

## WJEC 2019

5. Melibiose is a reducing sugar that is formed from galactose and glucose and is thirty times sweeter than sucrose.

(a) The diagram below shows the structure of melibiose.



- (i) **Complete the diagram above** to show how this molecule would be broken down into two monosaccharides. [2]
- (ii) State the type of reaction that occurs when this molecule is broken down into two monosaccharides. [1]
- .....

(iii) Describe a biochemical test that could be used to determine the presence of a reducing sugar such as melibiose. [2]

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|   |     |       |                                                                                                                                                                |   |   |  |   |  |   |
|---|-----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--|---|--|---|
| 5 | (a) | (i)   | Addition of water (1)<br>Galactose and glucose correctly drawn<br>(OH below C <sup>1</sup> on galactose + CH <sub>2</sub> OH at C <sup>6</sup> on glucose) (1) | 1 | 1 |  | 2 |  |   |
|   |     | (ii)  | Hydrolysis (1)                                                                                                                                                 | 1 |   |  | 1 |  |   |
|   |     | (iii) | Add Benedict's reagent/solution and heat (1) NOT ref to HCl<br>Change of colour <u>from blue</u> to<br>{yellow/green/orange/brown/(brick) red} (1)             | 2 |   |  | 2 |  | 2 |

6. The following extracts are adapted from articles published on the Internet.

**Hidden ocean found on Saturn's icy moon, Enceladus, could potentially support life.**

Enceladus harbours a big ocean of liquid water beneath its icy crust that may be capable of supporting life as we know it. Researchers said that the water is about 10 kilometres deep and lies beneath a shell of ice 30 to 40 km thick. Furthermore, it is in direct contact with a rocky sea floor, theoretically making possible all kinds of complex chemical reactions – such as the kind that may have led to the rise of life on Earth.

**NASA finds more evidence that the ocean on Enceladus could support alien life.**

NASA's Cassini spacecraft orbiting the moon has analysed the plumes of gas forced out through fissures in the ice. These plumes have been found to contain four of the six most important elements of life on Earth – carbon, hydrogen, nitrogen and oxygen – only phosphorus and sulfur have not been detected.

Describe the structure of water and explain how the properties of water would be essential to supporting life on Enceladus. Explain how the absence of phosphorus and sulfur would prevent the formation of biochemical molecules essential for life on Earth. [9 QER]

6

Indicative content

**Structure of water:**

Water has the chemical formula of H<sub>2</sub>O.

Water consists of one oxygen atom joined to two hydrogen atoms by covalent bonds.

The molecule is a dipole.

The oxygen atom has a slight negative charge / δ<sup>-</sup>, the hydrogen atoms have a slightly positive charge δ<sup>+</sup>.

| <b>Properties of water</b>      | <b>Essential to supporting life</b>                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Ice is less dense than water    | Ice floats insulating the water below allowing organisms to survive.                                                  |
| Polar solvent                   | All chemical reactions occur in aqueous solutions. Polar molecules and ions can dissolve and therefore be transported |
| Metabolite                      | Involved in chemical reactions                                                                                        |
| High specific heat capacity     | Requires a lot of energy to cause water to heat up / cool down therefore creating thermostable environments           |
| Cohesive / forms hydrogen bonds | Water molecules form hydrogen bonds allowing the movement of water                                                    |

**Chemical elements:**

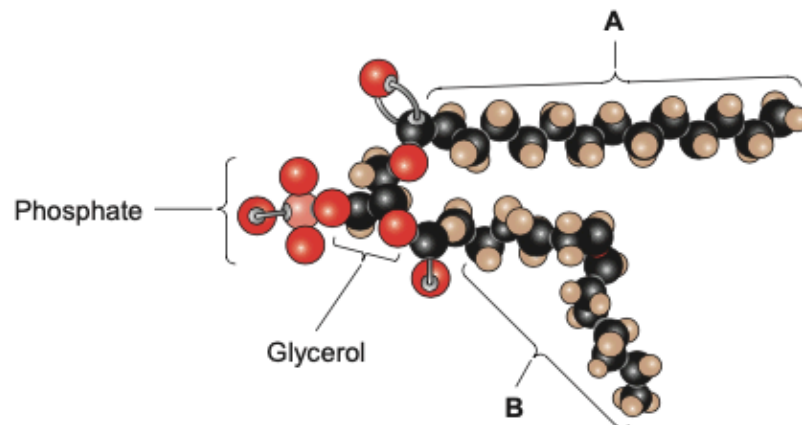
Phosphorous is required to synthesise phospholipids / nucleotides / ATP / DNA / RNA

Sulfur is required to synthesise some amino acids / methionine / cysteine / proteins.

4 5 0 9 0 0

2. (a) **Image 2.1** shows a molecular model of a phospholipid, a component of cell membranes.

**Image 2.1**



- (i) Structures **A** and **B** are both types of fatty acids. Identify both types and explain the difference in their shapes. [2]

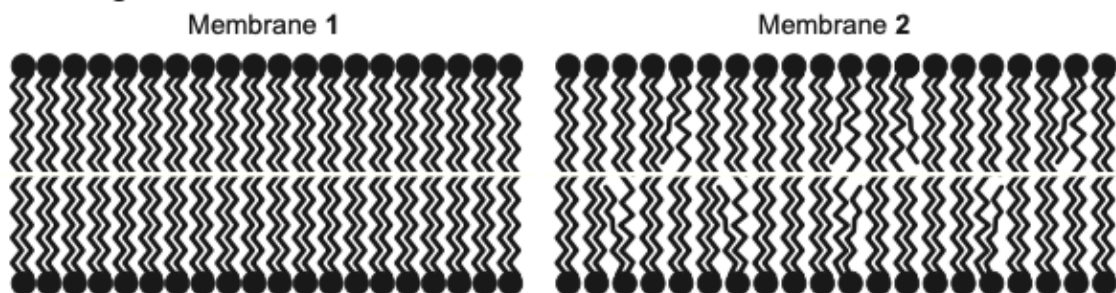
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- (ii) **Image 2.2** shows two examples of phospholipid bilayers. Membrane 1 has phospholipids containing only type **A** fatty acids and membrane 2 has phospholipids containing both type **A** and type **B**.

**Image 2.2**



Using the information provided and your own knowledge, explain which membrane you would expect to be more permeable to small molecules. [2]

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- (b) (i) Fatty acids are also found as part of triglycerides.

State how triglycerides differ from phospholipids in their structure and in their function in cells.

[2]

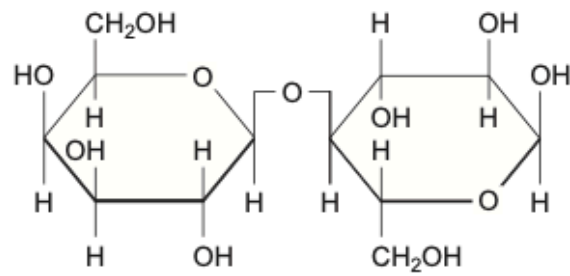
- (ii) State how the relative proportions of type A and B fatty acids in the diet can affect human health.

[1]

|               |                             |                                             |                                                                                                                                                                                                                                                                                                              |   |              |              |               |                            |                               |              |                             |                                             |   |  |  |   |  |  |
|---------------|-----------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------------|--------------|---------------|----------------------------|-------------------------------|--------------|-----------------------------|---------------------------------------------|---|--|--|---|--|--|
| 2.            | (a)                         | (i)                                         | A is a saturated fatty acid + B is unsaturated (1)<br>A contains only single bonds between C atoms +<br>B contains (at least) one double bonds between C atoms (1)                                                                                                                                           | 2 |              |              | 2             |                            |                               |              |                             |                                             |   |  |  |   |  |  |
|               |                             | (ii)                                        | <ul style="list-style-type: none"><li>• Example 2 (1)</li><li>• Because there are bigger spaces between {phospholipids/ fatty acid tails} the {phospholipids/ fatty acid tails} fit together less closely/ less densely packed (1)</li></ul>                                                                 |   |              | 2            | 2             |                            |                               |              |                             |                                             |   |  |  |   |  |  |
|               | (b)                         | (i)                                         | <table><tr><td></td><td>Phospholipid</td><td>Triglyceride</td></tr><tr><td>Structure (1)</td><td>2 fatty acids<br/>phosphate</td><td>3 fatty acids<br/>no phosphate</td></tr><tr><td>Function (1)</td><td>Component of cell membranes</td><td>Energy store/<br/>release of metabolic water</td></tr></table> |   | Phospholipid | Triglyceride | Structure (1) | 2 fatty acids<br>phosphate | 3 fatty acids<br>no phosphate | Function (1) | Component of cell membranes | Energy store/<br>release of metabolic water | 2 |  |  | 2 |  |  |
|               | Phospholipid                | Triglyceride                                |                                                                                                                                                                                                                                                                                                              |   |              |              |               |                            |                               |              |                             |                                             |   |  |  |   |  |  |
| Structure (1) | 2 fatty acids<br>phosphate  | 3 fatty acids<br>no phosphate               |                                                                                                                                                                                                                                                                                                              |   |              |              |               |                            |                               |              |                             |                                             |   |  |  |   |  |  |
| Function (1)  | Component of cell membranes | Energy store/<br>release of metabolic water |                                                                                                                                                                                                                                                                                                              |   |              |              |               |                            |                               |              |                             |                                             |   |  |  |   |  |  |
|               |                             | (ii)                                        | A higher proportion of {saturated/ type A} fatty acids can lead to {heart disease/ arteriosclerosis/ cardiovascular disease/ reference to increase in LDL leading to CVD} /ORA (1)                                                                                                                           | 1 |              |              | 1             |                            |                               |              |                             |                                             |   |  |  |   |  |  |
|               |                             |                                             | Question 2 total                                                                                                                                                                                                                                                                                             | 5 | 0            | 2            | 7             | 0                          | 0                             |              |                             |                                             |   |  |  |   |  |  |

- (b) The main carbohydrate in milk is lactose. **Image 5.2** shows a molecule of lactose.

**Image 5.2**



- (i) **Complete Image 5.2** to show the digestion of lactose.

[2]

- (ii) State:

I. the type of reaction;

[1]

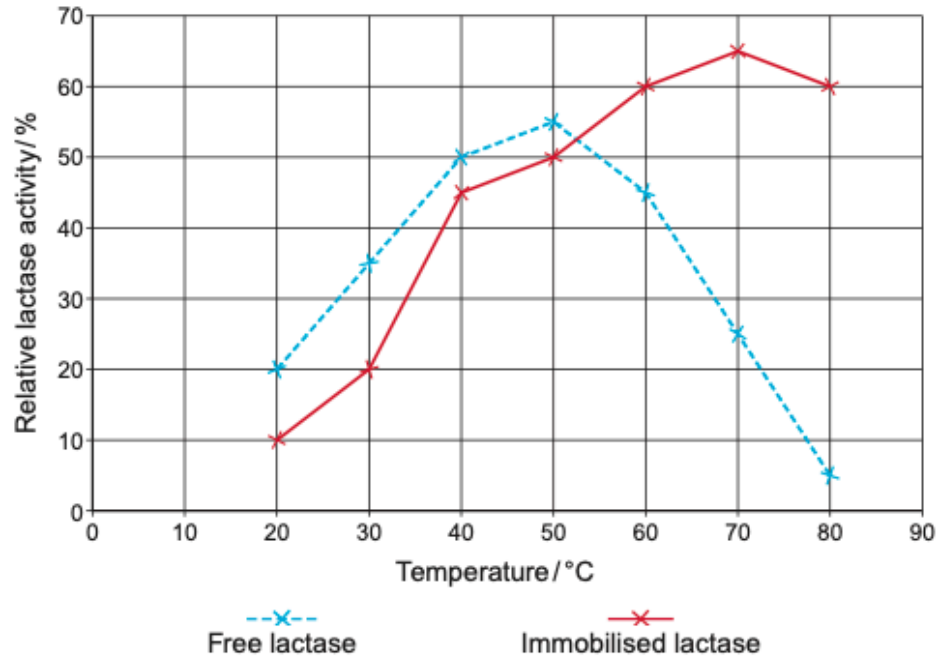
II. the names of the products of digestion of lactose.

[1]



- (c) Some people are lactose intolerant and in recent years it has been possible to produce lactose free milk and milk products using the enzyme lactase. One way to do this is to pour the milk through a column of alginate beads containing immobilised lactase. **Graph 5.3** shows the activity of free lactase and immobilised lactase at different temperatures.

**Graph 5.3**



- (i) Using **Graph 5.3** and your own knowledge, compare the relative lactase activity of the free and immobilised lactase at the following temperatures. Explain each answer.

I. Below 50 °C.

[2]

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II. Above 50 °C.

[3]

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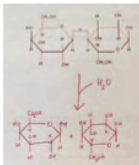
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(ii) Lactose is a reducing sugar. Explain why it is not possible to use Benedict's reagent to monitor the progress of the digestion of lactose by lactase.

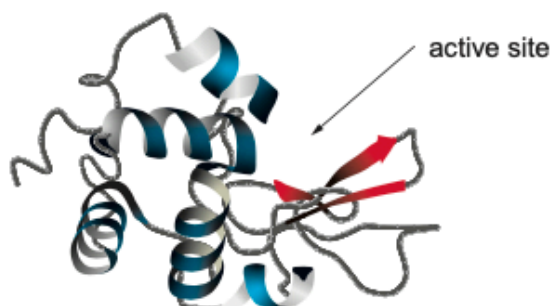
[1]

|     |      |    |                                                                                                                                                                                                                                                                                                                                                          |          |          |          |           |          |          |
|-----|------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
| (b) | (i)  |    |  <p>Water added (1)<br/>Products drawn correctly (1)</p>                                                                                                                                                                                                                | 2        |          |          | 2         |          |          |
|     | (ii) | I  | Hydrolysis (1)                                                                                                                                                                                                                                                                                                                                           | 1        |          |          | 1         |          |          |
|     |      | II | Galactose and Glucose (1)                                                                                                                                                                                                                                                                                                                                | 1        |          |          | 1         |          |          |
| (c) | (i)  | I  | <p><b>Any two (x1) from</b></p> <p>A. (Below 50°C) lactase {is more active/ has more kinetic energy} when free (1)</p> <p>B. (Because lactase and lactose can move freely) there is more chance of {successful collisions/ enzyme-substrate complexes forming} (1)</p> <p>C. Takes longer for lactose to diffuse through the alginate beads/ ORA (1)</p> |          | 1        | 1        | 2         |          |          |
|     |      | II | <p>A. lactase is more active when immobilised/ comparison of trends of immobilised and free (1) ORA</p> <p>B. the free lactase is {denaturing/denatured} (1)</p> <p>C. Because the immobilisation {stabilises lactase/ maintains shape of active site} (1)</p>                                                                                           |          | 2        | 1        | 3         |          |          |
|     | (ii) |    | <p>All give a (blue to orange) colour change with Benedict's so there would be no difference in reactants or products in the reaction/<br/>Reactants and products would both give a blue to orange colour change/<br/>All {monosaccharides/ glucose and galactose} are reducing sugars (1)</p>                                                           |          |          | 1        | 1         |          | 1        |
|     |      |    | <b>Question 5 total</b>                                                                                                                                                                                                                                                                                                                                  | <b>7</b> | <b>3</b> | <b>4</b> | <b>14</b> | <b>0</b> | <b>1</b> |

WJEC 2023

1. Lysozyme is an enzyme found in saliva and tears. It hydrolyses the carbohydrates in bacterial cell walls. Lysozyme, shown in **Image 1.1**, is a single polypeptide containing 129 amino acids.

**Image 1.1 Ribbon diagram of lysozyme**



- (a) Lysozyme shows primary, secondary and tertiary structure.

- (i) State what is meant by primary structure.

[1]

- (ii) Use the information given and your own knowledge to **complete Table 1.2** to explain how the secondary and tertiary structures of lysozyme are illustrated in **Image 1.1**.

[2]

**Table 1.2**

| Level of structure | Explanation    |
|--------------------|----------------|
| Secondary          | .....<br>..... |
| Tertiary           | .....<br>..... |

|   |     |      |                                                                                                                                                                           |   |   |  |   |  |  |
|---|-----|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--|---|--|--|
| 1 | (a) | (i)  | (Primary structure is) the {sequence/ order} of amino acids (1)                                                                                                           | 1 |   |  | 1 |  |  |
|   |     | (ii) | Secondary structure – alpha helices / beta pleated sheet (1)<br>Tertiary structure – (secondary structure) {folded into/ held in} a {globular /3D} {structure/ shape} (1) |   | 2 |  | 2 |  |  |

4. Starch is made up of two molecules, amylose and amylopectin, shown in **Image 4.1**.

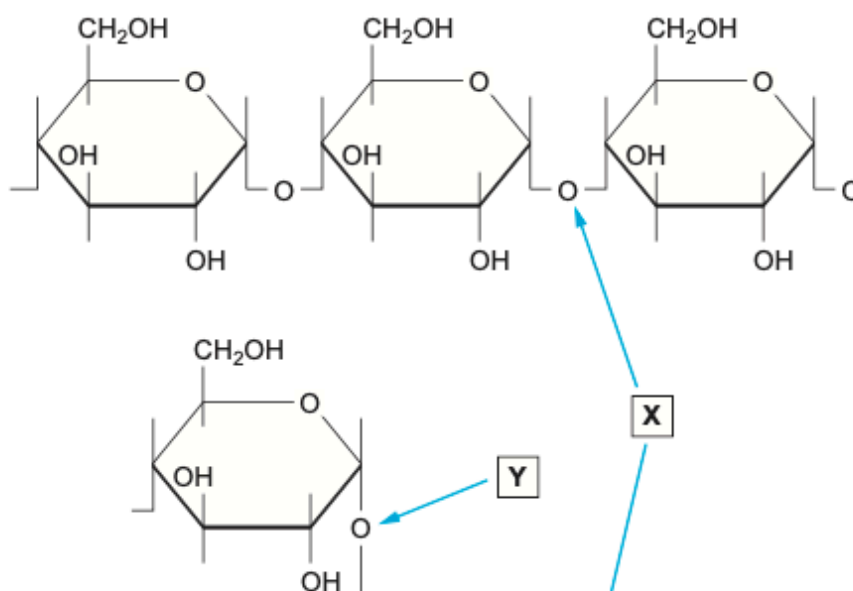
**Image 4.1**



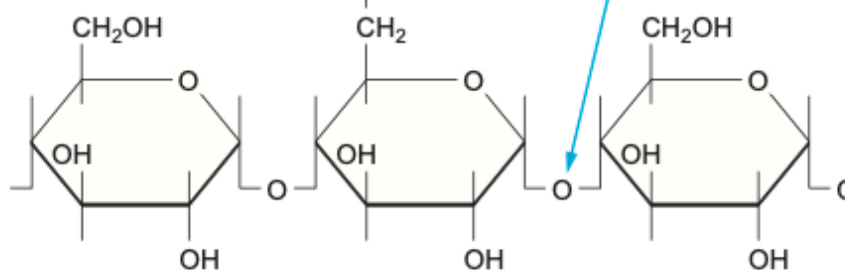
**Image 4.2** shows sections of amylose and amylopectin.

**Image 4.2**

**Amylose**



**Amylopectin**



- (a) (i) State the general name given to **bonds X** and **Y**.

[1]

- (ii) State **one** similarity and **one** difference between amylose and amylopectin, shown in **Images 4.1** and **4.2**. [2]

Similarity .....

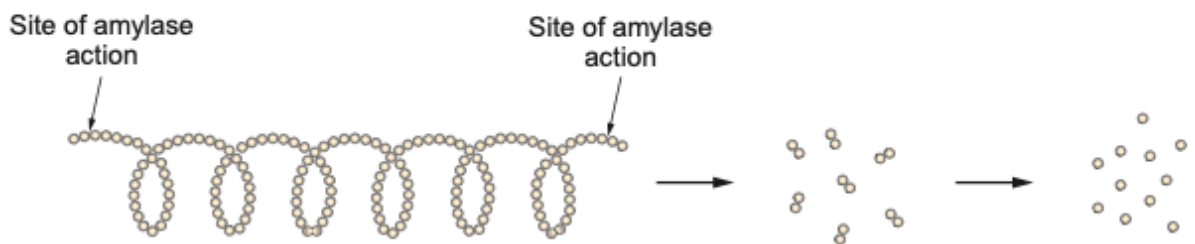
.....

Difference .....

.....

- (b) In order to be absorbed into the blood, starch has to be hydrolysed to produce glucose. The digestion of amylose happens in two stages as shown in **Image 4.3**.

**Image 4.3**



- (i) State what is meant by the term hydrolysis. [1]

.....

.....

Amylopectin is digested in a similar way but needs an additional enzyme, isomaltase, for complete hydrolysis.

- (ii) Explain the roles of the **three** enzymes involved in the complete hydrolysis of starch. [3]

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- (iii) One variety of corn used in animal feed has a lower amylose : amylopectin ratio than other varieties. With reference to **Images 4.2 and 4.3**, suggest why this lower ratio results in an increase in the rate of starch hydrolysis. [2]

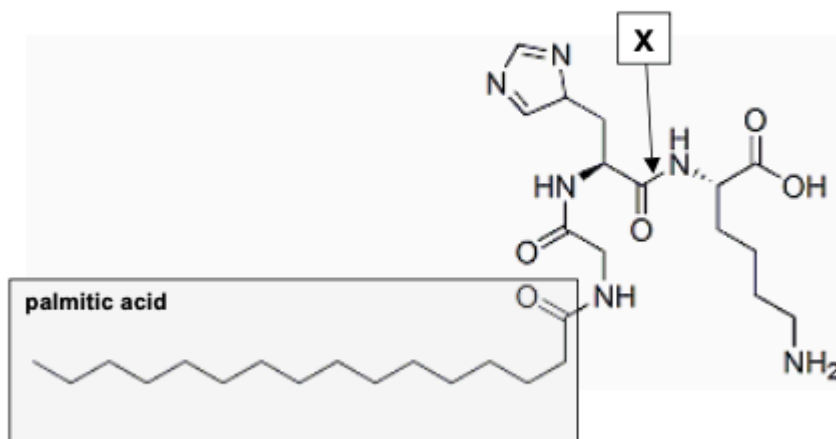
- (iv) Describe a chemical test that could be used to show that starch is no longer present at the end of hydrolysis. [2]

|   |     |       |                                                                                                                                                                                                                                                                                                                                   |          |          |          |           |          |          |
|---|-----|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
| 4 | (a) | (i)   | Glycosidic (bond) (1)                                                                                                                                                                                                                                                                                                             | 1        |          |          | 1         |          |          |
|   |     | (ii)  | <b>Similarity:</b><br>both made of alpha glucose/<br>both have (alpha)1,4 glycosidic bonds (1)<br><b>Difference:</b><br>Amylose straight chain and amylopectin branched/<br>amylose coiled and amylopectin branched/ amylopectin branched<br>amylose is not /<br>amylopectin has 1,6 glycosidic bonds amylose does not/ owtte (1) |          | 1<br>1   |          | 2         |          |          |
|   | (b) | (i)   | the {(chemical) insertion/ addition} of water to break the bond (1)                                                                                                                                                                                                                                                               | 1        |          |          | 1         |          |          |
|   |     | (ii)  | {Starch/ amylopectin/ amylose} is digested to maltose by amylase (1)<br>Maltose is digested to glucose by {maltase/ an enzyme} (1)<br>Isomaltase {debranches/ hydrolyses/ breaks} the amylopectin at the {branch points/ 1-6 glycosidic bonds/ bond Y} (1)                                                                        |          | 2        |          | 3         |          |          |
|   |     | (iii) | <b>Any two (x1) from</b><br>More amylopectin present (1)<br>It has more {ends/ sites} {to be digested / for the enzyme act on} (1)<br>It is less tightly packed so easier {to digest/ for enzyme to access} (1)<br>Accept reverse argument for all mark points.                                                                   |          |          | 1<br>1   | 2         |          |          |
|   |     | (iv)  | Iodine solution (1)<br>would stay brown (1)                                                                                                                                                                                                                                                                                       | 1        | 1        |          | 2         |          | 2        |
|   |     |       | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                           | <b>3</b> | <b>5</b> | <b>3</b> | <b>11</b> | <b>0</b> | <b>2</b> |

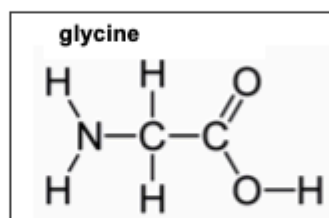
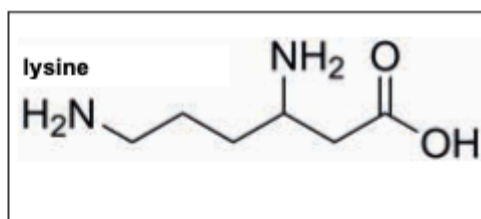
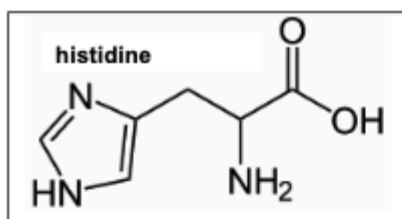
# Speciment

1. Palmitoyl tripeptide-1 is made of three amino acids bonded to a molecule of palmitic acid, a component of one form of a triglyceride. It is used in anti-ageing creams to stimulate collagen repair in skin.

The diagram below shows the structure of palmitoyl tripeptide-1.



The structural formulae of the amino acids present in this tripeptide are shown below.



- (i) Name the bond labelled **X** on the diagram. [1]

.....

- (ii) Use the diagrams of the individual amino acids to identify the primary structure of this tripeptide. [2]

**palmitic acid** – ..... " ..... ".....

- (iii) The molecule is claimed to be better at penetrating skin due to it having hydrophilic and hydrophobic properties. Name the part of the molecule which is hydrophobic. [1]

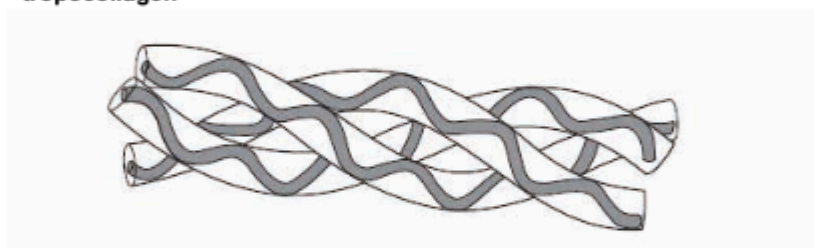
.....

- (b) Collagen is one of the main structural proteins found in skin and contains over 30% glycine. Each collagen molecule contains about 1000 amino acids. The fully functional protein is composed of three molecules of collagen and is called tropocollagen. The diagram below shows part of a single collagen molecule and a tropocollagen molecule.

**collagen**



**tropocollagen**



- (i) Using the diagrams above, explain why collagen is said to have a secondary structure but tropocollagen has a quaternary structure. [3]

.....  
 .....  
 .....

- (ii) Describe how the level of protein structure shown in the collagen diagram above would be different if the protein had a tertiary structure. [1]

.....

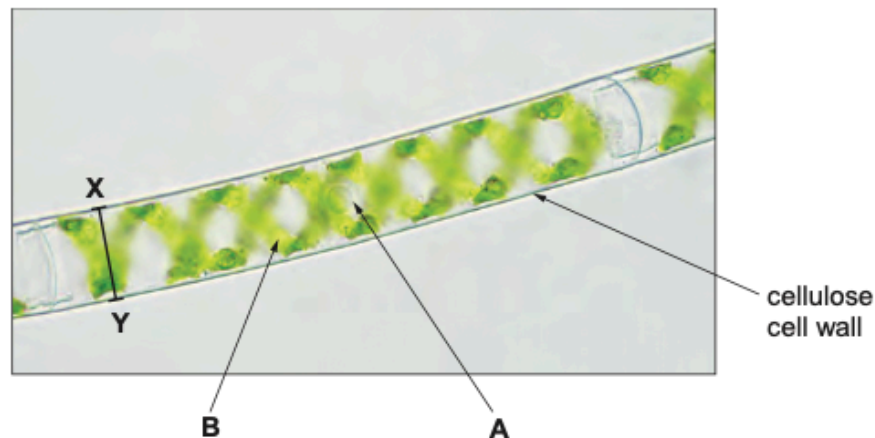


| Question |     |       |  | Marking details                                                                                                                                                                           | Marks available |          |          |          |          |          |
|----------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |       |  |                                                                                                                                                                                           | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 1        | (a) | (i)   |  | peptide bond (1)                                                                                                                                                                          | 1               |          |          | 1        |          |          |
|          |     | (ii)  |  | Palmitic acid – glycine - histidine - lysine (2)<br>Any two in correct order (1)                                                                                                          |                 | 2        |          | 2        |          |          |
|          |     | (iii) |  | palmitic acid (1)                                                                                                                                                                         | 1               |          |          | 1        |          |          |
|          |     |       |  |                                                                                                                                                                                           |                 |          |          |          |          |          |
|          | (b) | (i)   |  | collagen is a single polypeptide chain (1)<br>which has been coiled into an alpha helix (1)<br>tropocollagen is composed of {three / 2 or more} polypeptide<br>chains bonded together (1) | 3               |          |          | 3        |          |          |
|          |     | (ii)  |  | folding of molecule into a globular shape due to bonding/<br>interactions between R groups(1)                                                                                             | 1               |          |          | 1        |          |          |
|          |     |       |  | <b>Question 1 total</b>                                                                                                                                                                   | <b>6</b>        | <b>2</b> | <b>0</b> | <b>8</b> | <b>0</b> | <b>0</b> |

1.2

# WJEC 2016

3. The photomicrograph below is of *Spirogyra*; an autotrophic organism that inhabits fresh water ponds and ditches.



- (a) Identify the organelles labelled **A** and **B**. [2]

**A:** .....

**B:** .....

- (b) The actual width of the cell between points **X** – **Y** was 32.3  $\mu\text{m}$ . Calculate the magnification that was used to take the photomicrograph. [2]

magnification = .....

- (c) During the day the concentration of solutes in the pond water changes due to evaporation. However, the length of the *Spirogyra* cells remain almost constant. What can you conclude about how the structure of the cell wall enables *Spirogyra* to survive in different solute concentrations? [4]

.....

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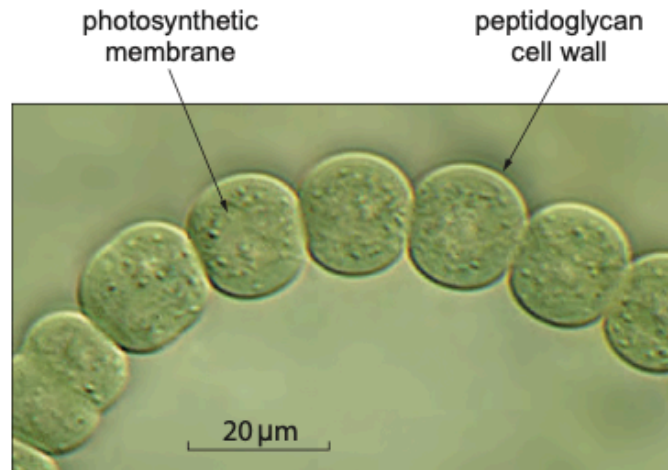
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- (d) Below is a photomicrograph of the bacteria *Nostoc*, which also inhabits fresh water ponds and ditches. They were once thought to belong to the same group of organisms as *Spirogyra* and thought to have the same cellular structure. Evidence from electron microscopy has now grouped these two separately.

Conclude what cell types are present in *Spirogyra* and *Nostoc*. Identify **two** differences and **one** similarity (not labelled in the photomicrographs) between these two species that would be revealed by electron microscopy. [3]



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| 3                                                            | (a)                                                 |  | A: nucleus (1) Accept nucleolus<br>B: chloroplast (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1                | 1             |                                                      | 2                                                   |                                      |                       |            |                            |                                                              |                       |                  |  |                              |                                  |                 |                  |  |                 |  |   |  |  |
|--------------------------------------------------------------|-----------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------|------------------------------------------------------|-----------------------------------------------------|--------------------------------------|-----------------------|------------|----------------------------|--------------------------------------------------------------|-----------------------|------------------|--|------------------------------|----------------------------------|-----------------|------------------|--|-----------------|--|---|--|--|
|                                                              | (b)                                                 |  | apparent length = 13mm / 13 x 1000 = 13000µm<br>magnification = 13000 / 32.3 µm or 13/0.0323 (1)<br>magnification = 402.48/ 402.5/ 403/ 402<br>(correct answer = 2 marks)<br>OR<br>apparent length = 13.5mm / 13.5 x 1000 = 13500µm<br>magnification = 13500 / 32.3 µm or 13.5/ 0.0323 (1)<br>magnification = 417.96/ 418 (correct answer = 2 marks)<br><br>Award 1 mark for evidence of figures showing image size/ actual size                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                  | 2             |                                                      | 2                                                   | 2                                    |                       |            |                            |                                                              |                       |                  |  |                              |                                  |                 |                  |  |                 |  |   |  |  |
|                                                              | (c)                                                 |  | 1. straight chains of β-glucose/ alternative molecules of β-glucose rotate through 180° / chains cross-linked / form microfibrils (1)<br>2. which provide {strength/rigidity/inelasticity} to the cell wall (1) support = neutral<br>3. When the {solute concentration/ solute potential} changes (water potential will change) causing water to {move into /out of} the cell (1)<br>4. cell wall prevents {osmotic lysis/ cell bursting}/ cell wall prevents cell shrinking (1) turgid/ plasmolysed = neutral<br>Reject if direction of water movement is incorrect                                                                                                                                                                                                                                                                                                                   | 1<br><br>1       |               | 1<br><br>1                                           | 4                                                   |                                      |                       |            |                            |                                                              |                       |                  |  |                              |                                  |                 |                  |  |                 |  |   |  |  |
| 3                                                            | (d)                                                 |  | <i>Spirogyra</i> – eukaryotic cells & <i>Nostoc</i> - prokaryotic cells (1)<br>Similarity: Both contain ribosomes / cell membranes / DNA/ genetic material (1)<br><br><b>Difference: any 2 from: (1 mark)</b> <table><tr><th><i>Spirogyra</i></th><th><i>Nostoc</i></th></tr><tr><td>{Membrane bound organelles/ named organelle} present</td><td>{Membrane bound organelles/ named organelle} absent</td></tr><tr><td>DNA enclosed within nuclear membrane</td><td>DNA free in cytoplasm</td></tr><tr><td>Linear DNA</td><td>Loop of DNA Accept plasmid</td></tr><tr><td>Larger/80s ribosomes<br/>Accept ribosomes are different sizes</td><td>Smaller/70s ribosomes</td></tr><tr><td colspan="2">Accept following</td></tr><tr><td>DNA associated with histones</td><td>DNA not associated with histones</td></tr><tr><td>Mesosome absent</td><td>Mesosome present</td></tr></table> | <i>Spirogyra</i> | <i>Nostoc</i> | {Membrane bound organelles/ named organelle} present | {Membrane bound organelles/ named organelle} absent | DNA enclosed within nuclear membrane | DNA free in cytoplasm | Linear DNA | Loop of DNA Accept plasmid | Larger/80s ribosomes<br>Accept ribosomes are different sizes | Smaller/70s ribosomes | Accept following |  | DNA associated with histones | DNA not associated with histones | Mesosome absent | Mesosome present |  | 1<br>1<br><br>1 |  | 3 |  |  |
| <i>Spirogyra</i>                                             | <i>Nostoc</i>                                       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |               |                                                      |                                                     |                                      |                       |            |                            |                                                              |                       |                  |  |                              |                                  |                 |                  |  |                 |  |   |  |  |
| {Membrane bound organelles/ named organelle} present         | {Membrane bound organelles/ named organelle} absent |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |               |                                                      |                                                     |                                      |                       |            |                            |                                                              |                       |                  |  |                              |                                  |                 |                  |  |                 |  |   |  |  |
| DNA enclosed within nuclear membrane                         | DNA free in cytoplasm                               |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |               |                                                      |                                                     |                                      |                       |            |                            |                                                              |                       |                  |  |                              |                                  |                 |                  |  |                 |  |   |  |  |
| Linear DNA                                                   | Loop of DNA Accept plasmid                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |               |                                                      |                                                     |                                      |                       |            |                            |                                                              |                       |                  |  |                              |                                  |                 |                  |  |                 |  |   |  |  |
| Larger/80s ribosomes<br>Accept ribosomes are different sizes | Smaller/70s ribosomes                               |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |               |                                                      |                                                     |                                      |                       |            |                            |                                                              |                       |                  |  |                              |                                  |                 |                  |  |                 |  |   |  |  |
| Accept following                                             |                                                     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |               |                                                      |                                                     |                                      |                       |            |                            |                                                              |                       |                  |  |                              |                                  |                 |                  |  |                 |  |   |  |  |
| DNA associated with histones                                 | DNA not associated with histones                    |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |               |                                                      |                                                     |                                      |                       |            |                            |                                                              |                       |                  |  |                              |                                  |                 |                  |  |                 |  |   |  |  |
| Mesosome absent                                              | Mesosome present                                    |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |               |                                                      |                                                     |                                      |                       |            |                            |                                                              |                       |                  |  |                              |                                  |                 |                  |  |                 |  |   |  |  |
|                                                              |                                                     |  | <b>Question 3 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>3</b>         | <b>6</b>      | <b>2</b>                                             | <b>11</b>                                           | <b>2</b>                             | <b>0</b>              |            |                            |                                                              |                       |                  |  |                              |                                  |                 |                  |  |                 |  |   |  |  |

# WJEC 2017

1. The images below show a longitudinal section through a tissue found lining the trachea of humans and a highly magnified image of the surface of this tissue.

Image 1

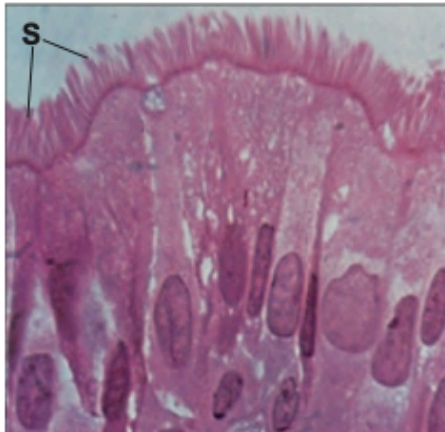
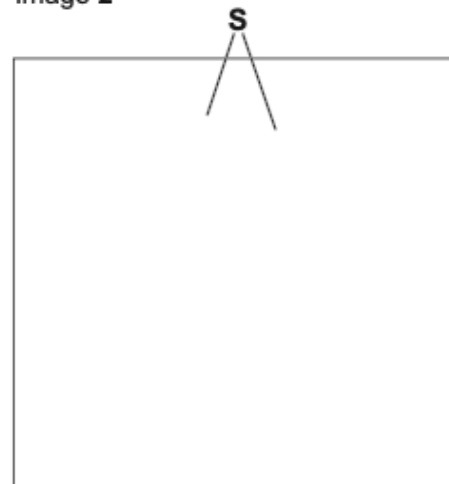


Image 2



- (a) (i) Name the structures labelled **S** and suggest their function in the human respiratory system. [2]

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.....

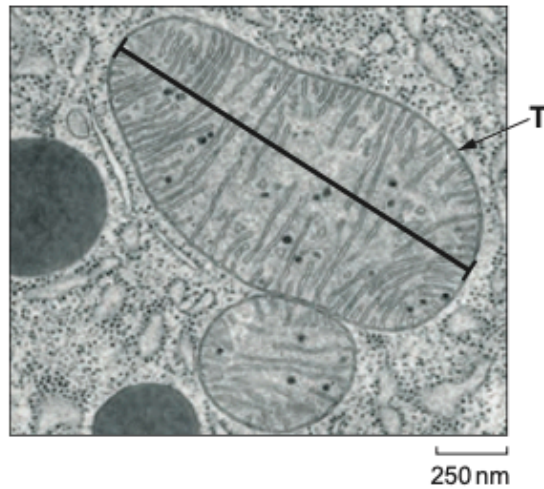
- (ii) Explain why Image 1 is referred to as a tissue rather than an organ. [1]

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The photo electron micrograph below shows two mitochondria found in this tissue.



- (b) (i) One reason for the different appearance of the mitochondria is that they have been cut in different planes. Suggest **one other** reason that could account for the difference in their appearance. [1]

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.....

- (ii) Calculate the actual size of mitochondrion T along the line shown on the image. Give your answer in micrometres. [3]

Length of organelle = .....  $\mu\text{m}$

- (iii) State and explain **one** way in which mitochondria are adapted to perform their function. [2]

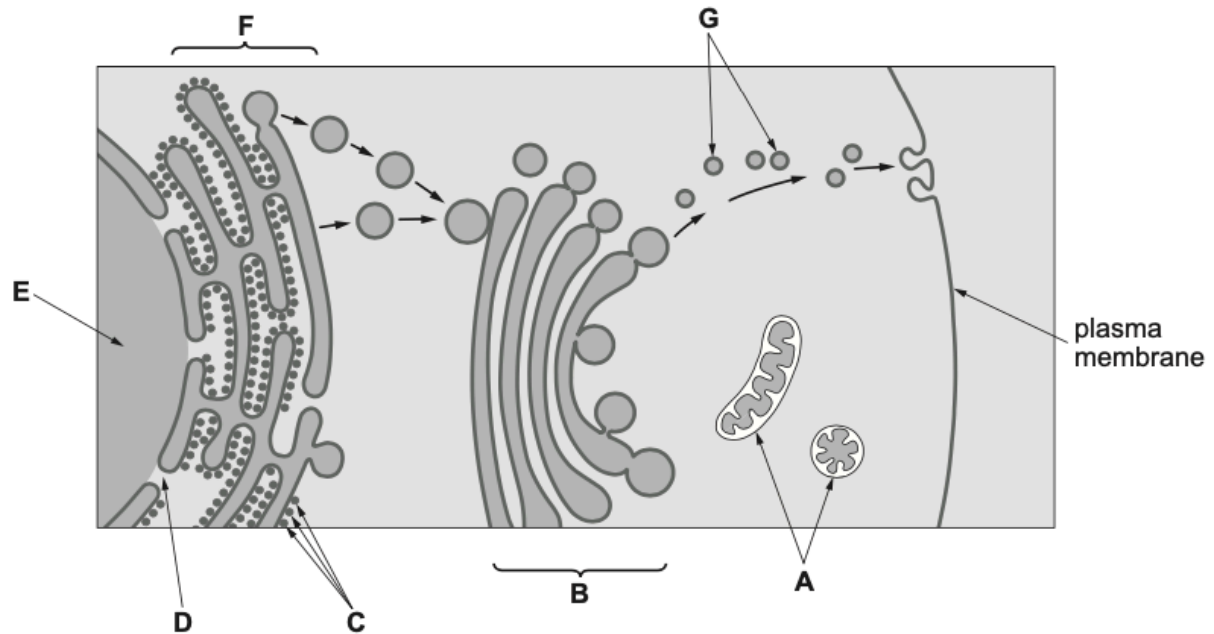
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|   |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                      |          |          |          |          |          |          |
|---|-----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|----------|----------|----------|
| 1 | (a) | (i)   | Cilia (1) Ignore ciliated cell<br>{Move/ sweep/ push/ clear} mucus (1)                                                                                                                                                                                                                                                                                                                                               | 2        |          |          | 2        |          |          |
|   |     | (ii)  | <ul style="list-style-type: none"> <li>An organ is an aggregation of several tissues but this only shows one/</li> <li>{It/ Image 1} only contains one type of {cell/ tissue}</li> </ul> Ignore similar cells                                                                                                                                                                                                        | 1        |          |          | 1        |          |          |
|   | (b) | (i)   | Self-replication/ Stage (of division) / stage of development / size/ not fully formed/ age                                                                                                                                                                                                                                                                                                                           |          | 1        |          | 1        |          |          |
|   |     | (ii)  | <b>25 x a measurement</b><br>1.45/ 1.475/ 1.48/ 1.5 um = 3 marks<br><b>If incorrect allow (250 x a measurement) for 2 marks</b><br>1450/1475/1480/1500 (nm)<br><b>If incorrect allow a measurement for 1 mark units must be shown</b><br>58/59/60mm or 5.8/5.9/6.0 cm                                                                                                                                                |          | 3        |          | 3        | 3        |          |
|   |     | (iii) | <b>Explanation must match adaptation</b><br>Increased/ larger surface area : volume (ratio)(1)<br>Increased gas exchange (1)<br><br>Cylindrical (1)<br>decreases diffusion distance / large surface area for gas exchange(1)<br><br>Folding of inner membrane/ cristae (1)<br>{Increased/ large} SA for {enzymes/ membrane proteins / (aerobic) respiration/ / ATP production} (1)<br>NOT energy production/ release | 2        |          |          | 2        |          |          |
|   |     |       | <b>Question 1 total</b>                                                                                                                                                                                                                                                                                                                                                                                              | <b>5</b> | <b>4</b> | <b>0</b> | <b>9</b> | <b>3</b> | <b>0</b> |



# WJEC 2018

1. The diagram below shows part of a cell taken from the pancreas, which is involved in the production of digestive enzymes.



- (a) Identify the structures labelled **A**, **B**, **C**, **D**.

[2]

**A:** .....

**B:** .....

**C:** .....

**D:** .....

- (b) (i) With reference to the structures labelled **C, D and E**, describe the sequence of events that lead to the production of digestive enzymes in this cell. [3]

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- (ii) With reference to the structures **B, F and G**, describe the sequence of events that lead to the secretion of digestive enzymes from this cell. [3]

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- (iii) Explain the role of organelle **A** in the production and secretion of digestive enzymes. [2]

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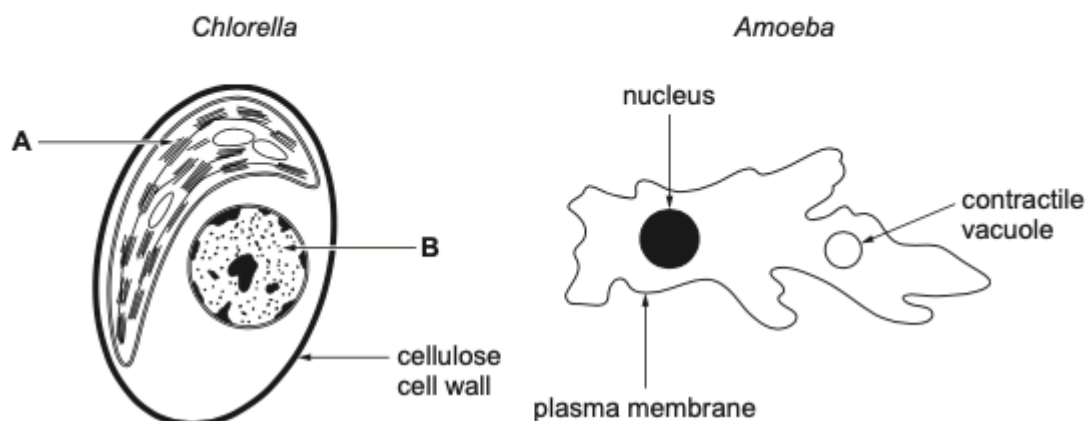
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|   |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |          |          |           |          |          |
|---|-----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
| 1 | (a) |       | A: Mitochondrion/mitochondria<br>B: Golgi {body/complex/apparatus}<br>C: Ribosome(s)<br>D: Nuclear pore<br>2 or 3 correct for 1 mark<br>All correct for 2 marks                                                                                                                                                                                                                                                                                                        | 2        |          |          | 2         |          |          |
|   | (b) | (i)   | Nucleus/ E: (contains DNA which) codes for the production of {proteins/polypeptides/sequence of amino acids in a polypeptide}/ transcription/ {pre-processing/ production/ synthesis} of mRNA (1)<br><br>Nuclear pores/ D: allow {mRNA/rRNA} to leave the nucleus (1)<br><br>Ribosome(s)/C: carry out translation/ protein synthesis/or description of (1)                                                                                                             | 3        |          |          | 3         |          |          |
| 1 | (b) | (ii)  | <b>Any 3 × (1) from:</b><br><br>Rough Endoplasmic Reticulum/ RER/ F: transports proteins {through the cell/ through the cytoplasm}/ transports proteins to golgi body/ package proteins into vesicles (1)<br><br>Golgi body/ B: {Packaging/Modification} of protein/description of/ activation of enzyme (1)<br><br>(transport) vesicle/ G: transports {proteins/enzymes} to the {cell membrane/ plasma membrane} (1)<br><br>Exocytosis (of enzymes from the cell) (1) | 3        |          |          | 3         |          |          |
|   |     | (iii) | <u>Provide ATP :</u><br>{for transcription / translation / protein synthesis} (1)<br><u>exocytosis</u> (1)                                                                                                                                                                                                                                                                                                                                                             |          | 2        |          | 2         |          |          |
|   |     |       | <b>Question 1 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>8</b> | <b>2</b> | <b>0</b> | <b>10</b> | <b>0</b> | <b>0</b> |

## WJEC 2019

2. The diagrams below show two unicellular prototists (not drawn to scale) that inhabit fresh water ponds. The presence of solutes in the cytoplasm of their cells makes them hypertonic to their environment.



- (a) Name organelles A and B.

[1]

organelle A: .....

organelle B: .....

- (ii) Identify **two other** organelles which must be present in large numbers in the *Amoeba* in order for this process to occur. Explain your answer for each organelle. [4]

Organelle 1 .....

Explanation .....

Organelle 2 .....

Explanation .....

- (c) Although *Chlorella* lives in the same habitat as *Amoeba*, it does not possess a contractile vacuole. Describe and explain how the structure of the cell wall of *Chlorella* allows it to survive in this environment. [4]

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- (d) Bacteria, such as *Cyanophyceae* also inhabit fresh water ponds. Describe **two** differences between the structure of *Chlorella* and *Cyanophyceae*. [2]

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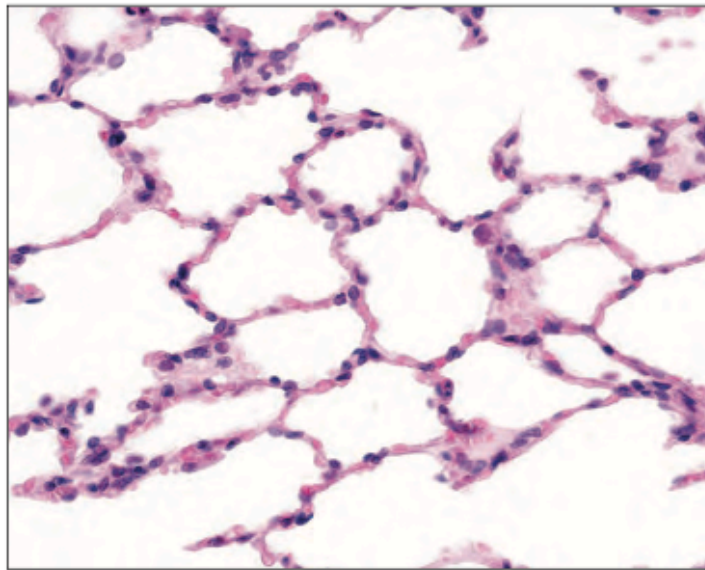
|   |     |  |                                          |   |  |  |   |  |   |
|---|-----|--|------------------------------------------|---|--|--|---|--|---|
| 2 | (a) |  | A chloroplast<br>B nucleus NOT nucleolus | 1 |  |  | 1 |  | 1 |
|---|-----|--|------------------------------------------|---|--|--|---|--|---|

| (ii)                                              |                                                |  | Mitochondria (1)<br>To provide the ATP for {active transport of <u>ions</u> / pumping of <u>ions</u> / exocytosis}(1)<br><br>Ribosomes(1)<br>Produce proteins for {transport of ions/ carrier proteins}(1)                                                                                                                                                                                                                                                                                                                                                                                                                         |                  | 4                   |                                                   | 4                                              |                            |                                          |                      |                       |                     |                                |                 |                  |                  |                   |                |                 |   |  |  |   |  |  |
|---------------------------------------------------|------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|---------------------------------------------------|------------------------------------------------|----------------------------|------------------------------------------|----------------------|-----------------------|---------------------|--------------------------------|-----------------|------------------|------------------|-------------------|----------------|-----------------|---|--|--|---|--|--|
| (c)                                               |                                                |  | <ul style="list-style-type: none"><li>• (The cell wall is) {strong / resists tension/ generates pressure potential/ owtte} <b>and</b> prevents {osmotic lysis/ bursting} (1)</li><li>• Beta glucose (1)</li><li>• It is made of (long) straight chains / alternate molecules {inverted/ rotated 180° }(1)</li><li>• {hydrogen bonds / cross-links} between (parallel) chains/ to form microfibrils (1)</li></ul>                                                                                                                                                                                                                   | 2                | 2                   |                                                   | 4                                              |                            |                                          |                      |                       |                     |                                |                 |                  |                  |                   |                |                 |   |  |  |   |  |  |
| (d)                                               |                                                |  | <table><tr><th><i>Chlorella</i></th><th><i>Cyanophyceae</i></th></tr><tr><td>Membrane bound organelle/ named organelle present</td><td>No membrane bound organelles / named organelle</td></tr><tr><td>Linear DNA/ plasmid absent</td><td>{Loop of/ circular} DNA/ plasmid present</td></tr><tr><td>Larger/80s ribosomes</td><td>Smaller/70s ribosomes</td></tr><tr><td>Cellulose cell wall</td><td>Murein/peptidoglycan cell wall</td></tr><tr><td>Mesosome absent</td><td>Mesosome present</td></tr><tr><td>Flagellum absent</td><td>Flagellum present</td></tr><tr><td>Capsule absent</td><td>Capsule present</td></tr></table> | <i>Chlorella</i> | <i>Cyanophyceae</i> | Membrane bound organelle/ named organelle present | No membrane bound organelles / named organelle | Linear DNA/ plasmid absent | {Loop of/ circular} DNA/ plasmid present | Larger/80s ribosomes | Smaller/70s ribosomes | Cellulose cell wall | Murein/peptidoglycan cell wall | Mesosome absent | Mesosome present | Flagellum absent | Flagellum present | Capsule absent | Capsule present | 2 |  |  | 2 |  |  |
| <i>Chlorella</i>                                  | <i>Cyanophyceae</i>                            |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |                     |                                                   |                                                |                            |                                          |                      |                       |                     |                                |                 |                  |                  |                   |                |                 |   |  |  |   |  |  |
| Membrane bound organelle/ named organelle present | No membrane bound organelles / named organelle |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |                     |                                                   |                                                |                            |                                          |                      |                       |                     |                                |                 |                  |                  |                   |                |                 |   |  |  |   |  |  |
| Linear DNA/ plasmid absent                        | {Loop of/ circular} DNA/ plasmid present       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |                     |                                                   |                                                |                            |                                          |                      |                       |                     |                                |                 |                  |                  |                   |                |                 |   |  |  |   |  |  |
| Larger/80s ribosomes                              | Smaller/70s ribosomes                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |                     |                                                   |                                                |                            |                                          |                      |                       |                     |                                |                 |                  |                  |                   |                |                 |   |  |  |   |  |  |
| Cellulose cell wall                               | Murein/peptidoglycan cell wall                 |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |                     |                                                   |                                                |                            |                                          |                      |                       |                     |                                |                 |                  |                  |                   |                |                 |   |  |  |   |  |  |
| Mesosome absent                                   | Mesosome present                               |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |                     |                                                   |                                                |                            |                                          |                      |                       |                     |                                |                 |                  |                  |                   |                |                 |   |  |  |   |  |  |
| Flagellum absent                                  | Flagellum present                              |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |                     |                                                   |                                                |                            |                                          |                      |                       |                     |                                |                 |                  |                  |                   |                |                 |   |  |  |   |  |  |
| Capsule absent                                    | Capsule present                                |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |                     |                                                   |                                                |                            |                                          |                      |                       |                     |                                |                 |                  |                  |                   |                |                 |   |  |  |   |  |  |

WJEC 2022

1. **Image 1.1** shows simple squamous epithelial tissue from the alveoli of the lungs.

**Image 1.1**

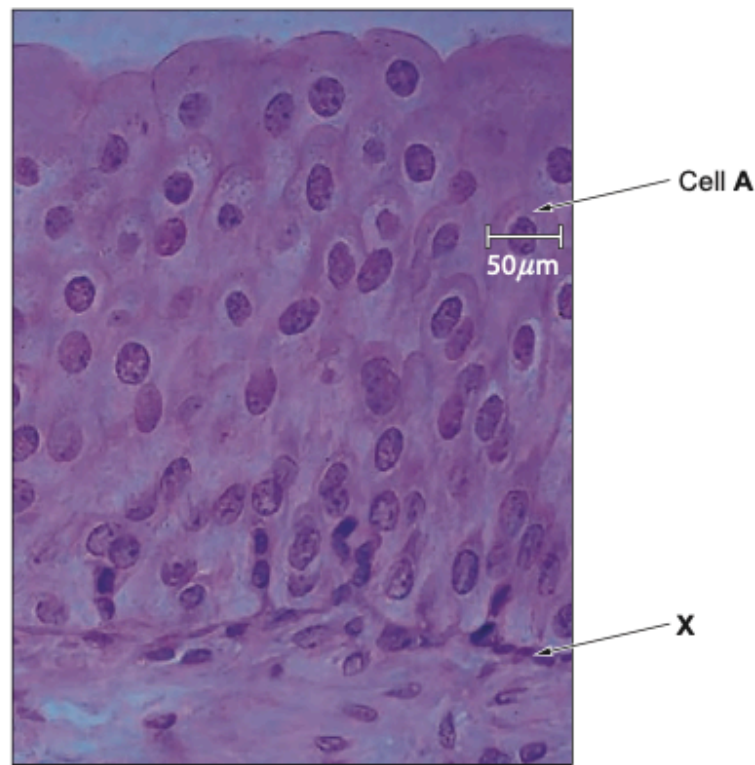


- (a) Explain how the shape of the cells is adapted to their function within the tissue. [1]

.....

- (b) The urinary bladder is lined with stratified epithelial tissue with a variable number of cell layers. **Image 1.2** shows this tissue when the bladder is relatively empty.

**Image 1.2**



- (i) Name the structure labelled **X** on **Image 1.2**. [1]
- .....
- (ii) State why the cells in **Image 1.2** are referred to as a "tissue". [1]
- .....
- .....
- .....
- (iii) Explain why some of the cells appear not to have a nucleus in **Image 1.2**. [1]
- .....
- .....
- .....
- (iv) With reference to **Image 1.1** and **Image 1.2**, describe the main difference between the two types of epithelial tissue. [1]
- .....

- (c) (i) When the bladder is full, the cells become wider and flatter. Suggest how this property of the epithelial tissue allows the bladder to perform its function. [1]

- (ii) The width of cell **A** in **Image 1.2** is  $50\mu\text{m}$ . Cell **A** is shown at 42% of its maximum width.  
Calculate the maximum width of this cell. Give your answer to three significant figures. [2]

Maximum width = .....  $\mu\text{m}$

- (iii) With reference to the actual width of cell **A**, calculate the magnification of **Image 1.2**. [2]

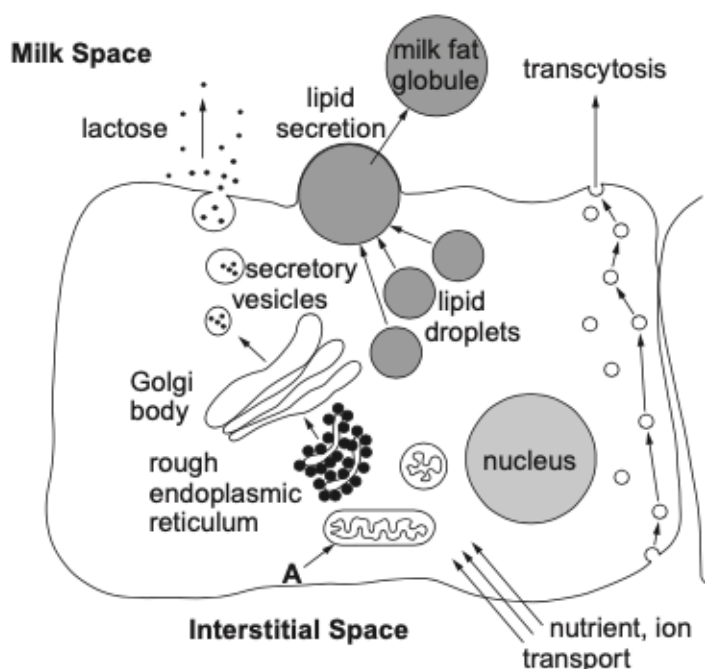
Magnification =  $\times$  .....

|    |     |       |                                                                                                                                                                                                           |          |          |          |           |          |          |
|----|-----|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
| 1. | (a) |       | {Flat/ thin/ large surface area/ one cell thick} +<br>{to reduce diffusion distance for gases/ gas exchange/ to make diffusion of gases more efficient}                                                   | 1        |          |          | 1         |          | 1        |
|    | (b) | (i)   | Basement membrane                                                                                                                                                                                         | 1        |          |          | 1         |          |          |
|    |     | (ii)  | A group of {similar/same/ specialised/ one type of} cell(s) working together to perform a function                                                                                                        | 1        |          |          | 1         |          |          |
|    |     | (iii) | The nucleus may have been in a different part of the cell/ the section has been taken from a part of the cell that does not include the nucleus/ the cells have been sectioned in different planes/ owtte |          | 1        |          | 1         |          | 1        |
|    |     | (iv)  | {Simple epithelial tissue/ one} has a single layer of cells +<br>{Stratified epithelial tissue/ the other} has more layers of cells                                                                       | 1        |          |          | 1         |          |          |
|    | (c) | (i)   | The bladder (can stretch when it is full (of urine) / changes size/ increase in volume) / so the bladder can fill with urine                                                                              |          |          | 1        | 1         |          |          |
|    |     | (ii)  | Maximum width = 119 = 2 marks<br>If incorrect award 1 mark for<br>119.0/ 119.05/ 119.047<br>50 x 100/42<br>50/42 x 100                                                                                    |          | 2        |          | 2         | 2        | 2        |
|    |     | (iii) | Magnification = 200 or 220 = 2 marks Accept 210<br>If incorrect award 1 mark for<br>Scale bar 10 or 11 mm = 10 000 $\mu\text{m}$ or 11000 $\mu\text{m}$<br>10000/50<br>Accept 10.5 mm = 10 500            |          | 2        |          | 2         | 2        | 2        |
|    |     |       | <b>Question 1 total</b>                                                                                                                                                                                   | <b>4</b> | <b>5</b> | <b>1</b> | <b>10</b> | <b>4</b> | <b>6</b> |



5. Human milk contains all the nutrients a young baby needs in exactly the right proportions. It is formed in the mammary glands by small groups of milk-producing cells in structures called lobules. These cells absorb molecules from the blood and use them to synthesise the lipids, carbohydrates and proteins found in milk. **Image 5.1** shows a milk producing cell.

**Image 5.1**



Lactose is synthesised in the Golgi body and is transported in vesicles through the cytoplasm. The vesicles increase in diameter as they move towards the plasma membrane. The vesicle membranes fuse with the plasma membrane and the vesicles empty their contents out of the cell. Lipids are also secreted in droplets.

- (a) Use the information provided and your own knowledge to answer the following questions.

- (i) State the name of the process by which the vesicles empty their contents out of the milk-producing cell. [1]

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- (ii) Identify the organelle labelled **A** and explain its role in the function of this cell. [2]

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 .....  
 .....

- (iii) Use **Image 5.1** to suggest how the cell avoids a continual increase in the area of its cell surface membrane. [1]

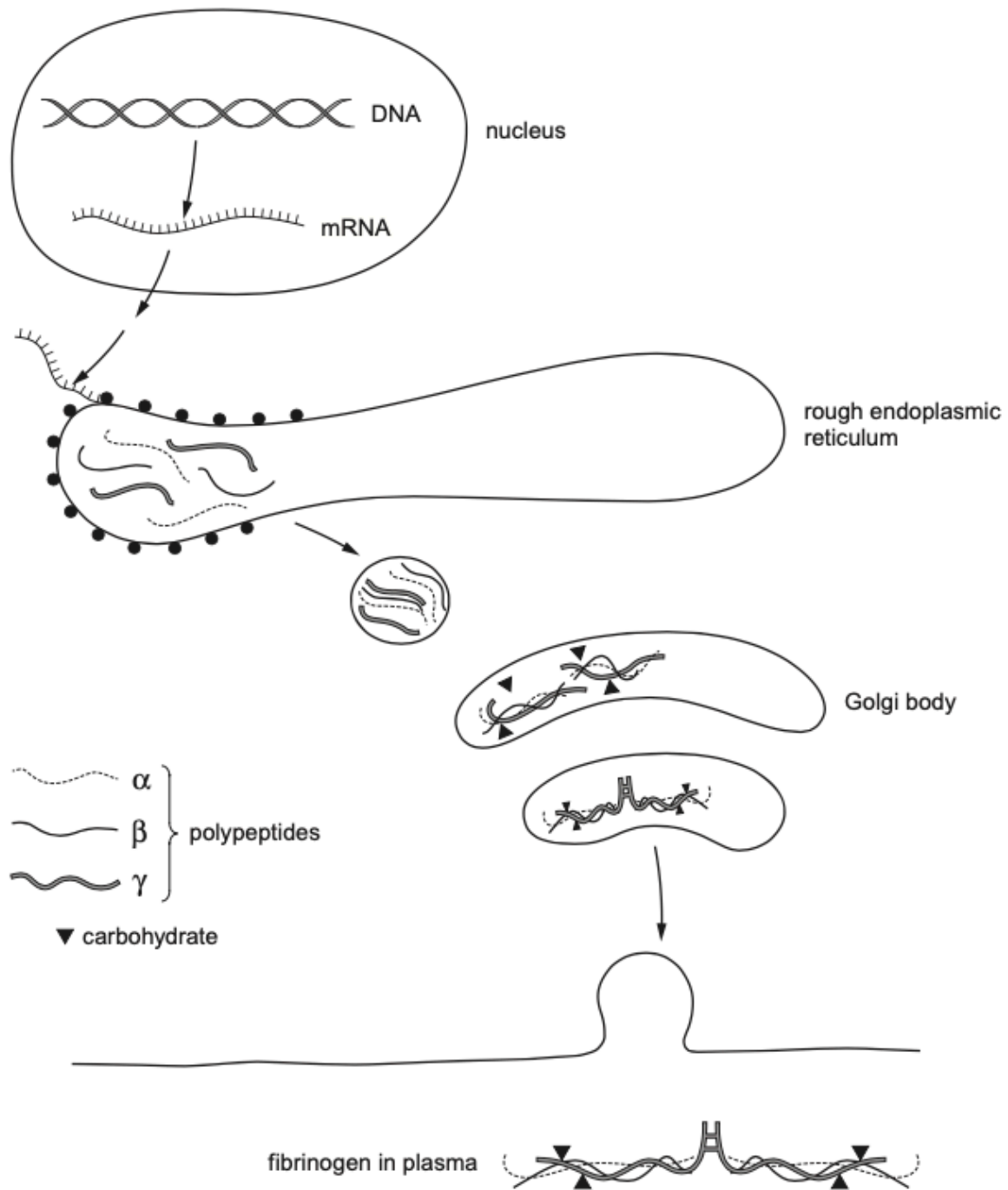
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|    |     |       |                                                                                                                                                                                                    |   |  |   |   |  |  |
|----|-----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|---|---|--|--|
| 5. | (a) | (i)   | Exocytosis (1)                                                                                                                                                                                     | 1 |  |   | 1 |  |  |
|    |     | (ii)  | Mitochondrion/ mitochondria (1)<br>provides ATP + {for synthesis of {lipid/ protein/ lactose} / for<br>exocytosis/active transport} (1)                                                            | 2 |  |   | 2 |  |  |
|    |     | (iii) | They take vesicles into the cell by endocytosis/ phagocytosis/<br>pinocytosis/ membrane lost surrounding lipid droplets (1)<br>Accept reference to transcytosis in context of removing<br>membrane |   |  | 1 | 1 |  |  |

6. Fibrinogen is a soluble glycoprotein produced in liver cells and secreted into the plasma. It contains three types of polypeptide chain.

**Image 6.1** shows the formation and secretion of fibrinogen.

**Image 6.1**



(a) Use **Image 6.1** and your knowledge of proteins and protein synthesis to answer the following questions.

(i) Name the highest level of protein structure shown by fibrinogen. [1]

.....

(ii) Explain the role of the rough endoplasmic reticulum in the production of fibrinogen. [2]

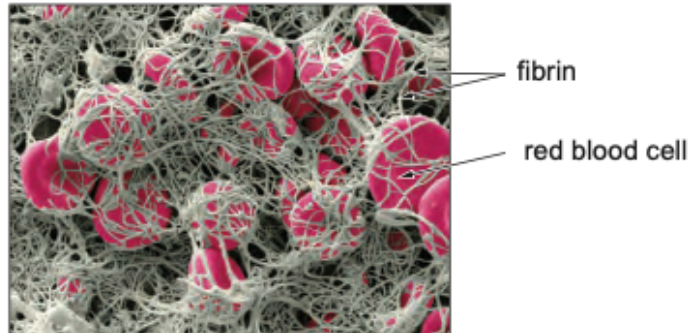
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(iii) Suggest the role of the Golgi body in the production of fibrinogen. [2]

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- (b) Thrombin is an enzyme which is secreted into blood plasma when tissues are damaged. It catalyses the conversion of fibrinogen, in plasma, into the insoluble, fibrous protein, fibrin. Fibrin seals wounds in tissue. This is shown in **Image 6.2**.

**Image 6.2**



Fibrin is a fibrous protein, fibrinogen is a globular protein.

Using the information provided, explain how the conversion of fibrinogen to fibrin allows the formation of a blood clot. [3]

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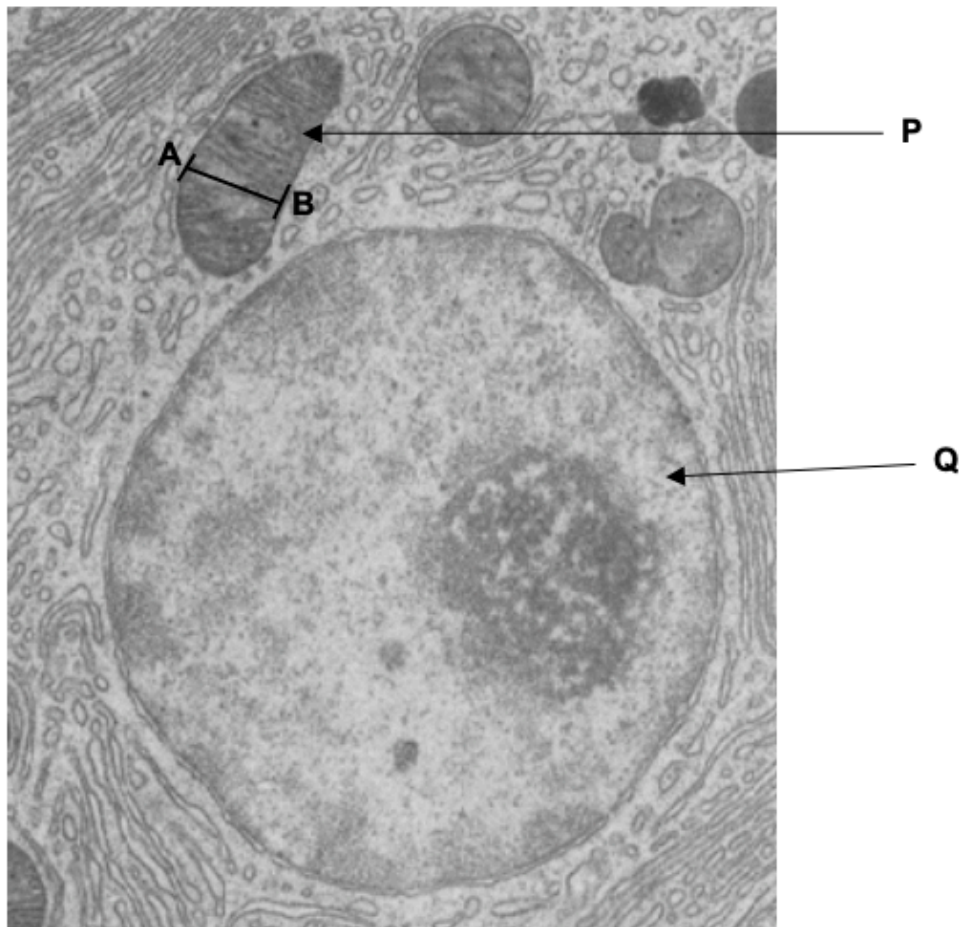
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|    |     |       |                                                                                                                                                                                                                                                                                                |   |   |   |   |  |  |
|----|-----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|--|--|
| 6. | (a) | (i)   | Quaternary                                                                                                                                                                                                                                                                                     | 1 |   |   | 1 |  |  |
|    |     | (ii)  | Translation of mRNA by ribosomes/ production of {polypeptides/ protein synthesis} on ribosomes (1)<br>(Packaging of polypeptide) into vesicles/ transport to Golgi (body) (1)                                                                                                                  |   | 2 |   | 2 |  |  |
|    |     | (iii) | <b>Any two (x1) from</b><br><ul style="list-style-type: none"> <li>{Assembly/ joining} of {polypeptides/ subunits/ protein} (into quaternary structure) (1)</li> <li>addition of carbohydrate/ glycosylation/ produce glycoprotein/ owtte (1)</li> <li>Packaged (into vesicles) (1)</li> </ul> |   | 2 |   | 2 |  |  |
|    | (b) |       | <ul style="list-style-type: none"> <li>Fibrinogen (is globular, so it) dissolves in plasma (1)</li> <li>When it is converted to Fibrin it {becomes insoluble/ forms fibres} (1)</li> <li>{fibres cross a wound and} {seal it/ trap blood cells/ owtte} (1)</li> </ul>                          |   | 2 | 1 | 3 |  |  |

# Speciment

4. The electron micrograph below shows a section through a eukaryote cell.



- (a) The organelles labelled **P** and **Q** are both surrounded by double membranes. Identify these organelles and describe **one** difference between them. [2]

**P** ..... **Q** .....

**Difference:** .....  
 .....

- (b) The magnification of the image above is  $\times 32\,500$ . Calculate the actual width of the organelle in micrometres between points **A** and **B**. Give your answer to three significant figures and show your working. [3]

Answer .....

- (c) The surface area of the organelle shown can be estimated by using the formula:

$$\text{surface area} = 2\pi r (l + r)$$

|                                    |   |                     |
|------------------------------------|---|---------------------|
| where: length (l) of the organelle | = | 9.8 $\mu\text{m}$ , |
| $\pi$                              | = | 3.14                |
| average diameter                   | = | 1.2 $\mu\text{m}$ . |

- (i) Estimate the surface area of the organelle shown.  
Show your working.

[3]

Answer .....  $\mu\text{m}^2$

- (ii) The surface area of a spherical organelle with the **same** volume is  $23.9\mu\text{m}^2$ . Scientists have concluded from the results from experiments that there are significant advantages to the cell if these organelles have a cylindrical shape. Evaluate this statement. [4]

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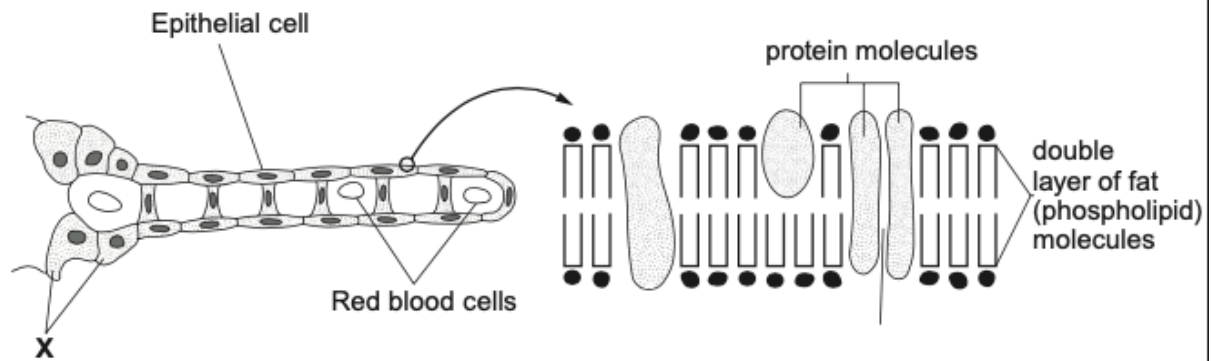
| Question                                                                                                          |                                                                                                   |      |  | Marking details                                                                                                                                                                                                                                                                                                                      | Marks available                                                                                                   |                                                                                                   |          |           |          |          |  |  |
|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------|-----------|----------|----------|--|--|
|                                                                                                                   |                                                                                                   |      |  |                                                                                                                                                                                                                                                                                                                                      | AO1                                                                                                               | AO2                                                                                               | AO3      | Total     | Maths    | Prac     |  |  |
| 4                                                                                                                 | (a)                                                                                               |      |  | P = Mitochondrion and Q = nucleus (1)<br>Any <b>one</b> from<br><table><tr><td><b>Mitochondrion / Organelle shown</b><br/><br/>inner membrane is folded / has cristae<br/><br/>no ribosomes attached</td><td><b>Nucleus</b><br/><br/>No folding of inner membrane / no cristae (1)<br/><br/>ribosomes attached (1)</td></tr></table> | <b>Mitochondrion / Organelle shown</b><br><br>inner membrane is folded / has cristae<br><br>no ribosomes attached | <b>Nucleus</b><br><br>No folding of inner membrane / no cristae (1)<br><br>ribosomes attached (1) | 2        |           |          | 2        |  |  |
| <b>Mitochondrion / Organelle shown</b><br><br>inner membrane is folded / has cristae<br><br>no ribosomes attached | <b>Nucleus</b><br><br>No folding of inner membrane / no cristae (1)<br><br>ribosomes attached (1) |      |  |                                                                                                                                                                                                                                                                                                                                      |                                                                                                                   |                                                                                                   |          |           |          |          |  |  |
|                                                                                                                   | (b)                                                                                               |      |  | 14/32 500 x1000 (1)<br>=0.43076.....µm(1)<br>0.431 µm (1) 3sig.fig.                                                                                                                                                                                                                                                                  |                                                                                                                   | 3                                                                                                 |          | 3         | 3        | 2        |  |  |
|                                                                                                                   | (c)                                                                                               | (i)  |  | Radius = 0.6 (1)<br>2 x 3.14 x 0.6 (9.8+0.6)(1)<br>=39.19 (µm <sup>2</sup> )(1)                                                                                                                                                                                                                                                      |                                                                                                                   | 3                                                                                                 |          | 3         | 3        |          |  |  |
|                                                                                                                   |                                                                                                   | (ii) |  | Diffusion distance to centre is smaller (1)<br>Larger surface area / SA:Vol bigger (1)<br>Accept converse<br>More {oxygen can diffuse in / CO <sub>2</sub> diffuse out} (1)<br><u>Aerobic</u> respiration is <u>more</u> efficient / <u>more</u> ATP produced (1)                                                                    |                                                                                                                   |                                                                                                   | 4        | 4         |          |          |  |  |
|                                                                                                                   |                                                                                                   |      |  | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                              | <b>2</b>                                                                                                          | <b>6</b>                                                                                          | <b>4</b> | <b>12</b> | <b>6</b> | <b>2</b> |  |  |



1.3

## WJEC 2016

5. The primary role of the gills of fish is gas exchange. Oxygen, a non-polar molecule, passes from the surrounding water into the blood of the gills. The gills are also permeable to water and solutes, such as sodium ions and chloride ions.



- (a) Use the information provided to describe how oxygen and sodium ions cross the membrane. [4]

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- (b) The data below shows the percentage ion composition of different aquatic habitats and of the blood plasma of two different fish.

|          |             | Percentage ion composition |         |
|----------|-------------|----------------------------|---------|
| Fish     | Habitat     | Blood plasma of fish       | Habitat |
| Flounder | Sea water   | 1.1                        | 3.5     |
| Carp     | Fresh water | 0.9                        | <0.1    |

Explain why it is important for the survival of the carp to produce large volumes of dilute urine. [4]

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- (c) (i) Specialised cells, labelled **X** on the diagram, are located in the epithelium of the gills. They are important in maintaining the ion composition of the blood plasma. Use the data to explain why this is necessary in the flounder. [3]

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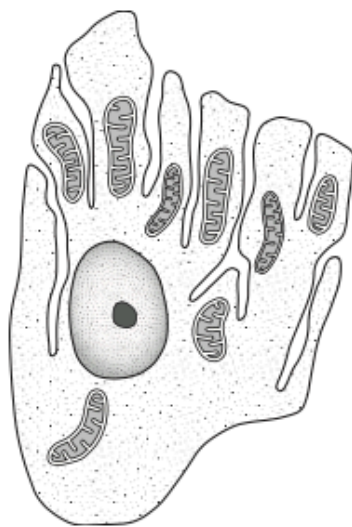
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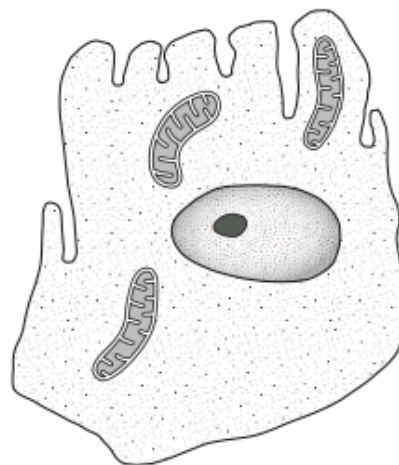
- (ii) Salmon spend the majority of their lives in sea water. In order to reproduce salmon must migrate into fresh water rivers. During these migrations these specialised cells undergo structural changes.

The diagrams below are of these cells, taken from the salmon in the two different habitats. Giving reasons for your answer, suggest which cell was taken from the salmon in fresh water, and which was taken from the salmon whilst in sea water.

[2]



Cell 1



Cell 2

.....

.....

.....

.....

- (d) Acidification of fresh-water lakes, due to acid rain, has been linked to the death of fish such as carp. Scientists concluded that one of the causes of death in these animals is their inability to maintain blood plasma ion concentrations. Use the information to explain how they arrived at this conclusion. [4]

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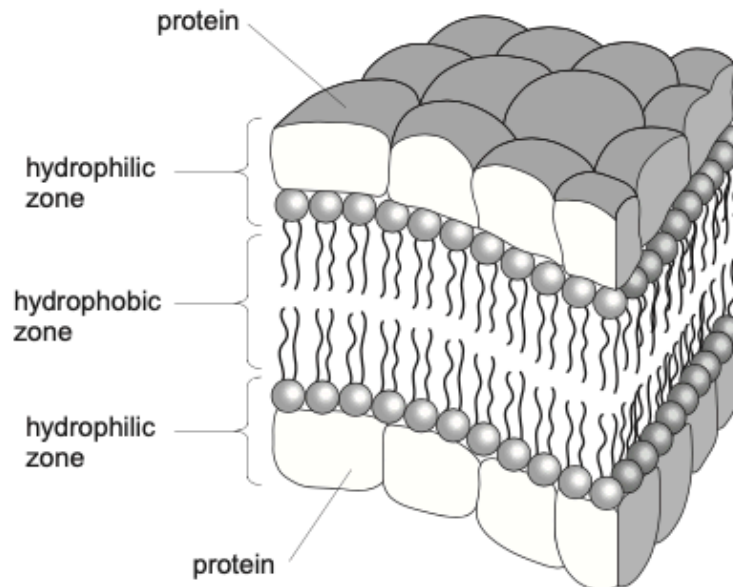
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|   |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |        |  |        |   |  |
|---|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------|--|--------|---|--|
| 5 | (a) | <p>1. Oxygen crosses the membrane by (simple) diffusion (1)<br/>NOT facilitated</p> <p>2. oxygen passes through the phospholipid (bilayer) (1)<br/>Membrane = neutral</p> <p>3. Sodium ions pass through {channel/carrier} proteins (1)<br/>intrinsic proteins = neutral</p> <p>4. by {facilitated diffusion/active transport/ cotransport} (1)<br/>Reject active transport in relation to channel proteins</p> <p>If oxygen and sodium are not treated separately, allow one mark max for correct linkage of diffusion with phospholipid bilayer or {facilitated diffusion/ active transport/ cotransport} with {channel/ carrier} proteins</p> | 4 |        |  | 4      |   |  |
|   | (b) | <p>1. The percentage ion composition of {fresh water/ habitat} is <u>lower</u> than that in the carp's blood plasma/ fresh water is hypotonic to the blood/ ORA (1)</p> <p>2. The water potential of fresh water is {higher/less negative} than the water potential of the carp's blood/ ORA (1)</p> <p>3. The carp takes in water (through the gills) by osmosis (1)</p> <p>4. (Producing large volumes of dilute urine) prevents {osmotic lysis/ bursting} of cells (1)</p> <p>5. Maintaining ion composition (1)<br/>Accept reduces the loss of ions</p>                                                                                      |   | 1<br>1 |  | 1<br>1 | 4 |  |

|     |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |   |  |  |
|-----|-----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|--|--|
| 5   | (c) | (i)  | <p>1. Percentage ion composition of sea water is <u>greater</u> than that of the blood plasma/ sea water is hypertonic/ water potential in the fish is <u>higher</u> than the sea water/ ORA (1)</p> <p>2. ions will move into the blood plasma, {by (facilitated) diffusion/ down a concentration gradient/ or description of}(1)</p> <p>3. therefore ions must be actively transported out (of the blood plasma into the surrounding sea water) (1)</p>                                                                                                                                                          | 1 | 1 | 3 |  |  |
|     | (c) | (ii) | <p>Cell 1 is from the salmon in sea water (1)</p> <p><b>Any one from:</b></p> <p>as it is {highly folded to give a larger surface} / contains {many/more} mitochondria to {provide ATP/ for respiration/ for active transport } (1)</p>                                                                                                                                                                                                                                                                                                                                                                            |   | 1 | 2 |  |  |
|     |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |   |  |  |
| (d) |     |      | <p>1. As the percentage ion composition of the blood plasma is higher than the surrounding water (1)</p> <p>2. Ions will move out of the {fish/gills/blood plasma} by {(facilitated) diffusion/down a concentration gradient} (1)</p> <p>3. Acidification will {denature/ inactivate/ change the shape of} the (carrier/ channel) proteins in the specialised cells/ denature/ inactivate enzymes (involved in ATP production) (1)</p> <p>4. Therefore {no/fewer} ions are (actively) transported <u>into</u> the {fish/gills/blood (plasma)}/ fewer ions can diffuse out (1) (in context of damaged proteins)</p> | 1 | 1 | 4 |  |  |

## WJEC 2017

5. In 1935, Hugh Davson and James Danielli proposed a model of the arrangement of molecules in the plasma membrane of a cell.



- (a) Compare Davson-Danielli's model shown in the diagram to the currently accepted fluid mosaic model of the plasma membrane. [4]

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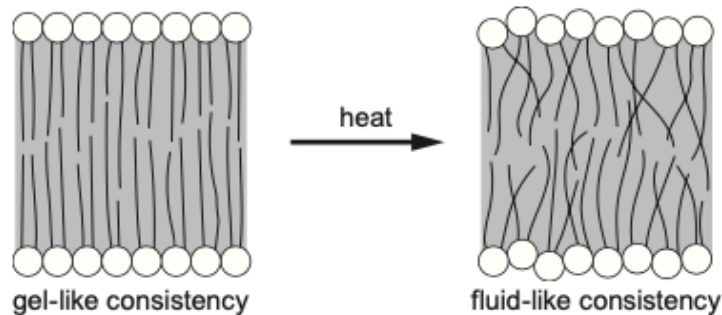
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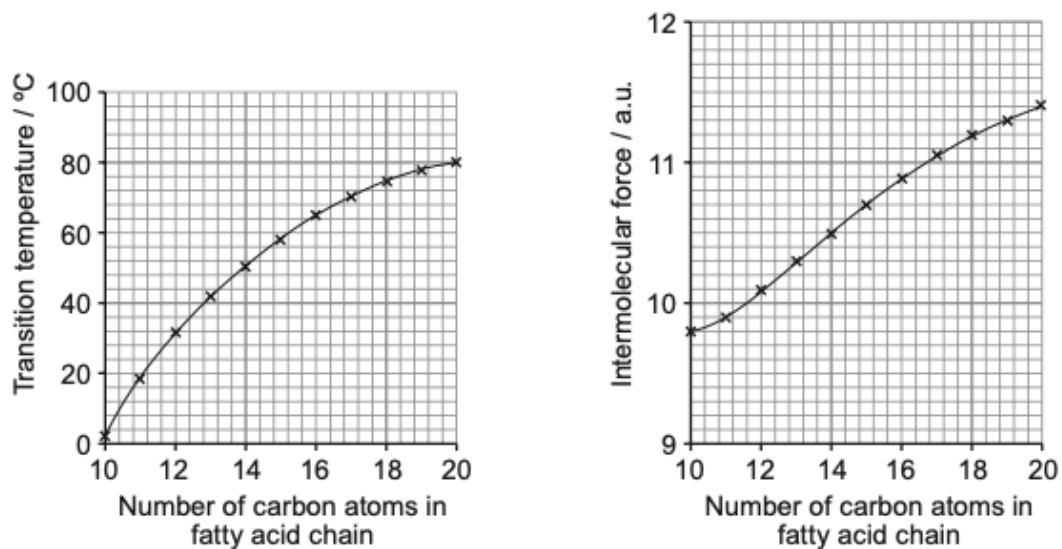
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Fluidity of the plasma membrane is due to the lateral movement of phospholipids. Increased fluidity occurs when the phospholipid bilayer changes from a more fixed gel-like consistency to a fluid-like consistency. Heating the molecules can result in increased fluidity.



The transition temperature, is the temperature at which the phospholipid bilayer changes from a gel like consistency to a fluid like consistency. This can also be affected by the length of the fatty acid chains which affect the strength of the intermolecular forces. These relationships are shown in the graphs below.





- (b) (i) With reference to the graphs, describe the relationship between fatty acid chain length and the temperature at which the phospholipid bilayer becomes fluid. Explain your answer. [3]

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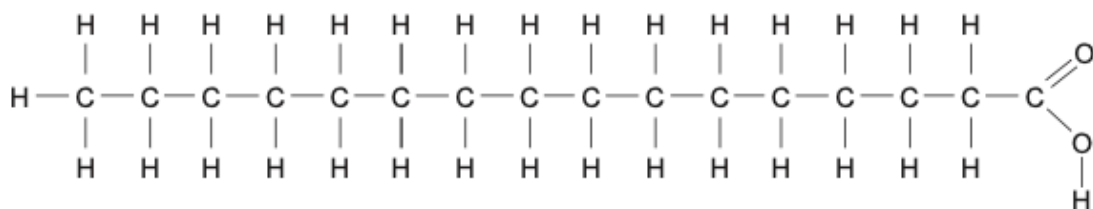
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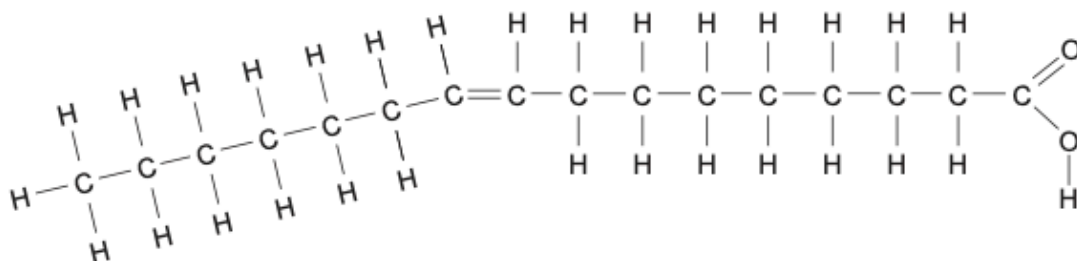
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The packing of hydrocarbons also plays a significant role in affecting fluidity. Two examples of fatty acids that can be found in phospholipids are shown below.

Saturated Fatty Acid



Unsaturated Fatty Acid



- (ii) Suggest why the presence of unsaturated fatty acids in the phospholipid would increase fluidity of the membrane. [4]

Cholesterol is a non-polar substance that is produced in the body and obtained from foods that come from animals (particularly egg yolks, meat, poultry, fish, and dairy products). The body needs this substance to build cell membranes. However, too much cholesterol increases a person's risk of developing heart disease.

(iii) Suggest the position of cholesterol molecules within the plasma membrane. [1]

(iv) State how a diet high in animal products and cholesterol can lead to a person developing heart disease. [2]

|   |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |          |          |           |          |          |
|---|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
| 5 | (a) |       | <p><b>Any 4 (x1) from:</b></p> <p>Similarities:</p> <p>A. Have phospholipid bilayer/ hydrophobic tails &amp; hydrophilic heads (1)</p> <p>B. Correct location of heads and tails: (hydrophobic tails) inside &amp; (hydrophilic heads) outside (1)</p> <p>C. Have protein (1)</p> <p>Differences:</p> <p>D. {Proteins not embedded in the lipid bilayer / proteins coat outer surface / continuous layer of protein / no intrinsic proteins} in Davson Danielli model (1)</p> <p>E. Ref to absence of {glycocalyx/ glycoproteins/ glycolipids/ cholesterol} in Davson Danielli model (1)</p> <p>Accept reverse for Fluid Mosaic</p> |          | 4        |          | 4         |          |          |
|   | (b) | (i)   | <p>A. As the number of carbon atoms in the chain increases, the (transition) temperature increases (1) Accept increased chain length</p> <p>B. Because the longer the chain length the greater the intermolecular forces (1) Accept increased number of C atoms</p> <p>C. Therefore more energy needed to break the {bonds / overcome the intermolecular forces}(1)</p> <p>Accept reverse argument</p>                                                                                                                                                                                                                              |          |          | 3        | 3         |          |          |
|   |     | (ii)  | <p>Unsaturated contain {double bonds between carbons/ C=C}(1)</p> <p><b>Any 3 (x1) from:</b></p> <ul style="list-style-type: none"> <li>Produces a kink / bend, in the side chain (1)</li> <li>Less packing is possible/ increased distance between the chains (1)</li> <li>Lowers intermolecular force /(phase) transition temperature (1)</li> <li>less energy needed {to break the bonds/ overcome the intermolecular forces} (1)</li> </ul>                                                                                                                                                                                     | 1        |          | 3        | 4         |          |          |
|   |     | (iii) | <p>Between the {fatty acids / hydrophobic tails}</p> <p>Accept labelled diagram</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          | 1        |          | 1         |          |          |
|   |     | (iv)  | <p>Raises level of LDLs/ causes LDLs to be made (1)</p> <p>Increase incidence of atheroma in arteries/ detailed description of atheroma (1)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2        |          |          | 2         |          |          |
|   |     |       | <b>Question 5 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>3</b> | <b>5</b> | <b>6</b> | <b>14</b> | <b>0</b> | <b>0</b> |

WJEC 2018

5. *Nereis virens* is a species of marine worm. A student wanted to investigate the effect of a change in solute concentration on these worms. The following method was used.

- Collect seawater and thirty worms.
- Make a 10% solution of seawater by placing 1 dm<sup>3</sup> of seawater and 9 dm<sup>3</sup> of freshwater into a fish tank.
- Record the mass of ten worms and place them into the fish tank of diluted seawater.
- Every 15 minutes remove the worms, blot dry, re-weigh and record their mass.
- Repeat this process, and after 75 minutes, return the worms to a container of undiluted seawater.
- Repeat the experiment a further two times using different worms for each trial.
- Calculate the mean percentage change in mass of the worms during the investigation.

The student's results are shown in the table below.

| Time / minutes | Mean percentage change in mass / % |
|----------------|------------------------------------|
| 0              | 0                                  |
| 15             | 15                                 |
| 30             | 18                                 |
| 45             | 20                                 |
| 60             | 22                                 |
| 75             | 22                                 |

(a) Explain why it was necessary for the student to calculate the percentage change in mass. [2]

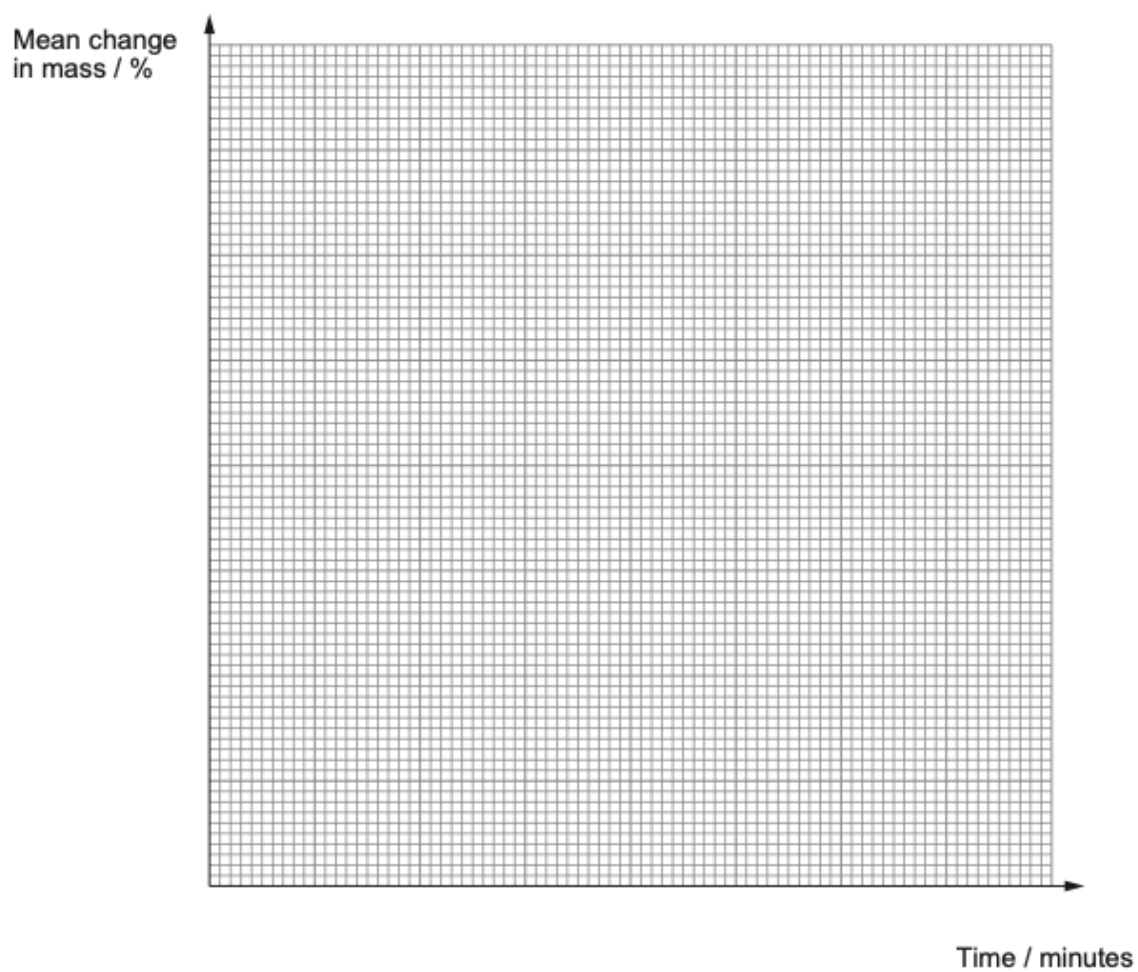
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(b) (i) Plot the student's results shown in the table on the grid below.

[3]



(ii) Describe and explain the changes in mass of the worms during the experiment. [4]

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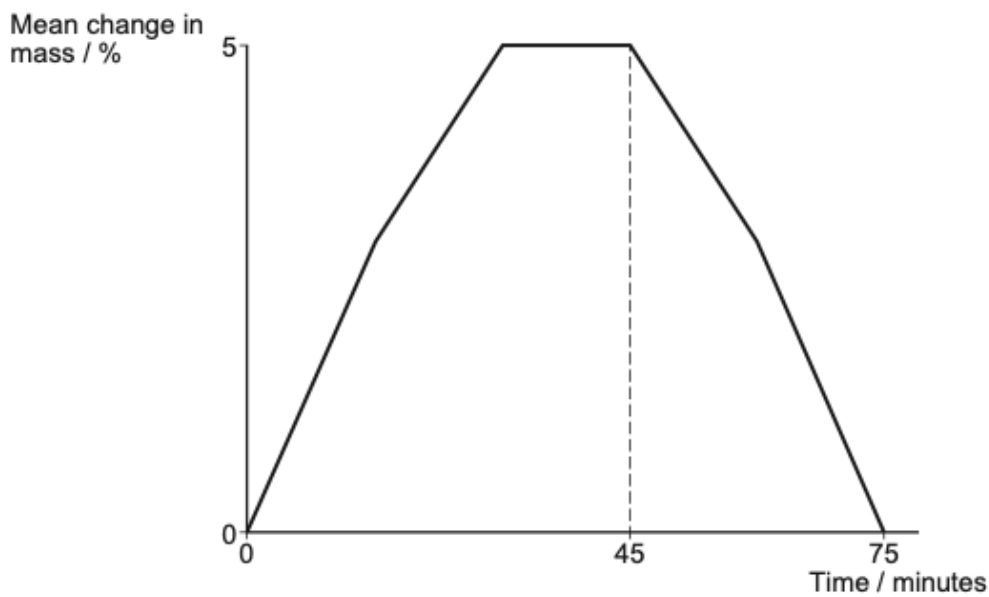
- (c) The student decided to repeat the experiment. She collected ten fresh worms, weighed them and placed them in a fish tank containing undiluted seawater. Explain why it was important for the student to carry out this second experiment. [2]

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- (d) A similar experiment was set up with a different species of marine worm, *Golfingia gouldi*. The results are shown below.



- (i) Describe how the results for *Golfingia* differ from those of *Nereis*. [2]

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- (ii) *Golfingia* can actively pump ions from its cells into the surrounding water. Explain the change in mass of *Golfingia* between 45 and 75 minutes. [2]

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|   |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |           |          |           |          |           |
|---|-----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|----------|-----------|----------|-----------|
| 5 | (a) |      | (calculating the percentage change) allows the results to be comparable (1)<br>Initial masses of worm would differ (1)                                                                                                                                                                                                                                                                                                                                                                                                                                       |          | 2         |          | 2         |          | 2         |
|   | (b) | (i)  | Linear scale (1) including figure at origin and no break line<br>Correct plots (1) $\pm \frac{1}{2}$ small square<br>Line drawn (with no extrapolation) (1)<br>extrapolation is acceptable with a curve of best fit                                                                                                                                                                                                                                                                                                                                          |          | 3         |          | 3         | 3        | 3         |
|   |     | (ii) | <b>Any four (x1) from:</b><br>1. Increase to 60 and then levels off(1)<br>2. (the body fluids of the marine) worms {are hypertonic to /have a lower water potential than} {the dilute seawater/ the solution}/ ORA(1)<br>3. The worms gain water by <u>osmosis</u> (resulting in an increase in mass)(1)<br>4. (during the first 15 mins) the increase in mass is greater due to the steepness of the water potential gradient(1)<br>5. (after 60 mins there is no change in mass as) equilibrium is reached / isotonic/ no net movement of water/ OWTTE (1) |          | 4         |          | 4         |          | 3         |
|   | (c) |      | As a control (1)<br>to show that any change in mass was due to the worms being placed in diluted seawater (and no other variable).(1)                                                                                                                                                                                                                                                                                                                                                                                                                        |          |           | 2        | 2         |          | 2         |
|   | (d) | (i)  | The percentage change in mass of <i>Golfingia</i> {returns to zero/ decreases} (after 45 mins) whereas the mass of <i>Nereis</i> {continued to increase/ plateaus} (1)<br><br>The maximum percentage increase in mass in <i>Golfingia</i> {is much lower / is reached sooner} than that of <i>Nereis</i> / equilibrium is reached quicker in <i>Golfingia</i> than in <i>Nereis</i> (1)                                                                                                                                                                      |          | 2         |          | 2         |          | 2         |
| 5 | (d) | (ii) | increase water potential of the cells of <i>Golfingia</i> (1)<br>therefore water moves out by <u>osmosis</u> (1)                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |           | 2        | 2         |          | 2         |
|   |     |      | <b>Question 5 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>0</b> | <b>11</b> | <b>4</b> | <b>15</b> | <b>3</b> | <b>15</b> |



## WJEC 2019

(b) *Amoeba* contains structures called contractile vacuoles which expel excess water from the cell. It can be seen that the contractile vacuole first increases in size, then migrates to and fuses with the plasma membrane. Scientists have recently discovered proteins within the membrane of these vacuoles that can pump ions into it.

(i) Use your knowledge of osmosis to explain why contractile vacuoles are necessary for the survival of the *Amoeba*. [4]

|     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |   |   |   |  |  |
|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|---|---|--|--|
| (b) | (i) | Any <b>four</b> (x1) from:<br>A. (The cytoplasm of) the amoeba has a {lower/ more negative} water potential than the {surrounding water / external environment}/ ORA (1)<br>B. Water will therefore move into the cell by osmosis. (1)<br>If not credited here, can be given with reference to vacuole<br>C. (By pumping ions into the vacuole) {it lowers the water potential of the vacuole / creates a water potential gradient between the cytoplasm and the vacuole} (1)<br>D. Water will move from the {cytoplasm/ cell} to the vacuole (by osmosis) (1)<br>E. (The water is then expelled from the cell) which prevents {osmotic lysis/ bursting} (1) Reject reference to turgidity |  |   |   |   |  |  |
|     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  | 2 | 2 | 4 |  |  |

## WJEC 2022

4. Beetroot contains a red pigment called betalain, which is stored within the cell vacuole. When the cell membranes are disrupted the pigment leaks out into the surrounding solution, colouring it red.

Some students investigated the permeability of the membranes of beetroot cells. A colorimeter was used to measure the absorbance of the surrounding solution. The absorbance of the solution is proportional to the concentration of betalain.

The students made the following prediction:

"The absorbance recorded will be directly proportional to the temperature of incubation."

The experiment was carried out as follows:

- 6 cubes of fresh beetroot were washed in cold water for 1 minute.
- The cubes were transferred into separate boiling tubes, each containing 20 cm<sup>3</sup> of water.
- The boiling tubes were incubated for 10 minutes in beakers at different temperatures which were maintained with the addition of hot or cold water.
- 3 cm<sup>3</sup> of water from around the beetroot cubes was removed and a colorimeter was used to measure the absorbance, at 540 nm, of each solution.

The experiment was repeated three times, the results are shown in **Table 4.1**.

**Table 4.1**

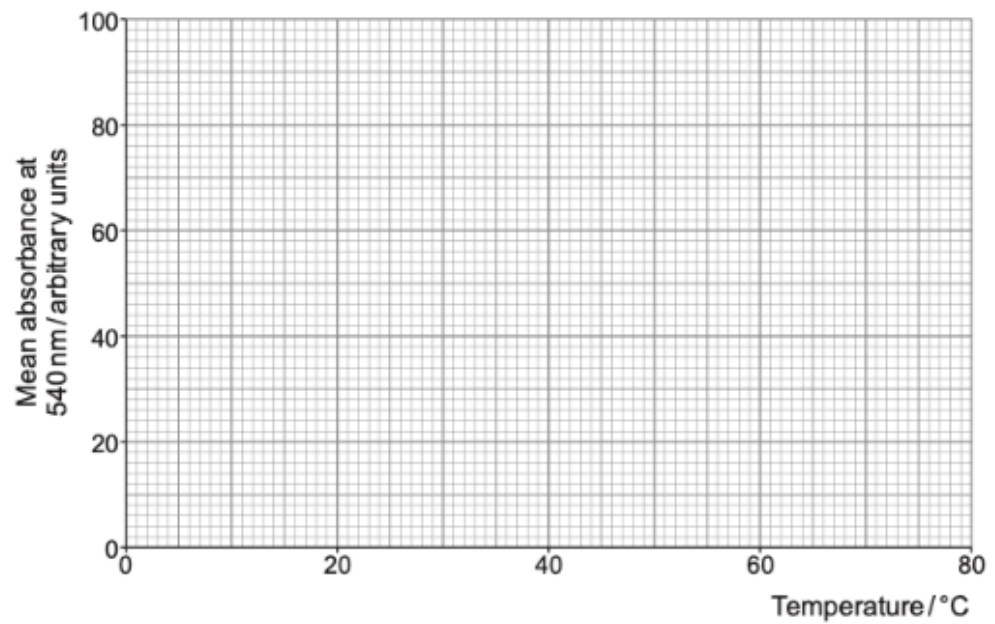
| Incubation temperature / °C | Absorbance at 540nm / a.u. |              |              |      |
|-----------------------------|----------------------------|--------------|--------------|------|
|                             | Experiment 1               | Experiment 2 | Experiment 3 | Mean |
| 10                          | 12                         | 15           | 21           | 16   |
| 30                          | 15                         | 20           | 19           | 18   |
| 40                          | 18                         | 22           | 32           | 24   |
| 60                          | 51                         | 68           | 71           | 63   |
| 70                          | 95                         | 96           | 97           | 96   |
| 80                          | 98                         | 98           | 98           | 98   |

- (a) (i) State which temperature provided the most reliable results and use your knowledge of cell structure to explain why this might be the case. [2]

- (ii) Plot the mean results from **Table 4.1** on **Graph 4.2**.

[3]

**Graph 4.2**



- (iii) Explain why the students' prediction was not entirely correct.

[1]

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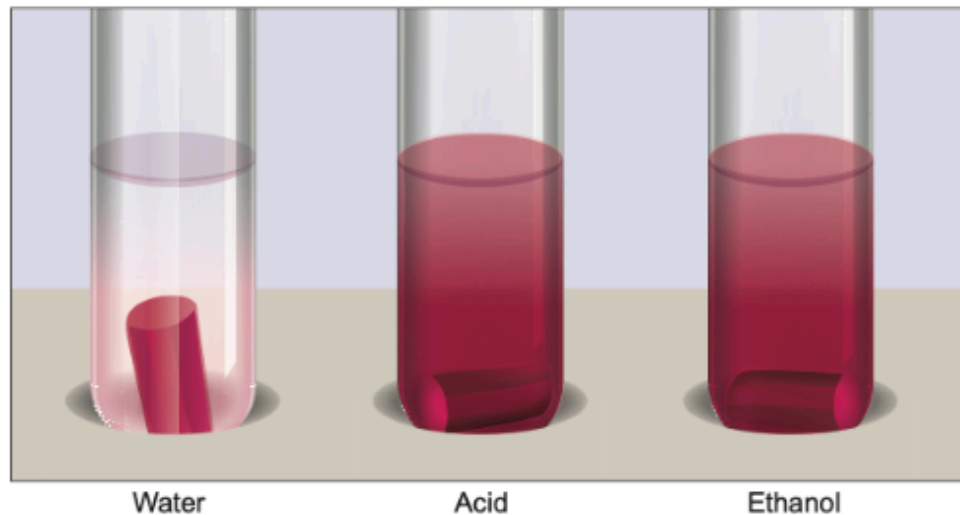
- (iv) Use your knowledge of cell membrane structure to explain the shape of **Graph 4.2**.

[2]

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- (b) Another group of students investigated how the type of surrounding solution affected the permeability of membranes in beetroot. They tested water, acid and ethanol. **Image 4.3** shows the experiment after incubation at 20 °C for 10 minutes.

**Image 4.3**



- (i) Using your knowledge of the structure of membranes, conclude how both **acid** and **ethanol** affect the membrane and explain the appearance of the solution in these tubes. [3]

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- (ii) The students also noted that the beetroot in the water had increased in size. Explain this observation. [3]

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|    |     |       |                                                                                                                                                                                                                                                                                                                                                                          |  |   |   |   |   |   |
|----|-----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|---|---|---|---|
| 4. | (a) | (i)   | 80°C (1)<br>because {the cells are ruptured/ membrane protein denatures} and (all) the pigment has leaked out of the cells/ (1)                                                                                                                                                                                                                                          |  | 2 |   | 2 |   | 2 |
|    |     | (ii)  | All plots correct = 2 marks<br>1 plot incorrect = 1 mark<br>>1 plot incorrect = 0 marks<br>½ small square tolerance<br>All points plotted correctly with a smooth curve drawn or a dot to dot plot (1)<br>No extrapolation with plot to plot /small amount allowed with line of best fit                                                                                 |  | 3 |   | 3 | 3 |   |
|    |     | (iii) | (The absorbance is not directly proportional to the temperature) {as it is not a straight line/ graph is a curve/ a description of changes in gradient} (1)                                                                                                                                                                                                              |  |   | 1 | 1 |   | 1 |
|    |     | (iv)  | <b>Any two (x1) from:</b><br>(Up to 40°C) As temperature rises the {membrane becomes more fluid/phospholipids gain more kinetic energy/ more gaps form} (1)<br><br>(Above 40°C) the proteins in the membrane start to denature (1)<br><br>(Above 70°C) membrane fully destroyed (so all pigment released) (1)<br>reject reference to membrane/ phospholipids denaturing. |  | 2 |   | 2 |   | 2 |

|  |     |      |                                                                                                                                                                                                            |   |    |   |    |   |    |
|--|-----|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|----|---|----|
|  | (b) | (i)  | Pigment {leaks/diffuses} {out of the <u>cells</u> / through cell membrane} (1)<br>Acid {denatures the proteins /changes shape of proteins} (1)<br>Ethanol dissolves the {(phospho)lipids/ fatty acids} (1) |   | 2  | 1 | 3  |   | 3  |
|  |     | (ii) | Water had entered the <u>cells</u> / moved in across <u>cell</u> membrane (1)<br>by osmosis (1)<br>as the beetroot <u>cells</u> had a lower water potential than the water (1)                             | 2 | 1  |   | 3  |   | 3  |
|  |     |      | <b>Question 4 total</b>                                                                                                                                                                                    | 2 | 10 | 2 | 14 | 3 | 11 |

## WJEC 2023

- (iii) The cytoplasm of bacterial cells is **hypertonic** to saliva and tears. Suggest how the destruction of the cell wall of bacteria by lysozyme results in the death of the bacteria. [3]

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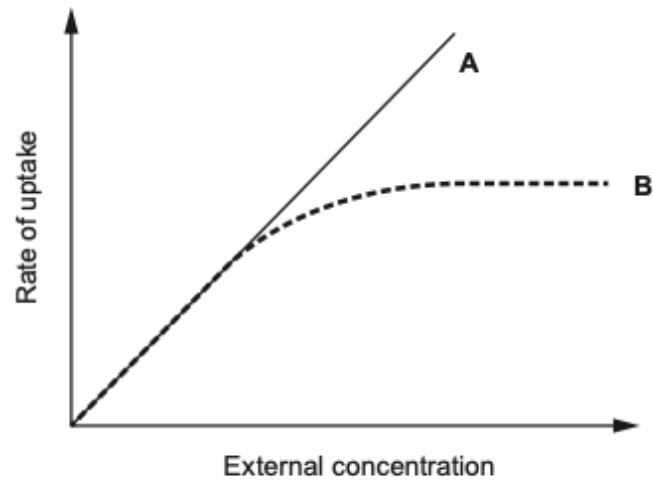
|  |  |       |                                                                                                                                                                                                                                                                                                                                                                                                                |  |   |  |   |  |  |
|--|--|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|--|---|--|--|
|  |  | (iii) | <ul style="list-style-type: none"> <li>{Bacteria absorb <u>water</u>/ <u>water</u> enters bacterial cells} by <u>osmosis</u> (1)</li> <li>{less negative/ higher} {water / solute} potential outside (than inside)/ lower solute concentration outside (than inside)/ reference to water potential gradient into cell(1) Accept ORA</li> <li>Bacterial cells burst (due to damage to cell wall) (1)</li> </ul> |  | 3 |  | 3 |  |  |
|--|--|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|--|---|--|--|

3. (a) Red blood cells need to take in glucose and oxygen from the blood plasma. Glucose is a polar molecule and oxygen is non-polar.

The uptake of glucose and oxygen by red blood cells was measured separately at increasing external concentrations of these two molecules.

**Graph 3.1** shows the rate of uptake of glucose and oxygen.

**Graph 3.1**



- (i) **Complete Table 3.2** to identify:
- which line on **Graph 3.1** represents glucose and which represents oxygen;
  - the type of transport used by each substance.
- [2]

**Table 3.2**

| Line     | Substance | Type of transport |
|----------|-----------|-------------------|
| <b>A</b> | .....     | .....             |
| <b>B</b> | .....     | .....             |

- (ii) Explain the difference in the shape of lines **A** and **B** shown on **Graph 3.1**. [4]

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- (b) Red blood cells are packed full of haemoglobin.  
When mature, they do not contain any of the organelles usually found in eukaryotic cells.

Using this information and your knowledge of **cell structure** conclude why mature red blood cells:

- (i) cannot make haemoglobin; [2]

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- (ii) can only transport substances across the cell membrane against a concentration gradient at very low rates. [2]

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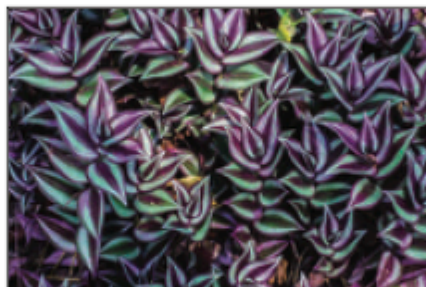
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|   |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |          |          |           |          |          |
|---|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
| 3 | (a) | (i)  | <p>A – Oxygen – Diffusion/ simple diffusion<br/>B – Glucose - Facilitated diffusion/ active transport</p> <p>1 mark for correct substances<br/>1 mark for transport mechanisms</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1        | 1        |          | 2         |          | 2        |
|   |     | (ii) | <ul style="list-style-type: none"> <li>• {Oxygen/ (substance) A} passes through {phospholipid/ bilayer} because it is non-polar (1)</li> <li>• {Line A/ oxygen/ diffusion}: {linear relationship/ positive correlation/ directly proportional} (between concentration gradient and rate of diffusion)/ or description (1)</li> <li>• {Glucose/ (substance) B} uses {channel/ carrier/ intrinsic/ transmembrane} proteins as it is polar (1)</li> <li>• {Line B/ glucose/ facilitated diffusion/ active transport}: the rate is eventually limited by the number of {channel/ carrier/ intrinsic/ cell membrane} proteins (1)</li> </ul> <p>Must refer to carrier proteins if use active transport</p> | 2        | 2        |          | 4         |          |          |
|   | (b) | (i)  | <p><b>Mark points in pairs</b></p> <ul style="list-style-type: none"> <li>• No {DNA/ genes} (1)</li> <li>• to code for haemoglobin (1)</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>• No ribosomes (1)</li> <li>• for translation/ protein synthesis. (1)</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>• No Golgi {body/ apparatus} (1)</li> <li>• so no post-translational modification (1)</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>• No mitochondria (1)</li> <li>• to make ATP for protein synthesis (1)</li> </ul>                                                                                                                              |          |          | 2        | 2         |          |          |
|   |     | (ii) | <p>No mitochondria (1)<br/>No ATP for Active transport (1)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |          | 2        | 2         |          |          |
|   |     |      | <b>Question 3 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>3</b> | <b>3</b> | <b>4</b> | <b>10</b> | <b>0</b> | <b>2</b> |



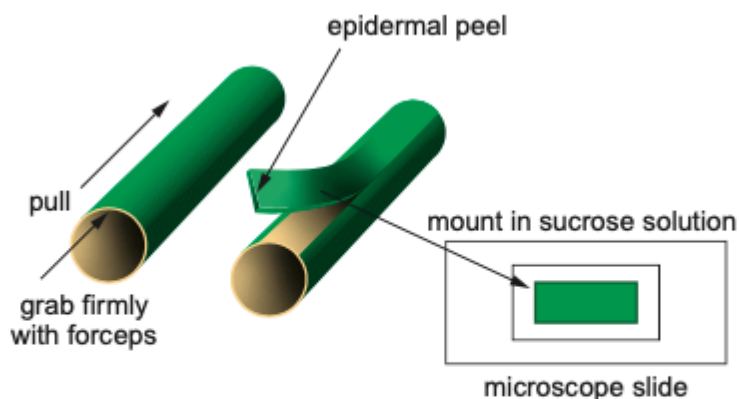
6. *Tradescantia*, shown in **Image 6.1**, is a house plant which is popular due to its ability to survive long periods without watering because its stems can store water.

**Image 6.1**



The solute potential of the cells of the stem was measured by using epidermal peels as shown in **Image 6.2**. Each peel was mounted on a different microscope slide in a different concentration of sucrose solution.

**Image 6.2**



After 10 minutes the peels were observed under a microscope. The number of plasmolysed and unplasmolysed cells in each sample was recorded. The results are shown in **Table 6.3**.

**Table 6.3**

| Sucrose concentration/<br>$\text{mol dm}^{-3}$ | Number of<br>plasmolysed<br>cells | Number of<br>unplasmolysed<br>cells | Total cells<br>counted | Percentage<br>of cells<br>plasmolysed |
|------------------------------------------------|-----------------------------------|-------------------------------------|------------------------|---------------------------------------|
| 0.0                                            | 0                                 | 165                                 | 165                    | 0.0                                   |
| 0.2                                            | 20                                | 174                                 | 194                    | 10.3                                  |
| 0.4                                            | 90                                | 105                                 | 195                    | 46.2                                  |
| 0.6                                            | 87                                | 69                                  | 156                    | 55.8                                  |
| 0.8                                            | 76                                | 54                                  | 130                    | 58.4                                  |
| 1.0                                            | 172                               | 73                                  | .....                  | .....                                 |

(a) (i) **Complete Table 6.3** for  $1.0 \text{ mol dm}^{-3}$  sucrose.

[2]

- (ii) Use the data in **Table 6.3** to draw a graph which shows the percentage of plasmolysed cells against sucrose concentration.

[4]



- (iii) I. Use your graph and the formula below to estimate the solute potential of the solution when 50% of the *Tradescantia* cells are plasmolysed. [2]

$$\text{Solute potential } (\Psi_s) = - (\text{sucrose concentration} \times 24.35)$$

Solute potential = ..... kPa

- II. State the pressure potential of the cells in the tissue of *Tradescantia* at this solute concentration and identify the term that describes their condition. [2]

Pressure potential ( $\Psi_p$ ) = ..... kPa

Term = .....

- (b) Suggest how the experiment could be altered to improve:

- confidence in the results
- the accuracy of the estimated solute potential in *Tradescantia* cells. [2]

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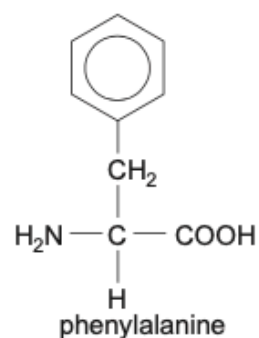
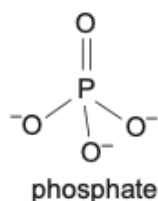
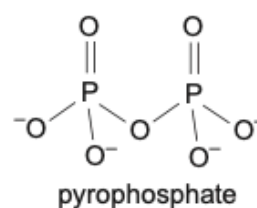
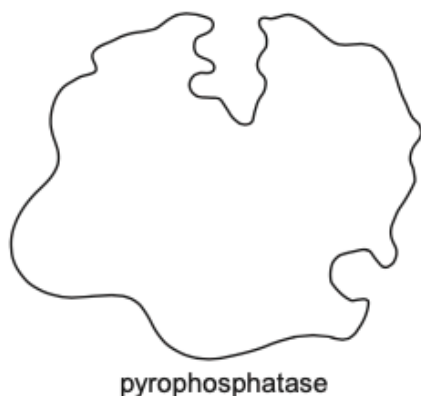
|   |     |         |                                                                                                                                                                                                                                            |          |                  |          |           |          |           |
|---|-----|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------|----------|-----------|----------|-----------|
| 6 | (a) | (i)     | 70.2 = 2 marks<br>If incorrect award 1 mark for<br>70.20.....<br>70<br>172/245 x 100                                                                                                                                                       |          | 2                |          | 2         | 2        |           |
|   |     | (ii)    | X axis labelled sucrose concentration/ mol dm <sup>-3</sup> <b>and</b><br>Y axis labelled % plasmolysed cells (1)<br>Linear scales on X and Y axes, including figure at origin (1)<br>Correct plots (tolerance $\pm 1/2$ small square) (2) |          | 1<br>1<br>1<br>1 |          | 4         | 4        | 4         |
|   |     | (iii) I | -11.7 or - 11.688 or -11.69 = 2 marks<br>If incorrect award 1 mark for<br>Calculation of $\Psi_s$ -( 0.48 x24.35)<br>11.7 or 11.688 or 11.69<br><br>Will need to check graph for $\Psi_s$ at 50% plasmolysis                               |          | 2                |          | 2         | 2        | 2         |
|   |     | II      | At this point pressure potential = 0 (1)<br>incipient plasmolysis (1)                                                                                                                                                                      | 2        |                  |          | 2         |          | 2         |
|   | (b) |         | Improve confidence: Repeat (all readings) / take more samples(1)<br>Improve accuracy: (Repeat with) sucrose solutions of concentration {between 0.4 and 0.6 / 0.5 / any value between 0.4 and 0.6 }(1) ecf from (iii) I                    |          |                  | 1<br>1   | 2         |          | 2         |
|   |     |         | <b>Question 6 total</b>                                                                                                                                                                                                                    | <b>2</b> | <b>8</b>         | <b>2</b> | <b>12</b> | <b>8</b> | <b>10</b> |

Speciment

1.4

## WJEC 2016

6. Pyrophosphatase is an enzyme found inside the nucleus of cells and is involved in DNA replication. The enzyme catalyses the conversion of a molecule of pyrophosphate to two phosphate ions. The diagrams below show the enzyme pyrophosphatase and its substrate pyrophosphate. Molecules of phenylalanine (an amino acid) and phosphate are also shown; both of these molecules are known to inhibit pyrophosphatase. (Drawings are not to the same scale).



Describe and explain why pyrophosphatase can only hydrolyse pyrophosphate and the mechanism by which phenylalanine and phosphate inhibit pyrophosphatase. [9 QER]

Indicative content

- Pyrophosphatase has tertiary structure
  - The active site has a specific shape
  - Pyrophosphate has a complementary shape
  - fits into the active site/ forms Enzyme Substrate Complex
  - lock and key theory/ induced fit
- 
- Phenylalanine must be a non-competitive inhibitor;
  - As it has a shape different to pyrophosphate;
  - Phenylalanine bonds to the allosteric site/site away from the active site;
  - Causes a change in the shape of the active site;
  - Pyrophosphate's active site is no longer complementary/prevents pyrophosphate binding
  - fewer enzyme-substrate complexes are formed;
  - The maximum rate of reaction cannot be reached at any concentration of pyrophosphate;
- 
- Phosphate must be a competitive inhibitor;
  - As it has a similar shape to {pyrophosphate/ substrate} / phosphate has a complementary shape the active site;
  - Phosphate {binds to/ competes for} the active site;
  - Prevents pyrophosphate binding / fewer enzyme-substrate complexes are formed;
  - The maximum rate of reaction can be reached at higher concentrations of pyrophosphate;

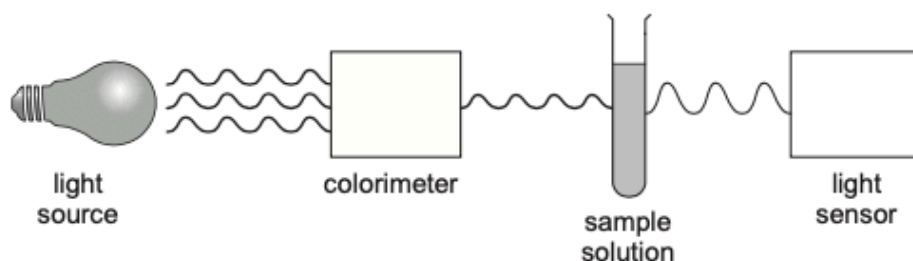
4. Lipases are involved in diverse biological processes, including the hydrolysis of triglycerides. Some lipases act in specific compartments inside cells, while others work in extracellular spaces.

(a) State the term given to enzymes which work within specific compartments inside cells.

[1]

.....

Triacetin is a colourless triglyceride and can be hydrolysed by bacterial lipase. When an indicator is added to alkaline triacetin solution a coloured solution is produced. A colorimeter can then be used to monitor the progress of the reaction. When the triacetin is completely digested, the solution will change from coloured to colourless, allowing more light to pass through, and produce an absorbance reading of zero.



The effect of temperature on the hydrolysis of triacetin was investigated.

The experiment was performed as follows:

1.  $2\text{ cm}^3$  of triacetin was mixed with  $3\text{ cm}^3$  of sodium hydrogen carbonate (pH10) in a test tube and 5 drops of indicator added.
2.  $1\text{ cm}^3$  of lipase was placed in a separate test tube.
3. Both test tubes were placed in a water bath ( $20^\circ\text{C}$ ) for 5 minutes.
4. The contents of the tubes were combined, mixed thoroughly and  $2\text{ cm}^3$  transferred to a cuvette.
5. The time taken for the absorbance to reach zero was measured.
6. The experiment was repeated to give a total of three readings.
7. Steps 1-6 were repeated at  $25$ ,  $35$ ,  $45$ , and  $50^\circ\text{C}$ .

(b) (i) Identify the independent and dependent variables in this investigation.

[2]

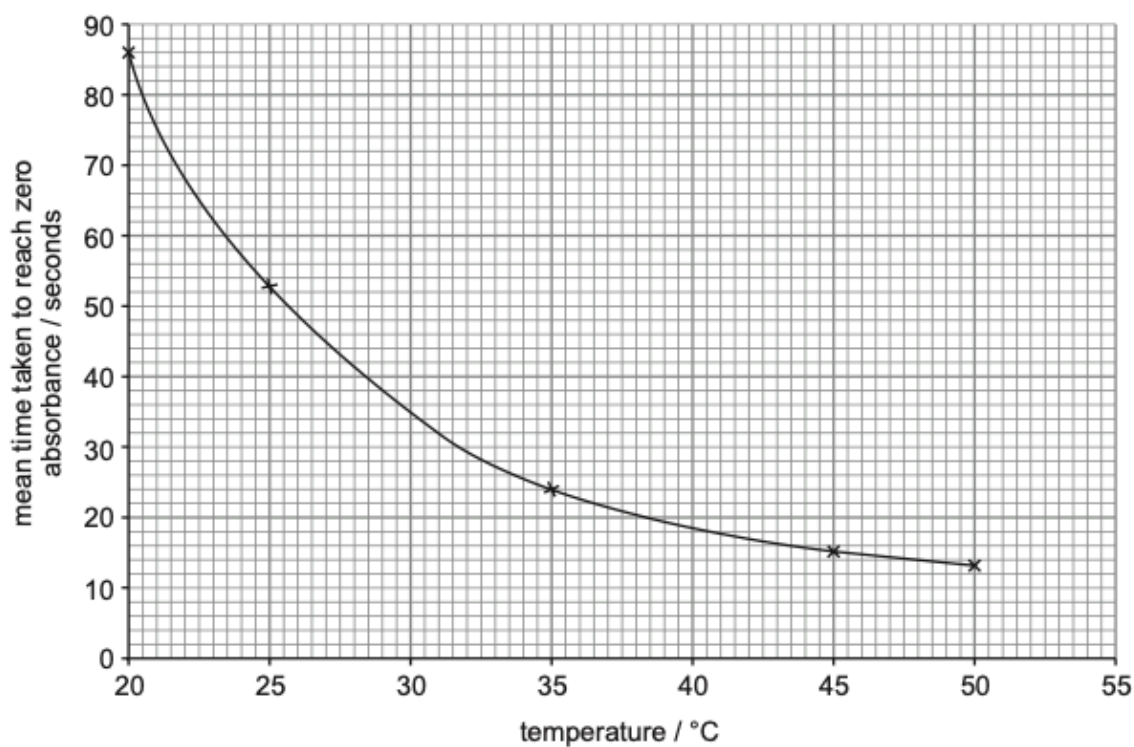
independent .....

dependent .....

- (ii) Explain why the solutions were left in the water bath for 5 minutes before combining and mixing. [1]

- (iii) Explain why it would be inappropriate to use a buffer solution in this experiment. [3]

The results of the investigation are shown below.





- (c) (i) In this experiment,  $2\text{ cm}^3$  of  $0.25\text{ g cm}^{-3}$  triacetin was used. Calculate the rate of digestion per second at  $40^\circ\text{C}$ .

Give your answer in standard form. [3]

Rate = .....  $\text{g s}^{-1}$

- (ii) Suggest **one** source of inaccuracy in this experimental method. [1]

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- (iii) It was not possible to determine the optimum temperature for this enzyme in this experiment. Explain why and suggest how the design might be improved. [4]

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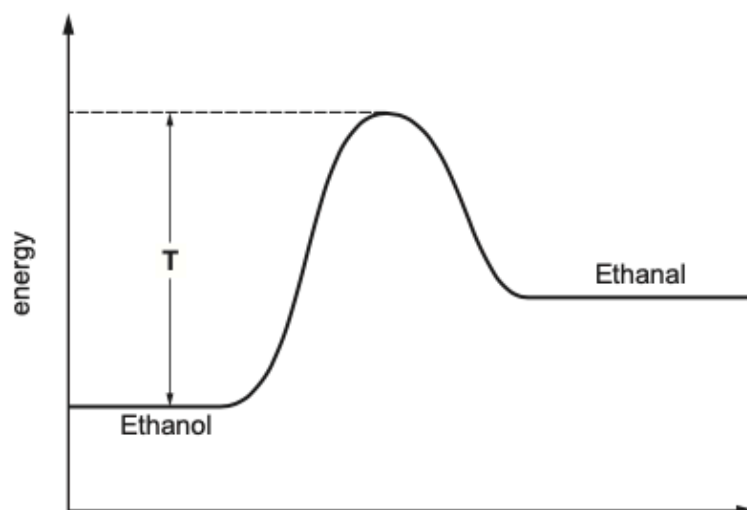
|   |     |       |                                                                                                                                                                                                                                                                         |   |   |  |   |  |   |
|---|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--|---|--|---|
| 4 | (a) |       | Intracellular                                                                                                                                                                                                                                                           | 1 |   |  | 1 |  |   |
|   | (b) | (i)   | <i>Independent:</i> temperature (1)<br><i>Dependent:</i> time (taken) to reach zero absorbance (1)                                                                                                                                                                      | 2 |   |  | 2 |  | 2 |
|   |     | (ii)  | Acclimatisation/ owtte<br>reject stated temperature                                                                                                                                                                                                                     |   | 1 |  | 1 |  | 1 |
|   |     | (iii) | A. {Colour change/ indicator} is dependent on a change in pH (1)<br>B. {Hydrolysis produces} fatty acids {that lower the pH/ make it acid} (1)<br>C. Buffer solution would {stabilise/ maintain/ control} pH (to pH10/ alkaline conditions) / 'mop-up' $\text{H}^+$ (1) |   | 3 |  | 3 |  | 3 |

|     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |          |          |           |          |           |
|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|-----------|
| (c) | (i)   | <p><b>Using 18 seconds</b><br/> <math>2.78 \times 10^{-2} / 2.8 \times 10^{-2} / 2.63 \times 10^{-2} / 2.6 \times 10^{-2} = 3</math> marks<br/> <math>0.0277777777777778 / 0.028 / 0.02631579 / 0.03 = 2</math> marks (not standard form)<br/> <math>2.7 \times 10^{-2} = 2</math> marks (incorrectly rounded)<br/> <math>2.78 \times 10^{-1} / 2.63 \times 10^{-1} = 2</math> marks (incorrect conversion)</p> <p><b>Using 19 seconds</b><br/> <math>2.63 \times 10^{-2} / 2.6 \times 10^{-2} = 3</math> marks<br/> <math>0.027 = 1</math> mark (not standard form and incorrect rounding)</p> <p><b>NOT taking volume into account</b><br/> <b>Using 18 seconds</b><br/> <math>1.39 \times 10^{-2} / 1.4 \times 10^{-2} = 2</math> marks<br/> <math>1.38 \times 10^{-2} = 1</math> mark (incorrectly rounded)<br/> <math>0.0139 / 0.014 = 1</math> mark ( not taking standard form into account )<br/> <math>1.3 \times 10^{-2} = 1</math> mark (incorrectly rounded)<br/> <b>Using 19 seconds</b><br/> <math>1.32 \times 10^{-2} / 1.3 \times 10^{-2} = 2</math> marks<br/> <math>0.0132 / 0.013 = 1</math> mark (not taking standard form into account)<br/> <math>1.31 \times 10^{-2} = 1</math> mark (incorrect rounding)</p> <p><b>OR</b><br/> Allow <math>2 \times 0.25 = 0.5</math> for 1 mark</p> |          | 3        |          | 3         | 3        |           |
|     | (ii)  | Cooling in the colorimeter/<br>Accuracy of equipment used to measure volumes/<br>not all volumes transferred (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |          | 1        | 1         |          | 1         |
|     | (iii) | <p>A. Inappropriate (temperature) range (1)<br/> B. {time taken/ graph} {is still decreasing/ has not levelled off/ all enzymes still active (1)<br/> <i>Improvement :</i><br/> C. Needs to test increase temperatures above <math>50^{\circ}\text{C}</math>, (to determine how well the enzyme is able to work at higher temperatures) (1) ignore any temperatures below <math>50^{\circ}\text{C}</math><br/> D. smaller increments (to determine the optimum accurately) (1)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |          | 4        | 4         |          | 4         |
|     |       | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>3</b> | <b>7</b> | <b>5</b> | <b>15</b> | <b>3</b> | <b>11</b> |

## WJEC 2018

6. Alcohol dehydrogenase is an enzyme that catalyses the conversion of ethanol into ethanal.

The graph below shows the change in energy state when ethanol is converted to ethanal when no enzyme is present.



- (a) (i) State what is represented by the region labelled **T**. [1]

- (ii) Draw a curve on the diagram to show the change in energy level when alcohol dehydrogenase is present. [1]

- (b) Describe how alcohol dehydrogenase can catalyse the conversion of ethanol into ethanal. [3]

.....

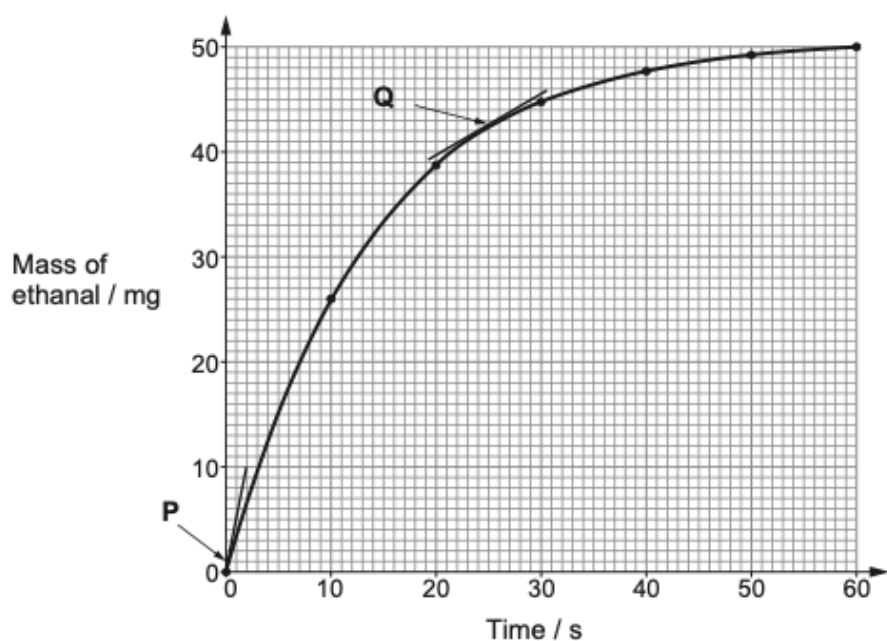
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- (c) A fixed mass of ethanol was added to a test tube containing alcohol dehydrogenase and a pH7 buffer solution. The test tube was incubated at 30°C and the mass of ethanal produced over time was recorded. The results are shown below.



- (i) The rate of reaction at **P** was  $5 \text{ mg s}^{-1}$ .  
Calculate the rate of reaction at **Q**.

[2]

rate = .....  $\text{mg s}^{-1}$

- (ii) What conclusions can be drawn that would account for the difference in rate at **P** and at **Q**? [4]

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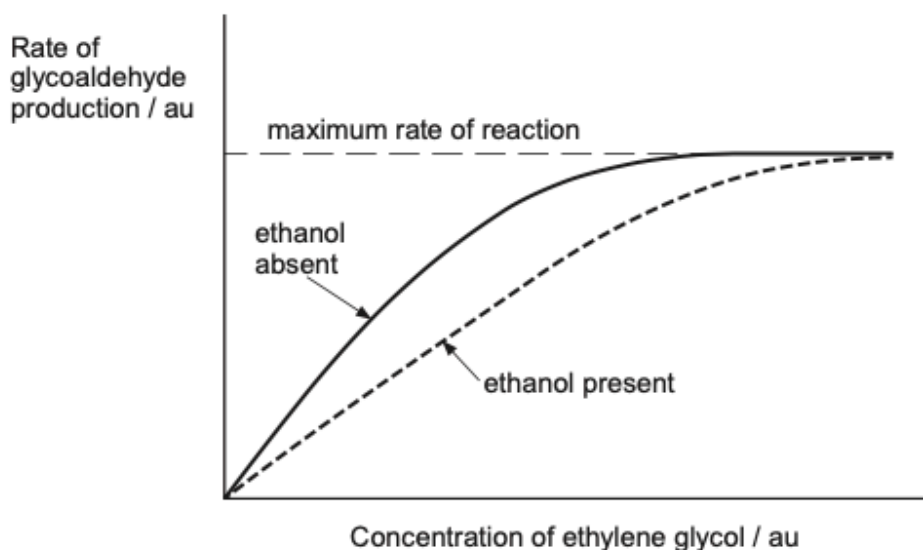
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- (d) Ethylene glycol is a colourless, odourless, sweet liquid, commonly found in antifreeze. It is highly toxic if ingested because once inside the body ethylene glycol is converted into glycoaldehyde. This reaction is also catalysed by alcohol dehydrogenase.

Treatment of ethylene glycol poisoning includes giving the patient ethanol, either intravenously or orally.

The graph below shows the rate of glycoaldehyde production in the presence, and absence of ethanol.



Use the information provided to explain why this treatment would reduce the effects of ethylene glycol poisoning. [5]

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|   |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |   |   |    |   |  |
|---|-----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|----|---|--|
| 6 | (a) | (i)  | Activation energy                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1 |   |   | 1  |   |  |
|   |     | (ii) | Curve drawn with a lower activation energy under existing curve – energy state at beginning and end must be the same.                                                                                                                                                                                                                                                                                                                                                    | 1 |   |   | 1  |   |  |
|   | (b) |      | {Alcohol dehydrogenase/ enzyme} has a <u>specific</u> {shaped active site/ tertiary structure/ OWTTE} (1)<br>{Ethanol/ substrate} has a <u>complementary</u> (shape) (1)<br>(The two fit together) to form an <u>enzyme-substrate complex</u> (1)                                                                                                                                                                                                                        | 3 |   |   | 3  |   |  |
|   | (c) | (i)  | Any answer between 0.57 - 0.63 = 2 marks<br>If incorrect award 1 mark for sight of :<br>attempted calculation of gradient                                                                                                                                                                                                                                                                                                                                                |   | 2 |   | 2  | 2 |  |
|   |     | (ii) | 1. P: the rate of reaction is higher as the concentration of {ethanol/ substrate} is high (1)<br>2. The concentration of {alcohol dehydrogenase/ enzyme} is limiting the rate of reaction; (1)<br>3. Q: the rate of reaction is lower the concentration of {ethanol/ substrate} decreases(1)<br>4. The concentration of {ethanol/ substrate} becomes the limiting factor. (1)                                                                                            |   | 2 | 2 | 4  |   |  |
| 6 | (d) |      | 1. Ethanol and ethylene glycol must have a similar structure / Ethanol and ethylene glycol must both be complementary to the shape of the active site of {alcohol dehydrogenase/ the enzyme} (1)<br>2. ethanol acts as a competitive inhibitor (1)<br>3. When ethanol binds to the active site it prevents ethylene glycol from attaching (1)<br>4. Fewer enzyme-substrate complexes form (1)<br>5. Which reduces the rate of production of {glycoaldehyde/ product} (1) |   | 3 | 2 | 5  |   |  |
|   |     |      | <b>Question 6 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 5 | 7 | 4 | 16 | 2 |  |

# WJEC 2019

3. (a) A student wanted to investigate the concentration of casein in milk powder. Casein is an insoluble protein found in milk. When milk powder is mixed with water it forms an opaque (cloudy) suspension. The enzyme trypsin breaks down the casein into soluble peptides, forming a transparent solution.

To begin with the student needed to construct a calibration curve from known concentrations of milk powder suspensions. He was provided with the following:

- 10 cm<sup>3</sup> suspensions made from milk powder at concentrations of 1%, 2%, 3%, 4% and 5% in separate test tubes with a buffer
- 1% trypsin solution

The milk suspensions and trypsin solution were placed separately in a thermostatically controlled water bath at 30 °C. Once both the milk suspensions and trypsin solution were at the correct temperature, he transferred 2 cm<sup>3</sup> of trypsin to each of the suspensions of milk powder and timed how long it took for the suspensions to become transparent.

- (i) Using your knowledge of enzymes, explain why:

- I. it was necessary for a buffer to be added to the solutions; [2]

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 .....  
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- II. all test tubes were placed in a thermostatically controlled water bath. [2]

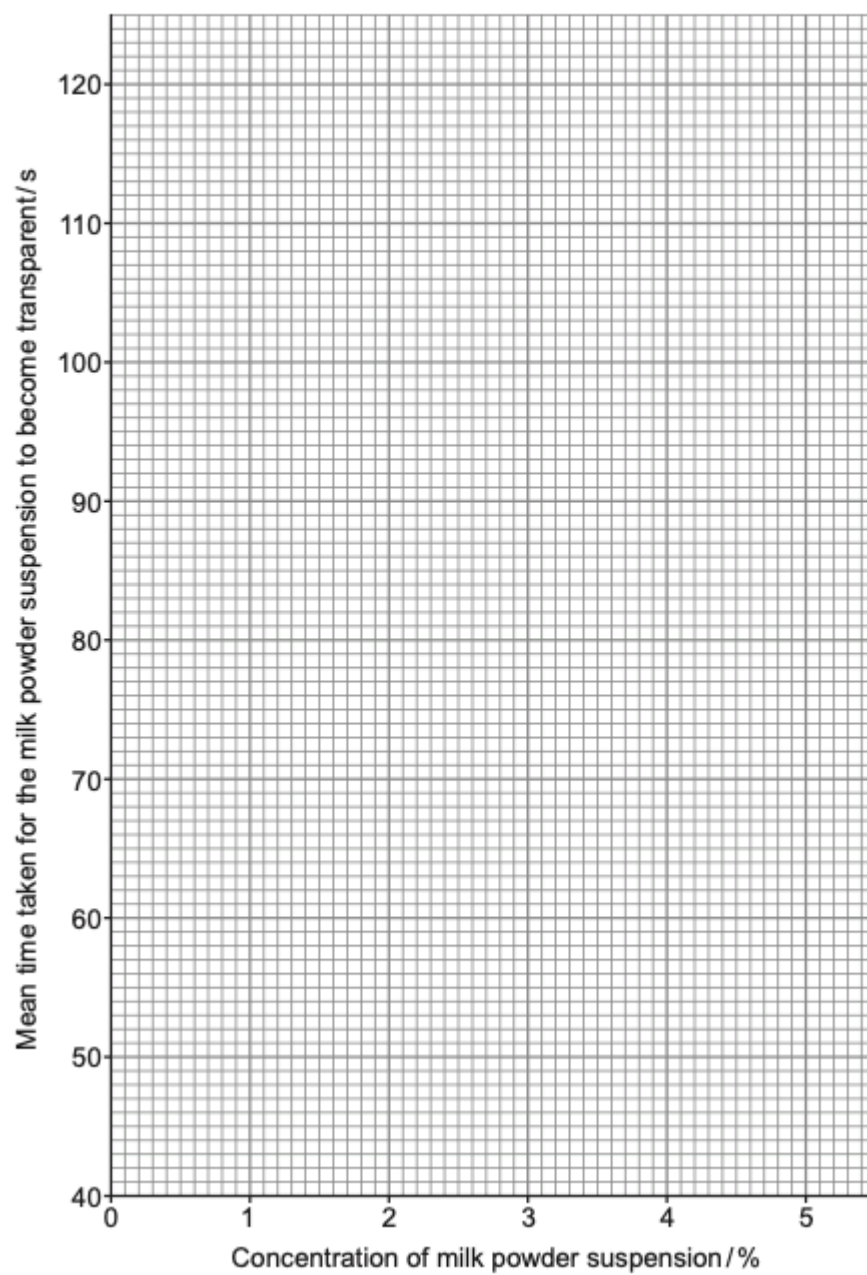
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The table below shows the results.

| Concentration of milk powder suspension / % | Time taken for milk powder suspension to become transparent / s |         |         |         |         |       |
|---------------------------------------------|-----------------------------------------------------------------|---------|---------|---------|---------|-------|
|                                             | Trial 1                                                         | Trial 2 | Trial 3 | Trial 4 | Trial 5 | mean  |
| 1                                           | 49                                                              | 51      | 55      | 52      | 53      | 52.0  |
| 2                                           | 66                                                              | 64      | 62      | 61      | 63      | 63.2  |
| 3                                           | 88                                                              | 77      | 73      | 81      | 84      | 80.6  |
| 4                                           | 91                                                              | 87      | 93      | 86      | 99      | ..... |
| 5                                           | 96                                                              | 109     | 108     | 111     | 105     | 105.8 |

- (ii) Calculate the mean time taken for the 4% milk powder suspension to become transparent. **Write your answer in the table.** [1]

- (iii) Plot the mean data on the graph paper provided. **Range bars should also be added to the graph.** [4]





(b) The student was then provided with two different suspensions of milk powder, **X** and **Y**, of unknown casein concentration. He carried out the same procedure and obtained the following mean results: **X** took 61 seconds to clear and **Y** took 97 seconds to clear.

(i) Estimate the percentage concentration of casein in these two suspensions. [1]

**X:** .....

**Y:** .....

(ii) The student had more confidence in the estimate for suspension **X** than the estimate for suspension **Y**. Use the graph to explain the reason for this statement. [2]

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(iii) Suggest **two** sources of inaccuracy in this experiment. [2]

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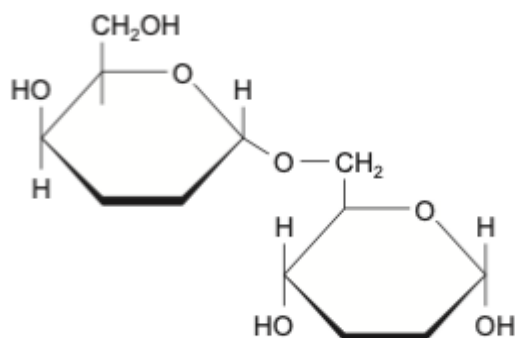
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|   |     |       |    |                                                                                                                                                                                                                                                                                                                                                                         |   |   |   |   |
|---|-----|-------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|
| 3 | (a) | (i)   | I  | {Maintain/ control/ prevent changes in} pH (1)<br><br>keeping pH at optimum/<br>which would alter the {rate of reaction/ time taken}/<br>which could denature the enzymes/<br>prevents coagulation of milk (1)                                                                                                                                                          | 1 | 1 | 2 | 2 |
|   |     |       | II | {Maintain/ control/ prevent changes in} temperature/ acclimatise to the same temperature (1)<br>so that all of the enzymes and substrates had the same <u>kinetic</u> energy /<br>if some test tubes were incubated at a higher temperature the enzymes (and substrates) would have more <u>kinetic</u> energy /<br>ORA (1)<br>Ignore references to denaturing/ optimum | 1 | 1 | 2 | 2 |
|   |     | (ii)  |    | 91.2                                                                                                                                                                                                                                                                                                                                                                    |   | 1 | 1 | 1 |
|   |     | (iii) |    | Correct plots (2) $\pm \frac{1}{2}$ small square<br>5 correct = 2 marks<br>4 correct = 1 mark<br>0/1/2/3 = 0 marks<br>Centre of plots joined with a ruler, no extrapolation/<br>line of best fit with ruler (allow extrapolation) (1)<br>Range bars correct (1) $\pm \frac{1}{2}$ small square                                                                          |   | 4 | 4 | 4 |

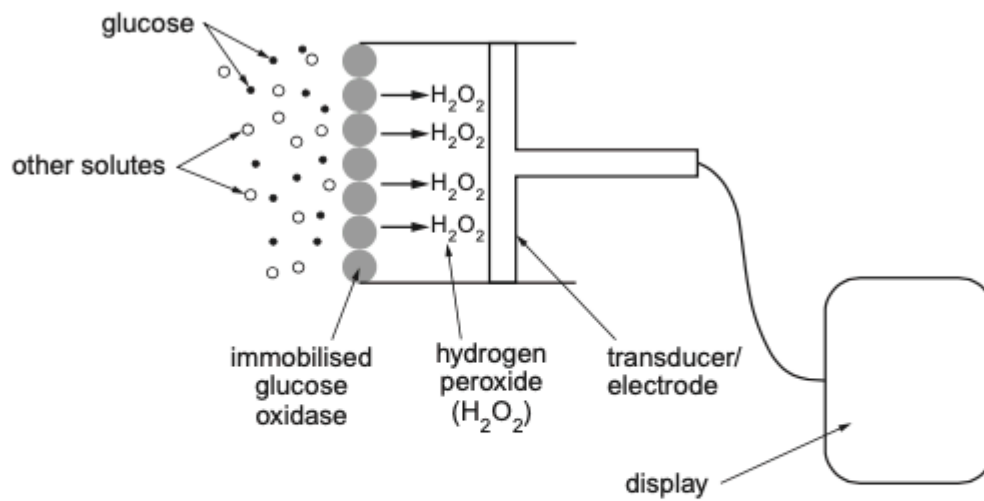
|     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |          |          |           |          |           |
|-----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|-----------|
| (b) | (i)   | X = 1.6 - 1.8% <u>and</u> Y = 4.3 – 4.5%<br>Take from candidates graph                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          | 1        |          | 1         | 1        | 1         |
|     | (ii)  | Any <b>two</b> (x1) from:<br><ul style="list-style-type: none"> <li>• Comparison of {range bars/ data} for {1%/2%/ X/ lower concentrations} (smaller) compared to {3%/4%/ 5%/ Y/ higher concentrations} (larger) (1)</li> <li>• Comparison of degree of overlap of the range bars e.g. at 3%, 4% and 5% all overlap and no overlap between 1 and 2% (1)</li> <li>• Correct reference to {reliability/ repeatability/ consistency} linked to {length of range bar/ spread of data}<br/>OR there would be less confidence in the trend for the overlapping results. (1)<br/>Reject reference to accuracy/ reproducibility</li> </ul> |          |          | 2        | 2         |          | 2         |
|     | (iii) | {Judging/ subjectivity of} {when the solution becomes transparent/ the end point } (1)<br>As the milk powder forms a suspension, the powder may form sediment at the bottom off the test tube altering the time taken for the milk to clear / not mixed evenly/ not all transferred from one test tube to another (1)                                                                                                                                                                                                                                                                                                              |          |          | 2        | 2         |          | 2         |
|     |       | <b>Question 3 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>2</b> | <b>8</b> | <b>4</b> | <b>14</b> | <b>7</b> | <b>14</b> |

5. Melibiose is a reducing sugar that is formed from galactose and glucose and is thirty times sweeter than sucrose.

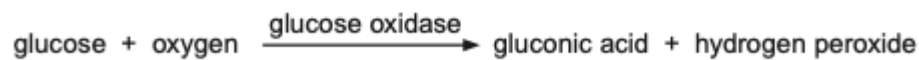
The diagram below shows the structure of melibiose.



- (b) Biosensors make use of immobilised enzymes to detect specific molecules in a mixture. The diagram below shows a possible structure of a biosensor that could be used to detect the presence of glucose in a urine sample.



Within the biosensor, the glucose oxidase catalyses the following reaction:



The concentration of hydrogen peroxide is detected by the transducer and converted into an electrical signal, shown as a measurement on the display.

- (i) Using your knowledge of enzymes explain why this biosensor will only detect the presence of glucose. [3]

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- (ii) With reference to the information given, explain how the biosensor could be used to give a measurement of the glucose concentration. [2]

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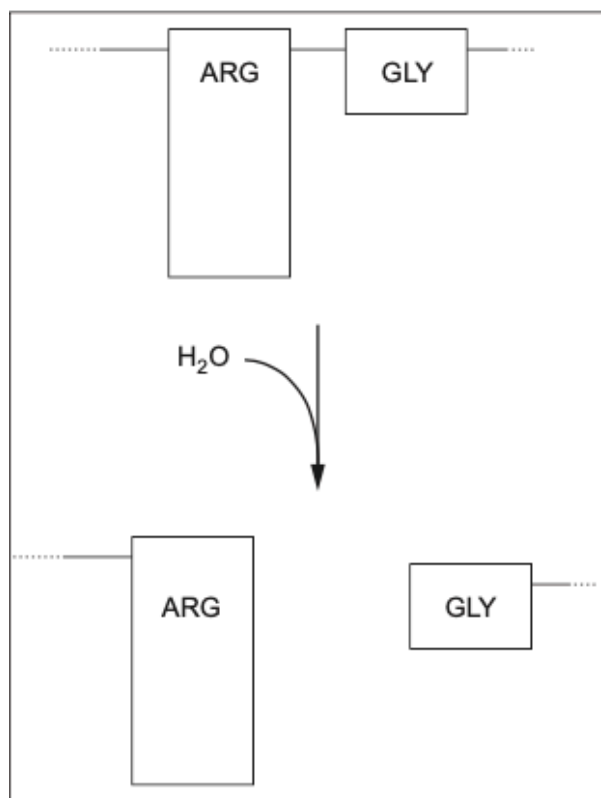
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|     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |          |          |           |          |          |
|-----|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
| (b) | (i)  | Glucose oxidase has a <u>specific shaped</u> active site (1)<br>Only glucose molecules have a complementary (shape) (1)<br>(Therefore glucose fits) to form an enzyme-substrate complex (1)                                                                                                                                                                                                                                                                                                                                                                      | 1        | 2        |          | 3         |          |          |
|     | (ii) | Any <b>two</b> (x1) from:<br><ul style="list-style-type: none"> <li>More glucose the more hydrogen peroxide is produced (1)</li> <li>Equation shows a one to one relationship / is directly proportional (1)</li> <li>Concentration of hydrogen peroxide is converted to an electrical signal/ detected by the electrode (1)</li> </ul>                                                                                                                                                                                                                          |          |          | 2        | 2         |          |          |
| (c) |      | Any <b>four</b> (x1) from:<br>A. Arabinose has a similar {shape/ structure} to <u>glucose/</u> complementary to <u>active site</u> of glucose oxidase (1)<br>Reject same shape<br>B. So would act as a competitive inhibitor (of glucose oxidase)/ {competes for/ binds to} the active site (1)<br>C. preventing the formation of enzyme-substrate complexes (1)<br>D. Therefore, less hydrogen peroxide produced (1)<br>E. The display would give a lower reading (1)<br><br>Answer including description of non-competitive inhibition negates mark points A-C |          |          | 4        | 4         |          |          |
|     |      | <b>Question 5 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>5</b> | <b>3</b> | <b>6</b> | <b>14</b> | <b>0</b> | <b>2</b> |

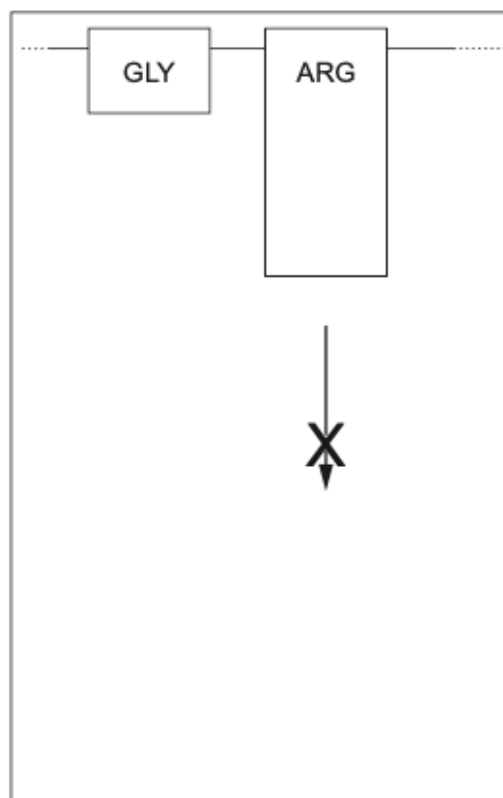
## WJEC 2022

- (c) Thrombin catalyses the hydrolysis of an arginine-glycine bond in fibrinogen as shown in **Image 6.3**.

**Image 6.3**



**Image 6.4**

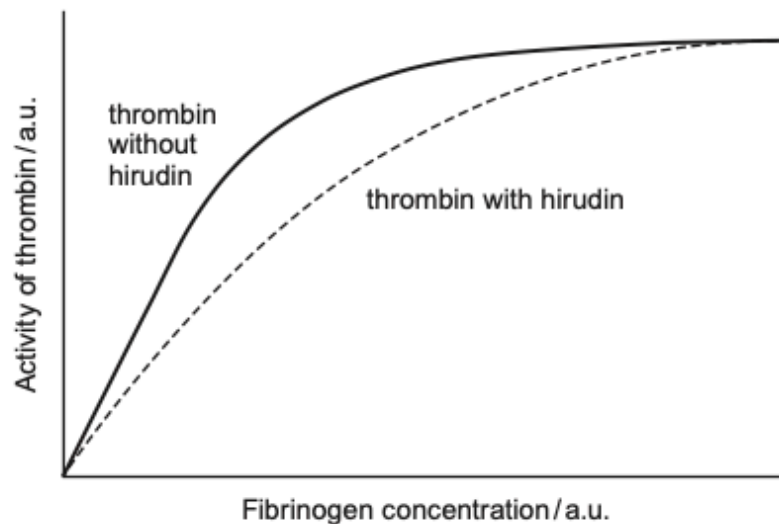


Name the type of bond broken in **Image 6.3** and explain why thrombin can catalyse this reaction, but it would not catalyse the breaking of a glycine-arginine bond as shown in **Image 6.4**. [2]

- (d) Leeches are organisms which feed by sucking blood. Their saliva contains many components, including a protein called hirudin.

**Graph 6.5** shows the inhibitory effect of hirudin on the activity of thrombin.

**Graph 6.5**



- (i) Using your knowledge of enzymes, identify and explain the type of inhibition shown by hirudin and suggest how hirudin enables the leech to feed effectively. [4]

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- (ii) Leeches for medical use are farmed in South Wales as a source of hirudin. Suggest a use for hirudin in medicine. [1]

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|     |      |  |                                                                                                                                                                                                                                                                                                                  |   |   |   |   |  |  |
|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|--|--|
| (c) |      |  | Peptide bond (1)<br>The active site of thrombin is (only) complimentary to {the arginine-glycine bond/ the enzyme}/ ORA<br>thrombin is specific to the arginine - glycine bond (1)                                                                                                                               | 1 | 1 |   | 2 |  |  |
| (d) | (i)  |  | A. {Hirudin/ it} acts as a competitive inhibitor (1)<br>B. {Fits into/ binds to/ complimentary to/ competes for} active site (on thrombin) (1)<br>C. {Preventing/ reducing} {ESCs/ or description of} (1)<br>D. (Hirudin slows down the action of Thrombin) so that blood {clots more slowly/ does not clot} (1) | 2 | 1 | 1 | 4 |  |  |
|     | (ii) |  | Used to {prevent/ reduce risk of/ slow down formation of} blood clots/ DVT/ anticoagulant/ owtte                                                                                                                                                                                                                 |   |   | 1 | 1 |  |  |

## WJEC 2023

(iv) Lysozyme works by an induced fit mechanism. Explain what this means.

[1]

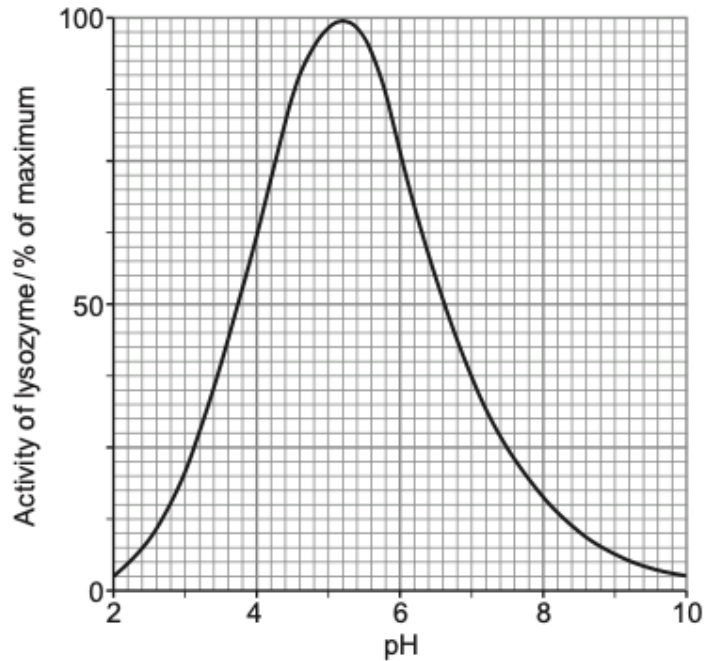
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- (b) The effect of pH on the activity of lysozyme is shown in **Graph 1.3**.

**Graph 1.3**



- (i) Describe and explain the effect of pH on the activity of lysozyme.

[3]

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- (ii) Lysozyme has a wider range of pH activity than many other enzymes.

- I. Use **Graph 1.3** to find the range of pH over which the activity of lysozyme is more than 50% of the maximum. [1]

Range = ..... to .....

- II. Use the information in the question to suggest **one** advantage of lysozyme having a wider range of pH activity than many other enzymes. [1]

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|  |  |      |                                                                                                                                                                                 |   |  |  |   |  |  |
|--|--|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|--|---|--|--|
|  |  | (iv) | substrate changes the shape of the <u>active site</u> (as it enters it) / (enzyme/ lysozyme) changes shape to allow <u>substrate</u> to fit into <u>active site</u> / OWTTE (1) | 1 |  |  | 1 |  |  |
|--|--|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|--|---|--|--|



|     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |  |   |   |  |
|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--|---|---|--|
| (b) | (i) | <p><b>Any three (×1) from</b></p> <p>A. Decreases away from optimum/ description of shape of full graph/ owtte (1)</p> <p>B. Of pH 5.2 (1)</p> <p>C. (changing pH) changes the {secondary/tertiary} structure/ (changing pH) {changes/ breaks} the bonds that hold the enzyme in shape/ owtte (1) Ignore reference to denaturing</p> <p>D. Disruption of active site prevents substrate binding/ fewer enzyme substrate complexes/ active site is no longer complementary to the substrate (1)</p> | 2 | 1 |  | 3 | 1 |  |
|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--|---|---|--|

|  |      |    |                                                                                                                                      |          |          |          |           |          |          |
|--|------|----|--------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
|  | (ii) | I  | {3.7/ 3.8} -6.6                                                                                                                      |          | 1        |          | 1         | 1        |          |
|  |      | II | {It is active/ can work} in a range of {areas/ pHs} {in and out of the body/ around the body}/ owtte (1) Must be in relation to body |          |          | 1        | 1         |          |          |
|  |      |    | <b>Question 1 total</b>                                                                                                              | <b>4</b> | <b>7</b> | <b>1</b> | <b>12</b> | <b>2</b> | <b>0</b> |

# Speciment

3. The scientists Devulapalle and Mooser carried out experiments using the enzyme glucosyl transferase that catalyses the polymerisation of glucose molecules from dietary sucrose. Polymers of glucose form part of the plaque on teeth that can lead to dental disease.

- (a) (i) Name two theories that can explain how glucosyl transferase can combine with its substrate, sucrose. [1]

..... and .....

- (ii) Explain why the rate of a chemical reaction is increased by the glucosyl transferase when acting as a biological catalyst. [1]

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Devulapalle and Mooser showed that the presence of iron (II) sulphate affected the activity of glucosyl transferase. Some of their results are shown in the table below.

| <i>Iron II sulphate concentration (mM)</i> | <i>glucosyl transferase activity (labelled product <math>\mu</math>moles)</i> |
|--------------------------------------------|-------------------------------------------------------------------------------|
| 0.0                                        | 60.0                                                                          |
| 1.0                                        | 25.2                                                                          |
| 2.0                                        | 12.0                                                                          |
| 3.0                                        | 5.2                                                                           |
| 4.0                                        | 2.4                                                                           |
| 5.0                                        | 1.6                                                                           |
| 6.0                                        | 1.2                                                                           |

- (b) They concluded that iron (II) sulphate was acting as an enzyme inhibitor. Further investigations showed that the effect of iron (II) sulphate was reduced at higher concentrations of sucrose. Explain how iron (II) sulphate could inhibit glucosyl transferase. [4]

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(c) Suggest a possible use of Devulapalle and Mooser's findings.

[1]

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| 7 |
|---|

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                             | Marks available |          |          |          |          |          |
|----------|-----|------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                             | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 3        | (a) | (i)  |  | Lock and key <b>and</b> induced fit (1)                                                                                                                                                                                                                                                     | 1               |          |          | 1        |          |          |
|          |     | (ii) |  | Lower the activation energy/eq (1)                                                                                                                                                                                                                                                          | 1               |          |          | 1        |          |          |
|          | (b) |      |  | competitive inhibition (1)<br>iron sulphate could have a shape similar to the substrate/complementary to the active site of glucosyl transferase (1)<br>Fit/ bind into the active site (1)<br>Prevent the {substrate molecule/ sucrose} entering the active site/blocks the active site (1) | 2               | 2        |          | 4        |          |          |
|          | (c) |      |  | (Add iron sulphate to toothpaste / mouthwash / sugary drinks.) to prevent formation of plaque / tooth decay. (1)                                                                                                                                                                            |                 | 1        |          | 1        |          |          |
|          |     |      |  | <b>Question 3 total</b>                                                                                                                                                                                                                                                                     | <b>4</b>        | <b>3</b> | <b>0</b> | <b>7</b> | <b>0</b> | <b>0</b> |

6. A student carried out an experiment to investigate the effect of temperature on cell membranes. Using a borer, equal sized pieces of beetroot were cut, washed, and blotted with a paper towel. Each piece was placed into a test tube containing 25 cm<sup>3</sup> of 70 % ethanol (an organic solvent) and incubated at 15 °C. A red pigment called betacyanin found in the vacuoles of the beetroot cells began to leak out into the ethanol turning it red. The experiment was repeated at 30 °C and 45 °C and the time taken for the ethanol to turn red was recorded below.

| Temperature<br>(°C) | Time taken for the ethanol to turn red (s) |         |         |       |
|---------------------|--------------------------------------------|---------|---------|-------|
|                     | Trial 1                                    | Trial 2 | Trial 3 | Mean  |
| 15                  | 450                                        | 427     | 466     | 447.7 |
| 30                  | 322                                        | 299     | 367     | 329.3 |
| 45                  | 170                                        | 99      | 215     | 161.3 |

- (a) Calculate the percentage decrease in mean time taken for the ethanol to turn red between 15 and 30 °C. [2]

Answer .....

- (b) Using your knowledge of the structure of cell membranes, explain why ethanol causes the pigment to leak out of the beetroot cells. [2]

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- (c) What conclusions could be drawn from this experiment regarding the effect of temperature on cell membranes? [3]

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- (d) Comment on the validity of your conclusion. [2]

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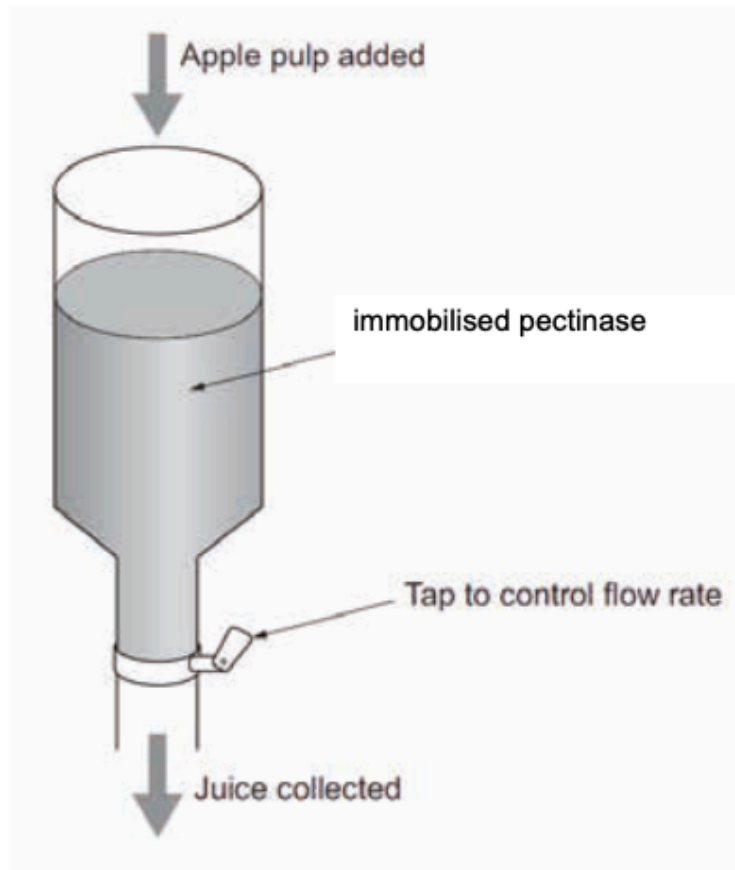
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| 9 |
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| Question |     |  |  | Marking details                                                                                                                                                                                                                               |  | Marks available |          |          |          |          |          |
|----------|-----|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------|----------|----------|----------|----------|----------|
|          |     |  |  |                                                                                                                                                                                                                                               |  | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 6        | (a) |  |  | 447.7 - 329.3 = 118.4 (1)<br>118.4 / 447.7 x 100 = 26.4 % (Acc 26) (1)                                                                                                                                                                        |  |                 | 2        |          | 2        | 2        |          |
|          | (b) |  |  | Ethanol dissolves phospholipids / denatures proteins (1)<br>Creates gaps in the membrane (1)                                                                                                                                                  |  |                 | 2        |          | 2        |          |          |
|          | (c) |  |  | Increased temperature increases KE of membrane and dye molecules(1)<br>Increased movement of membrane molecules increases the number and size of gaps in membrane (1)<br>More dye molecules can escape from cells (1)                         |  |                 |          | 3        | 3        |          |          |
|          | (d) |  |  | Any <b>two</b> from<br>At 45°C {repeat results vary greatly / range from 99 to 215 seconds} (1)<br>Only three temperatures were investigated (1)<br>Could be several shades of red/ no standard red colour used/ red colour not quantified(1) |  |                 |          | 2        | 2        |          | 2        |
|          |     |  |  | <b>Question 6 total</b>                                                                                                                                                                                                                       |  | <b>0</b>        | <b>4</b> | <b>5</b> | <b>9</b> | <b>2</b> | <b>2</b> |

7. Pectin is a structural polysaccharide found in plant cell walls and in the middle lamella between cells, where it acts to bind cells together. Pectinases are enzymes that are used routinely in industry to increase the volume and clarity of fruit juice extracted from apples. The enzyme is immobilised and then placed inside a column. Apple pulp is added at the top, and juice is collected at the bottom. The process is shown in the diagram below.



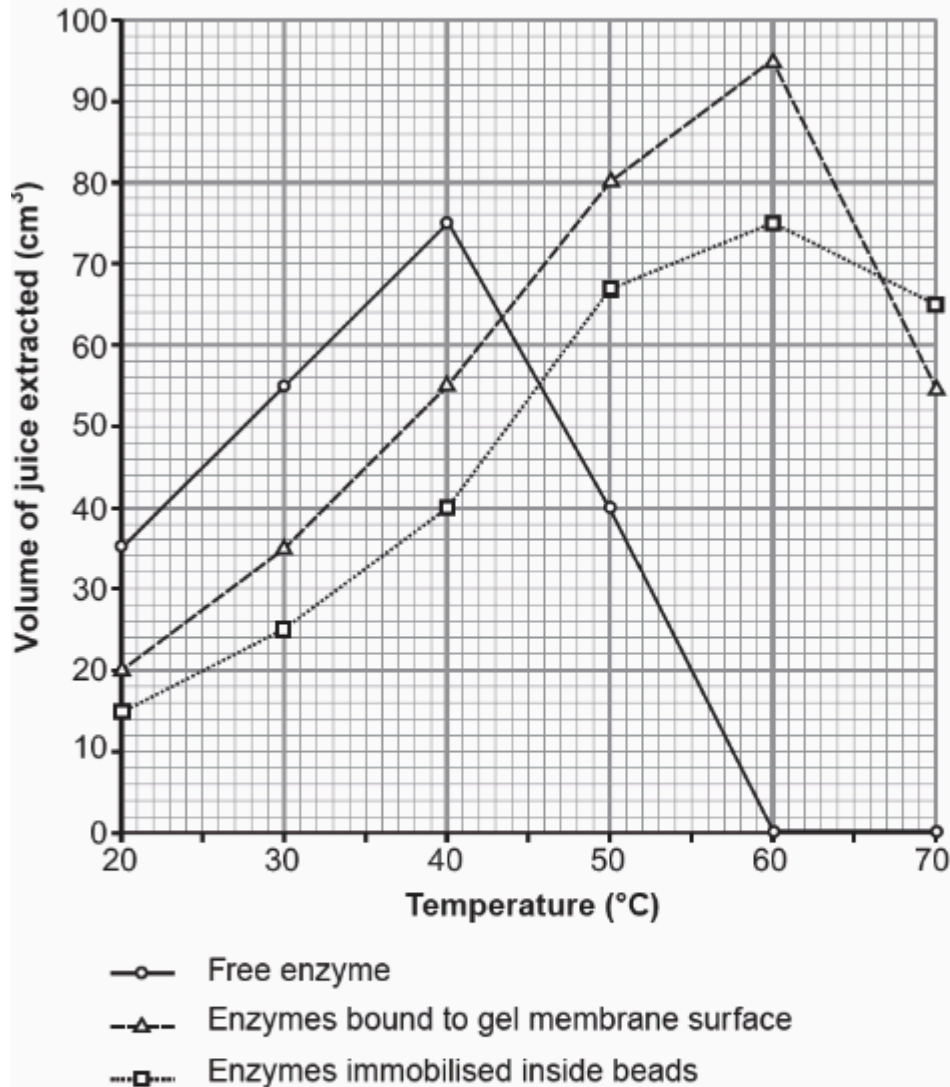
- (a) Explain why reducing the flow rate of material through the column would result in increased juice collected. [2]

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- (b) The extraction of juice at different temperatures was compared using equal volumes and concentrations of free enzyme, enzymes bound to the surface of a gel membrane and enzymes encapsulated inside alginate beads. The results are shown in the graph below.



Using the graph and your own knowledge of enzymes, answer the following questions.

- (i) Describe and explain the results for the free enzymes at temperatures **above 40°C**. [4]

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- (ii) Explain why a higher yield of juice was obtained when using free enzymes between temperatures of 20°C and 40°C than when using either kind of immobilised enzyme. [2]

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- (iii) Explain the differences seen in the results for the enzymes bound to the gel membrane surface with those immobilised inside the beads, between temperatures of 20°C and 60°C. [2]

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- (iv) A Welsh apple juice producer wishes to increase their yield of apple juice. Use the information given to outline how they could adapt the basic method shown to extract a higher volume of apple juice. Explain the recommended modifications. [4]

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| Question         |     |       | Marking details                                                                                                                                                                                                                                                                                                                                                                      | Marks available |     |     |       |       |      |
|------------------|-----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|                  |     |       |                                                                                                                                                                                                                                                                                                                                                                                      | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 7                | (a) |       | Increases (contact) time between enzymes and substrate (1)<br>More successful collisions/more enzyme substrate complexes formed (1)                                                                                                                                                                                                                                                  |                 | 2   |     | 2     |       | 2    |
|                  | (b) | (i)   | 40°C to 60°C decrease in volume of fruit juice extracted (1)<br>Above/at 60°C no juice extracted (1)<br>Between 40°C to 60°C enzymes are denaturing / at 60°C they are denatured/ above 40 °C hydrogen bonds breaking (1)<br>{Tertiary structure deformed / active site changes shape} {so substrate can no longer fit active site/{no/fewer} enzyme substrate complexes formed} (1) |                 | 4   |     | 4     |       |      |
|                  |     | (ii)  | (Free enzymes) can move (1)<br>Increased chance of successful collision / more enzyme substrate complexes formed(1)<br>allow converse                                                                                                                                                                                                                                                |                 | 2   |     | 2     |       |      |
|                  |     | (iii) | (Increased juice extracted with membrane bound enzymes) because membrane bound enzymes are directly exposed to substrate (1)<br>(Enzymes immobilised inside bead) substrate has to {diffuse/pass} into bead (1)                                                                                                                                                                      |                 | 2   |     | 2     |       |      |
|                  |     | (iv)  | Use enzymes bound to a gel membrane at 60°C(1)<br>Highest yield compared to free and encapsulated enzyme(1)<br>Use many pieces of gel membrane to increase surface area(1)<br>Slow flow rate to increase contact time for formation of enzyme substrate complexes (1)                                                                                                                |                 |     | 4   | 4     |       | 4    |
| Question 7 total |     |       |                                                                                                                                                                                                                                                                                                                                                                                      | 0               | 10  | 4   | 14    | 0     | 6    |

1.5

WJEC 2016

2. Influenza, commonly known as “the flu”, is an infectious disease caused by the influenza virus, which infects the epithelial cells of the upper respiratory tract. The virus is composed of a protein capsid which surrounds the enzyme, RNA polymerase and its genetic material, RNA.

(a) Describe the differences between the genetic material of the Influenza virus and the genetic material found within the nucleus of the epithelial cell it infects. [4]

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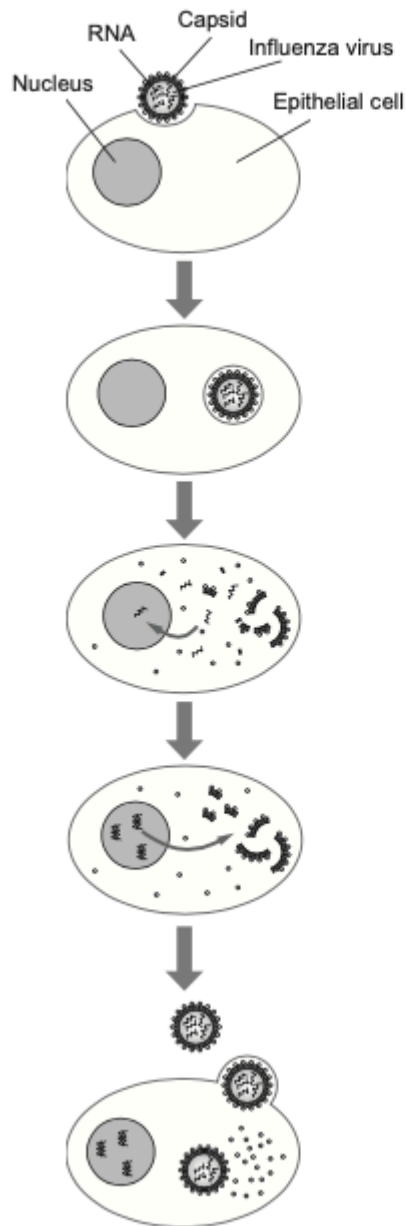
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- (b) The virus can only replicate in living (host) cells, where it utilises nutrients and organelles within the cell to multiply quickly. The diagram below shows the stages in the replication of the virus.



① Influenza virus becomes attached to a target epithelial cell.

② The cell engulfs the virus.

③ Viral contents are released. Viral RNA enters the nucleus where it is replicated by the viral RNA polymerase.

④ Viral mRNA is used to make viral proteins.

⑤ New viral particles are made and released into the extracellular fluid. The cell, which is not killed in the process, continues to make a new virus.

(i) State the name of the process occurring between stages 1 and 2. [1]

(ii) In stage 3, the capsid has broken down and the viral RNA and RNA polymerase enter the nucleus, where the viral RNA is replicated. Apply your knowledge of RNA polymerase to describe how the viral RNA is replicated. [4]

(iii) To complete the replication of the virus, capsids need to be produced (stage 4). Describe how the proteins in the capsid are produced. [5]

(iv) Explain why the virus is unable to synthesise its own capsid.

[2]

| 2                                       | (a)                                                             |                              | genetic material in the epithelial cell is DNA (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                      |            |                              |                     |             |            |                  |                     |                         |                               |                                         |                                                                 |   |  |  |   |  |  |
|-----------------------------------------|-----------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------|------------------------------|---------------------|-------------|------------|------------------|---------------------|-------------------------|-------------------------------|-----------------------------------------|-----------------------------------------------------------------|---|--|--|---|--|--|
|                                         |                                                                 |                              | <b>Any 3 ( x1) from:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                      |            |                              |                     |             |            |                  |                     |                         |                               |                                         |                                                                 |   |  |  |   |  |  |
|                                         |                                                                 |                              | <table><tr><th>DNA/ epithelial cell</th><th>RNA/ virus</th></tr><tr><td>double stranded/double helix</td><td>single stranded (1)</td></tr><tr><td>deoxyribose</td><td>ribose (1)</td></tr><tr><td>contains thymine</td><td>contains uracil (1)</td></tr><tr><td>Larger/ longer molecule</td><td>Smaller/ shorter molecule (1)</td></tr><tr><td>(Genetic material contained in nucleus)</td><td>Genetic material not {contained in nucleus/ membrane bound} (1)</td></tr></table>                                                                                                                                                                                   | DNA/ epithelial cell | RNA/ virus | double stranded/double helix | single stranded (1) | deoxyribose | ribose (1) | contains thymine | contains uracil (1) | Larger/ longer molecule | Smaller/ shorter molecule (1) | (Genetic material contained in nucleus) | Genetic material not {contained in nucleus/ membrane bound} (1) | 4 |  |  | 4 |  |  |
|                                         |                                                                 | DNA/ epithelial cell         | RNA/ virus                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                      |            |                              |                     |             |            |                  |                     |                         |                               |                                         |                                                                 |   |  |  |   |  |  |
|                                         |                                                                 | double stranded/double helix | single stranded (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                      |            |                              |                     |             |            |                  |                     |                         |                               |                                         |                                                                 |   |  |  |   |  |  |
| deoxyribose                             | ribose (1)                                                      |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |            |                              |                     |             |            |                  |                     |                         |                               |                                         |                                                                 |   |  |  |   |  |  |
| contains thymine                        | contains uracil (1)                                             |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |            |                              |                     |             |            |                  |                     |                         |                               |                                         |                                                                 |   |  |  |   |  |  |
| Larger/ longer molecule                 | Smaller/ shorter molecule (1)                                   |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |            |                              |                     |             |            |                  |                     |                         |                               |                                         |                                                                 |   |  |  |   |  |  |
| (Genetic material contained in nucleus) | Genetic material not {contained in nucleus/ membrane bound} (1) |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |            |                              |                     |             |            |                  |                     |                         |                               |                                         |                                                                 |   |  |  |   |  |  |
|                                         | Answers must be comparative (apart from fifth MP)               |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |            |                              |                     |             |            |                  |                     |                         |                               |                                         |                                                                 |   |  |  |   |  |  |
|                                         | Reference to prokaryotic cells is neutral                       |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |            |                              |                     |             |            |                  |                     |                         |                               |                                         |                                                                 |   |  |  |   |  |  |
|                                         | (b)                                                             | (i)                          | {Endocytosis/phagocytosis} Accept micropinocytosis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1                    |            |                              | 1                   |             |            |                  |                     |                         |                               |                                         |                                                                 |   |  |  |   |  |  |
|                                         |                                                                 | (ii)                         | <b>Any four (x1) from:</b><br>1. (Viral) RNA acts as a template (1)<br>2. (Free RNA nucleotides attach to their) complementary base pairs (1) Reject DNA<br>3. Adenine with uracil <u>and</u> cytosine with guanine (1)<br>Accept A:U, C:G<br>4. RNA polymerase catalyses the formation of the new RNA molecule (1)<br>5. Forming the {sugar phosphate backbone/phosphodiester bond} (1)                                                                                                                                                                                                                                                                           |                      | 4          |                              | 4                   |             |            |                  |                     |                         |                               |                                         |                                                                 |   |  |  |   |  |  |
|                                         |                                                                 | (iii)                        | <b>Any five (x1) from:</b><br>1. Translation (1)<br>2. mRNA attaches to a ribosome (1)<br>3. tRNA brings a (specific) amino acid (1)<br>4. Complementary base pairing between mRNA codon and tRNA anticodon/ codon anticodon interaction (or description of) (1)<br>5. Ribosome has room for two tRNA molecules/Brings two amino acids into close proximity/ or description of (1)<br>6. Condensation reaction occurs/peptide bond formed (between amino acids)(1)<br>7. Ribosome moves to the next codon (or description of)/ reference to stop codon (1)<br>8. Post translation processing/ or description of e.g. folding of polypeptide chain role of Golgi(1) | 5                    |            |                              | 5                   |             |            |                  |                     |                         |                               |                                         |                                                                 |   |  |  |   |  |  |
|                                         |                                                                 | (iv)                         | It does not possess {any ribosomes/tRNA / the organelles for protein synthesis} (1) Reject: organelles unqualified<br>All other organelles = neutral<br>It does not possess mitochondria / cannot produce its own ATP for the process (1)                                                                                                                                                                                                                                                                                                                                                                                                                          |                      | 2          |                              | 2                   |             |            |                  |                     |                         |                               |                                         |                                                                 |   |  |  |   |  |  |
|                                         |                                                                 |                              | <b>Question 2 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>10</b>            | <b>6</b>   | <b>0</b>                     | <b>16</b>           | <b>0</b>    | <b>0</b>   |                  |                     |                         |                               |                                         |                                                                 |   |  |  |   |  |  |

# WJEC 2017

## 2. Meselson and Stahl investigated the replication of DNA prior to cell division.

In their experiments, bacteria were cultured in a medium containing only the  $^{15}\text{N}$  isotope (heavy nitrogen). The bacteria were then transferred to  $^{14}\text{N}$  (light nitrogen) culture medium and allowed to divide. Following each generation, samples of DNA were taken, separated using an ultracentrifuge and analysed.

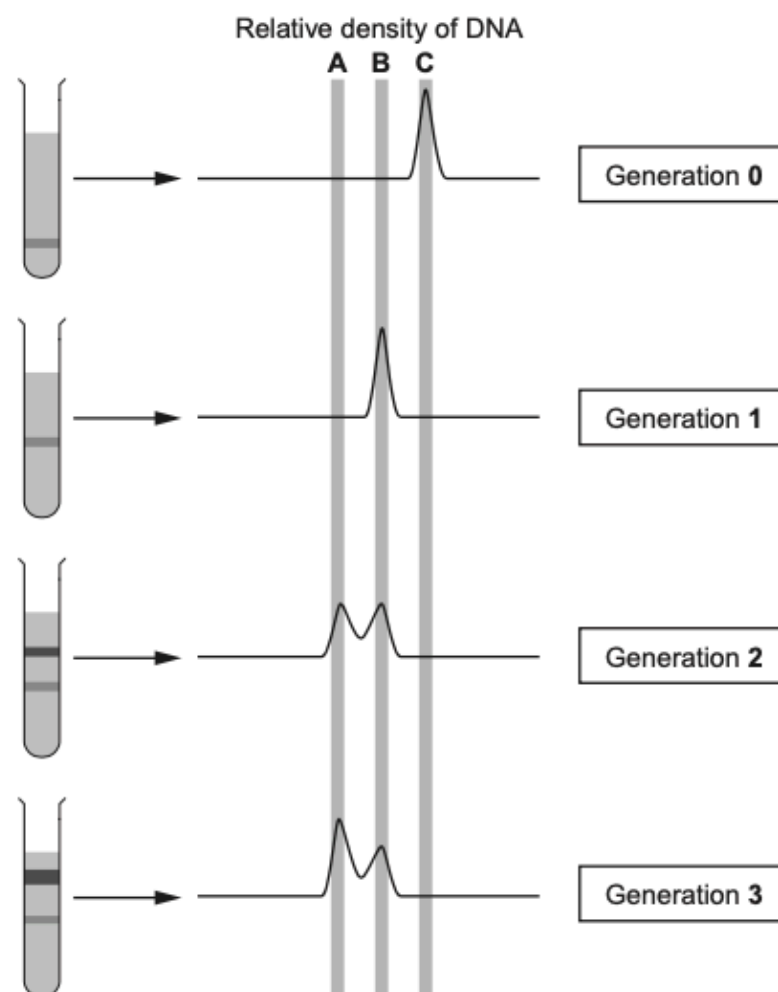
- (a) Identify **three** variables which would need to be controlled during centrifugation of the samples. [2]

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Photographs were taken of the extracted bands of DNA. The relative density of each band was scanned and measured; the more dense the DNA, the higher the reading. Shown below are some of their experimental results.





- (b) (i) State which isotopes of nitrogen would be found in the DNA at **A**, **B** and **C** in the scans. [1]

**A** .....

**B** .....

**C** .....

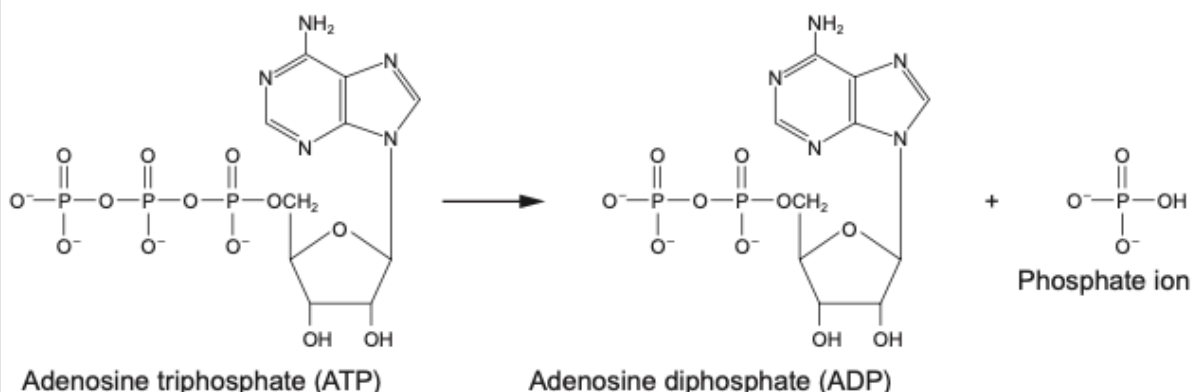
- (ii) Explain how the results in generations 0 and 1 support the theory of semi-conservative replication. [3]

- (iii) Explain the height of the peaks in the scans for generations 2 and 3. [4]

- (iv) Name **two** enzymes now known to be involved in DNA replication and explain their functions. [3]

|   |      |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |          |          |           |          |          |
|---|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
| 2 | (a)  |       | Any three (x1) from:<br>Temperature<br>Time<br>Rate of spin<br>Volume/ mass/ amount of DNA/ sample<br>relevant condition of tube ie {density / concentration / volume / height of/ type of } medium Not solution unqualified<br>3 correct = 2 marks, 2/1 correct = 1 mark, 0 correct = 0 marks                                                                                                                                                                                        |          | 2        |          | 2         |          | 2        |
|   | (b)  | (i)   | A: $^{14}\text{N}$ /light/14<br>B: $^{14}\text{N}$ + $^{15}\text{N}$ /light and heavy/14 and 15<br>C: $^{15}\text{N}$ / heavy/ 15<br>All correct = 1 mark                                                                                                                                                                                                                                                                                                                             |          | 1        |          | 1         |          | 1        |
|   |      | (ii)  | A. {Generation 0 /{band/peak} at C} is all {heavy isotope/ $\text{N}^{15}$ } (1)<br>B. <b>Generation 1</b> {band/ peak} at B is {a mixture of heavy and light isotope/ an intermediate}(1)<br>C. Because each molecule retains one strand of heavy isotope and one strand of newly formed DNA of light isotope (1)<br>Accept hybrid DNA for intermediate DNA                                                                                                                          | 2        |          |          | 3         |          |          |
|   |      |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          | 1        |          |           |          |          |
|   |      | (iii) | A. (Only) { $^{14}\text{N}$ / light isotope} (nucleotides ) are available for replication (1)<br>B. <b>Generation 2</b> has equal {height scans/ peaks / density} (1)<br>C. <b>because</b> one strand of heavy and one strand of light DNA have been used as templates for the formation of new DNA molecules/ owtte (1)<br>D. <b>Generation 3</b> has a higher peak of { $^{14}\text{N}$ / light isotope/ A} <b>because</b> more light DNA (strands) are used as template/ owtte (1) |          | 3        |          | 4         |          |          |
|   |      |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |          | 1        |           |          |          |
|   | (iv) |       | Any two for 1 mark from:<br>DNA polymerase / (DNA) helicase / (DNA) ligase (1)<br>Any two (x1) from:<br>• DNA polymerase – join free nucleotides (to the ends of the new DNA strand)(1)<br>• (DNA) helicase – {unzip/ unwind} {DNA/ helix} / breaks H-bonds {between nucleotides/ within the double strand/ between the bases} / separate (DNA) strands (1)<br>• (DNA) ligase – joins fragment of DNA to an existing fragment(1)                                                      | 3        |          |          | 3         |          |          |
|   |      |       | <b>Question 2 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>5</b> | <b>7</b> | <b>1</b> | <b>13</b> | <b>0</b> | <b>3</b> |

3. Adenosine triphosphate (ATP) belongs to a group of molecules that are made from nucleotides. ATP is an unstable molecule which possesses two high energy phosphate bonds and is often referred to as the universal energy currency. The diagram below shows the conversion of ATP into ADP.



- (a) (i) State the parts of the molecule that are common to DNA, RNA and ATP nucleotides. [1]

.....

.....

- (ii) State what is meant by the term *universal energy currency*. [1]

.....

.....

.....

- (iii) State whether the reaction shown above is exergonic or endergonic. Explain your answer. [2]

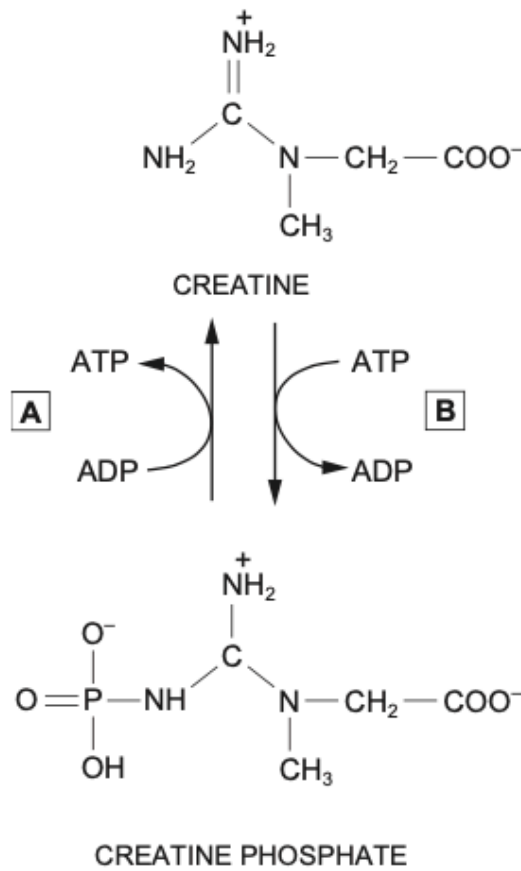
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.....

- (b) It is estimated that the mass of ATP that is found within a resting human at any one time is approximately 5g, yet a human uses an estimated 50 kg of ATP each day. Calculate the percentage of the total ATP used per day that is available at any one time. [2]

Percentage of the total ATP available = ..... %

Creatine phosphate is a naturally occurring chemical that works in partnership with ATP to increase the ability of muscle tissue to perform. It is not an energy source itself, but is used to increase the availability of ATP. It is stored inside muscle tissue.



- (c) Under what circumstances would each of the reactions **A** and **B** take place? Explain your answer. [4]

.....

.....

.....

.....

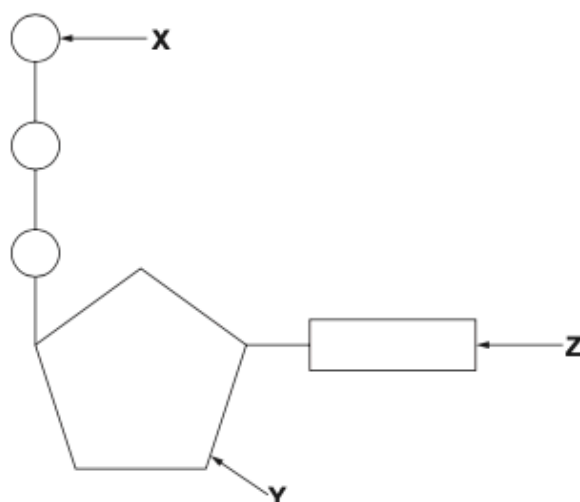
.....

.....

|   |     |       |                                                                                                                                                                                                                                                          |          |          |          |           |          |          |
|---|-----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
| 3 | (a) | (i)   | Pentose/ 5C (sugar) + phosphate (group) + nitrogenous / organic base NOT named base                                                                                                                                                                      | 1        |          |          | 1         |          |          |
|   |     | (ii)  | {All reactions/ source of energy} in all {cells/ organisms/ living things} NOT energy produced                                                                                                                                                           | 1        |          |          | 1         |          |          |
|   |     | (iii) | Exergonic (1)<br>{Energy/ 30.6kJ mol} {released/ given out} (1)<br>NOT produced                                                                                                                                                                          | 2        |          |          | 2         |          |          |
|   | (b) |       | 0.01 % = 2 marks<br>Allow correct conversion of units for one mark<br>5 / 1000 OR 50 x 1000 (1)                                                                                                                                                          |          | 2        |          | 2         | 2        |          |
|   | (c) |       | A: exercise / contracting <b>and</b> B: relaxing/ at rest (1)<br><br>At rest: CP is created to be a store of phosphate (1)<br>During exercise ATP can be generated quickly (1)<br>(When) phosphate group from CP is given to ADP/ is used to make ATP(1) |          | 1        | 3        | 4         |          |          |
|   |     |       | <b>Question 3 total</b>                                                                                                                                                                                                                                  | <b>4</b> | <b>3</b> | <b>3</b> | <b>10</b> | <b>2</b> | <b>0</b> |

WJEC 2018

2. The diagram below shows a molecule of ATP which is produced during respiration and is an energy carrier molecule used in cells. When ATP is broken down into ADP and  $P_i$ ,  $30.6 \text{ kJ mol}^{-1}$  of energy is released, which can be used for cellular activities.



- (a) (i) State the names of the parts of the ATP molecule labelled X, Y and Z. [2]

X: .....

Y: .....

Z: .....

- (ii) State **two** uses for the energy released from ATP in a plant cell. [1]

.....

.....

- (b) One mole of glucose releases  $2870 \text{ kJ}$  of energy when completely oxidised during aerobic respiration. Synthesis of one mole of ATP from ADP and  $P_i$  requires  $30.6 \text{ kJ}$  of energy. Calculate the percentage (%) efficiency of respiration if 38 moles of ATP are produced from one mole of glucose. Give your answer to three significant figures. [3]

% efficiency = .....

|   |     |      |                                                                                                                                                                                    |   |   |   |   |   |  |
|---|-----|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|--|
| 2 | (a) | (i)  | X: phosphate<br>Y: ribose<br>Z: adenine NOT adenosine<br>(2 correct = 1 mark, 3 correct = 2 marks)                                                                                 | 2 |   |   | 2 |   |  |
|   |     | (ii) | <b>Any two for one mark</b><br>Active transport / DNA replication / protein synthesis/ cell division (1)<br>Accept any correct function for a plant cell                           | 1 |   |   | 1 |   |  |
|   | (b) |      | 40.5 = 3 marks<br>If incorrect sig fig<br>$1162.8/2870 \times 100 = 40.5156794$ = 2 marks Accept any correct rounding<br>If answer incorrect<br>$38 \times 30.6 = 1162.8$ = 1 mark |   | 3 |   | 3 | 3 |  |
|   |     |      | <b>Question 2 total</b>                                                                                                                                                            | 3 | 3 | 0 | 6 | 3 |  |

4. Haemoglobin exhibits quaternary structure and consists of four subunits, two subunits called alpha-globin and two subunits called beta-globin.

(a) Define quaternary structure.

[1]

- (b) (i) The beta-globin molecule consists of 146 amino acids. State the minimum number of mRNA nucleotides required to code for this molecule.

[1]

- (ii) The genetic code is a triplet code and is degenerate. The table below shows the mRNA codons, that code for specific amino acids.

|              |   | Second letter                                                   |                                    |                                                         |                                                         |              |                  |
|--------------|---|-----------------------------------------------------------------|------------------------------------|---------------------------------------------------------|---------------------------------------------------------|--------------|------------------|
|              |   | U                                                               | C                                  | A                                                       | G                                                       |              |                  |
| First letter | U | UUU Phenyl-alanine<br>UUC<br>UUA Leucine<br>UUG                 | UCU Serine<br>UCC<br>UCA<br>UCG    | UAU Tyrosine<br>UAC<br>UAA Stop codon<br>UAG Stop codon | UGU Cysteine<br>UGC<br>UGA Stop codon<br>UGG Tryptophan | Third letter | U<br>C<br>A<br>G |
|              | C | CUU Leucine<br>CUC<br>CUA<br>CUG                                | CCU Proline<br>CCC<br>CCA<br>CCG   | CAU Histidine<br>CAC<br>CAA Glutamine<br>CAG            | CGU Arginine<br>CGC<br>CGA<br>CGG                       |              | U<br>C<br>A<br>G |
|              | A | AUU Isoleucine<br>AUC<br>AUA<br>AUG Methionine initiation codon | ACU Threonine<br>ACC<br>ACA<br>ACG | AAU Asparagine<br>AAC<br>AAA Lysine<br>AAG              | AGU Serine<br>AGC<br>AGA Arginine<br>AGG                |              | U<br>C<br>A<br>G |
|              | G | GUU Valine<br>GUC<br>GUA<br>GUG                                 | GCU Alanine<br>GCC<br>GCA<br>GCG   | GAU Aspartic acid<br>GAC<br>GAA Glutamic acid<br>GAG    | GGU Glycine<br>GGC<br>GGA<br>GGG                        |              | U<br>C<br>A<br>G |

Use the table to explain why it is necessary that the code is a triplet code and why the code is said to be degenerate.

[2]



- (iii) The table below shows part of the DNA sequence that codes for beta-globin and the corresponding mRNA codons, tRNA anticodons and the amino acid sequence of the molecule.

**Complete the table.**

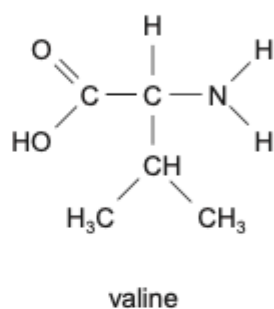
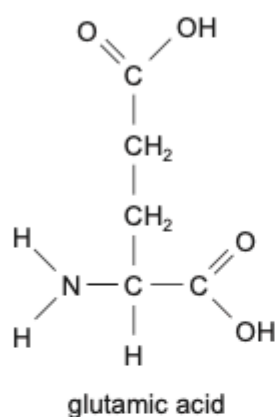
[4]

|                     | Codon position number |     |     |               |               |        |
|---------------------|-----------------------|-----|-----|---------------|---------------|--------|
|                     | 3                     | 4   | 5   | 6             | 7             | 8      |
| DNA coding strand   | GAC                   |     |     | CTC           | CTC           |        |
| mRNA codons         |                       | ACU |     | GAG           | GAG           |        |
| tRNA anticodons     |                       |     | GGA | CUC           | CUC           |        |
| Amino acid sequence |                       |     |     | Glutamic acid | Glutamic acid | Lysine |

- (c) Describe the events which take place in the production of the beta-globin molecule after the ribosome has attached to the mRNA. [5]

- (d) A mutation, where a single base was substituted by a different base, resulted in glutamic acid being replaced by valine at position 6 in the beta-globin molecule. Use the information provided to determine the mutated DNA base sequence at position 6 on the coding strand of DNA. [1]

- (e) The diagrams below show the structures of glutamic acid and valine.

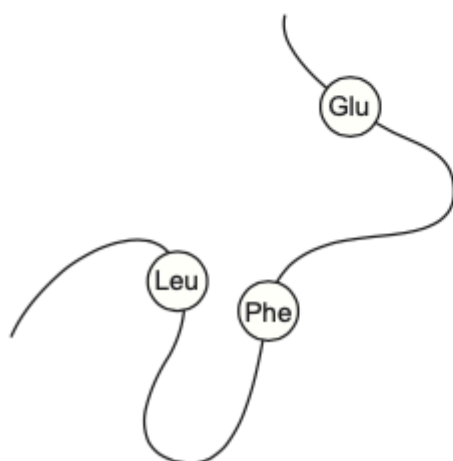


- (i) **Draw a circle** around the R-groups of these **two** amino acids. [1]

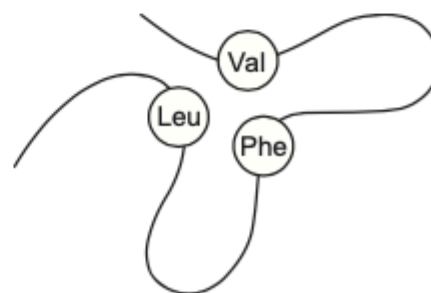
The R-group of glutamic acid is polar whereas the R-group of valine is non-polar. This affects the 3D structure of the resulting beta-globin molecule, as shown in the diagram below.

**Key:**

Glu – glutamic acid  
Val – valine  
Leu – leucine  
Phe – phenylalanine



Section of beta-globin molecule in a person **without** mutation



Section of beta-globin molecule in a person **with** mutation

- (ii) Valine interacts with phenylalanine and leucine in the beta-globin molecule. Phenylalanine and leucine both have non-polar R-groups. Explain how the properties of these amino acids mean that valine interacts with phenylalanine and leucine but glutamic acid does not. [3]

| 4                   | (a)                   |           | Two or more polypeptide chains {bonded/ joined} together (to form a functional protein)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1             |                       |          | 1         |          |          |  |  |  |   |   |   |   |   |   |  |                   |     |     |     |     |     |     |     |             |     |     |     |     |     |     |     |                 |     |     |     |     |     |     |     |                     |         |           |         |               |               |        |        |  |   |  |   |  |  |
|---------------------|-----------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------|----------|-----------|----------|----------|--|--|--|---|---|---|---|---|---|--|-------------------|-----|-----|-----|-----|-----|-----|-----|-------------|-----|-----|-----|-----|-----|-----|-----|-----------------|-----|-----|-----|-----|-----|-----|-----|---------------------|---------|-----------|---------|---------------|---------------|--------|--------|--|---|--|---|--|--|
|                     | (b)                   | (i)       | 438 (Accept 441/444)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               | 1                     |          | 1         | 1        |          |  |  |  |   |   |   |   |   |   |  |                   |     |     |     |     |     |     |     |             |     |     |     |     |     |     |     |                 |     |     |     |     |     |     |     |                     |         |           |         |               |               |        |        |  |   |  |   |  |  |
|                     |                       | (ii)      | A triplet code can produce 64 different combinations, which is enough to code for the 20 different amino acids/ 2 bases would only give 16 combinations which is not enough for 20 amino acids (1)<br>(As there are more than 20 codes) some amino acids are coded for by more than one mRNA codon. (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2             |                       |          | 2         |          |          |  |  |  |   |   |   |   |   |   |  |                   |     |     |     |     |     |     |     |             |     |     |     |     |     |     |     |                 |     |     |     |     |     |     |     |                     |         |           |         |               |               |        |        |  |   |  |   |  |  |
|                     |                       | (iii)     | <table border="1"><thead><tr><th></th><th colspan="7">Codon position number</th></tr><tr><th></th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th colspan="2">8</th></tr></thead><tbody><tr><td>DNA coding strand</td><td>GAC</td><td>TGA</td><td>GGA</td><td>CTC</td><td>CTC</td><td>TTC</td><td>TTT</td></tr><tr><td>mRNA codons</td><td>CUG</td><td>ACU</td><td>CCU</td><td>GAG</td><td>GAG</td><td>AAG</td><td>AAA</td></tr><tr><td>tRNA anticodons</td><td>GAC</td><td>UGA</td><td>GGA</td><td>CUC</td><td>CUC</td><td>UUC</td><td>UUU</td></tr><tr><td>Amino acid sequence</td><td>leucine</td><td>threonine</td><td>proline</td><td>Glutamic acid</td><td>Glutamic acid</td><td>lysine</td><td>lysine</td></tr></tbody></table> <p>Two marks for 6 correct cells in columns 3,4 and 5.<br/>one mark for 4/5 correct cells (2)<br/>One mark for correct amino acid sequence (1)<br/>One mark for column 8 correct (1)</p> |               | Codon position number |          |           |          |          |  |  |  | 3 | 4 | 5 | 6 | 7 | 8 |  | DNA coding strand | GAC | TGA | GGA | CTC | CTC | TTC | TTT | mRNA codons | CUG | ACU | CCU | GAG | GAG | AAG | AAA | tRNA anticodons | GAC | UGA | GGA | CUC | CUC | UUC | UUU | Amino acid sequence | leucine | threonine | proline | Glutamic acid | Glutamic acid | lysine | lysine |  | 4 |  | 4 |  |  |
|                     | Codon position number |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |                       |          |           |          |          |  |  |  |   |   |   |   |   |   |  |                   |     |     |     |     |     |     |     |             |     |     |     |     |     |     |     |                 |     |     |     |     |     |     |     |                     |         |           |         |               |               |        |        |  |   |  |   |  |  |
|                     | 3                     | 4         | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 6             | 7                     | 8        |           |          |          |  |  |  |   |   |   |   |   |   |  |                   |     |     |     |     |     |     |     |             |     |     |     |     |     |     |     |                 |     |     |     |     |     |     |     |                     |         |           |         |               |               |        |        |  |   |  |   |  |  |
| DNA coding strand   | GAC                   | TGA       | GGA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | CTC           | CTC                   | TTC      | TTT       |          |          |  |  |  |   |   |   |   |   |   |  |                   |     |     |     |     |     |     |     |             |     |     |     |     |     |     |     |                 |     |     |     |     |     |     |     |                     |         |           |         |               |               |        |        |  |   |  |   |  |  |
| mRNA codons         | CUG                   | ACU       | CCU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | GAG           | GAG                   | AAG      | AAA       |          |          |  |  |  |   |   |   |   |   |   |  |                   |     |     |     |     |     |     |     |             |     |     |     |     |     |     |     |                 |     |     |     |     |     |     |     |                     |         |           |         |               |               |        |        |  |   |  |   |  |  |
| tRNA anticodons     | GAC                   | UGA       | GGA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | CUC           | CUC                   | UUC      | UUU       |          |          |  |  |  |   |   |   |   |   |   |  |                   |     |     |     |     |     |     |     |             |     |     |     |     |     |     |     |                 |     |     |     |     |     |     |     |                     |         |           |         |               |               |        |        |  |   |  |   |  |  |
| Amino acid sequence | leucine               | threonine | proline                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Glutamic acid | Glutamic acid         | lysine   | lysine    |          |          |  |  |  |   |   |   |   |   |   |  |                   |     |     |     |     |     |     |     |             |     |     |     |     |     |     |     |                 |     |     |     |     |     |     |     |                     |         |           |         |               |               |        |        |  |   |  |   |  |  |
|                     | (c)                   |           | Any <b>five</b> (x1) from:<br>A. Translation (1)<br>B. tRNA brings an amino acid to the ribosome (1)<br>C. Each tRNA has a {specific/ particular} amino acid (1)<br>D. Complementary base pairing occurs between the tRNA anticodon and the mRNA codon/ description of codon anticodon interaction (1)<br>E. A second amino acid is brought to the ribosome / two tRNA molecules can attach to the mRNA (at the ribosome)/ reference to two binding sites. (1)<br>F. A condensation reaction occurs/ a peptide bond forms between the amino acids. (1)<br>G. The ribosome moves along the mRNA {by one codon/until a stop codon is reached.} (1)                                                                                                                                                                                                                                                                                      | 5             |                       |          | 5         |          |          |  |  |  |   |   |   |   |   |   |  |                   |     |     |     |     |     |     |     |             |     |     |     |     |     |     |     |                 |     |     |     |     |     |     |     |                     |         |           |         |               |               |        |        |  |   |  |   |  |  |
|                     | (d)                   |           | {Thymine/ T} is replaced with {adenine/ A} / CAC. (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               | 1                     |          | 1         |          |          |  |  |  |   |   |   |   |   |   |  |                   |     |     |     |     |     |     |     |             |     |     |     |     |     |     |     |                 |     |     |     |     |     |     |     |                     |         |           |         |               |               |        |        |  |   |  |   |  |  |
|                     | (e)                   | (i)       | <b>Both</b> R groups circled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1             |                       |          | 1         |          |          |  |  |  |   |   |   |   |   |   |  |                   |     |     |     |     |     |     |     |             |     |     |     |     |     |     |     |                 |     |     |     |     |     |     |     |                     |         |           |         |               |               |        |        |  |   |  |   |  |  |
|                     |                       | (ii)      | Phenylalanine, leucine and valine are all non- polar <b>and</b> therefore are attracted to each other (1)<br>Glutamic acid is polar <b>and</b> will therefore {be repelled from/ not attracted to} phenylalanine and leucine (1)<br>any correct reference to hydrophobic/ hydrophilic (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |               |                       | 3        | 3         |          |          |  |  |  |   |   |   |   |   |   |  |                   |     |     |     |     |     |     |     |             |     |     |     |     |     |     |     |                 |     |     |     |     |     |     |     |                     |         |           |         |               |               |        |        |  |   |  |   |  |  |
|                     |                       |           | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>9</b>      | <b>6</b>              | <b>3</b> | <b>18</b> | <b>1</b> | <b>0</b> |  |  |  |   |   |   |   |   |   |  |                   |     |     |     |     |     |     |     |             |     |     |     |     |     |     |     |                 |     |     |     |     |     |     |     |                     |         |           |         |               |               |        |        |  |   |  |   |  |  |

3. (a) In 1958 Meselson and Stahl conducted an investigation into DNA replication.
- Bacteria were grown in a culture medium that contained only the heavy isotope of nitrogen,  $^{15}\text{N}$ . After many generations, the bacterial DNA contained only the heavy isotope of nitrogen.
  - Some of the bacteria were then transferred to another culture medium containing only the lighter isotope of nitrogen,  $^{14}\text{N}$ .
  - DNA was extracted from the bacteria and centrifuged at the **start** (before cell division), and then after **1, 2 and 3** cell divisions. Heavier DNA was found towards the bottom of the tube after centrifuging.

**Image 3.1** shows the distribution of DNA in the tubes at the start and after Division 1 and Division 2. The graphs represent the relative densities of DNA but they are not in the correct order.

- (i) **Draw lines on Image 3.1** to match up the tubes with the correct graphs. [1]

**Image 3.1**

**Tube**



Start

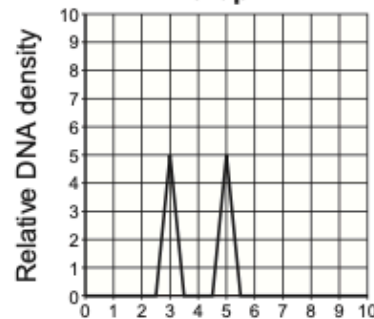


Division 1

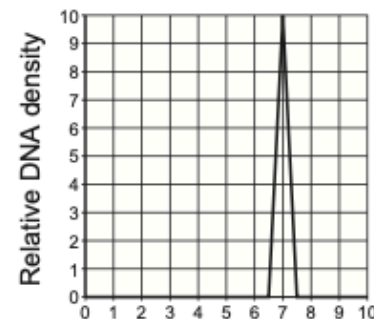


Division 2

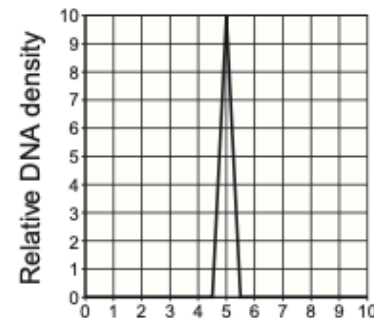
**Graph**



Top of tube → Bottom of tube



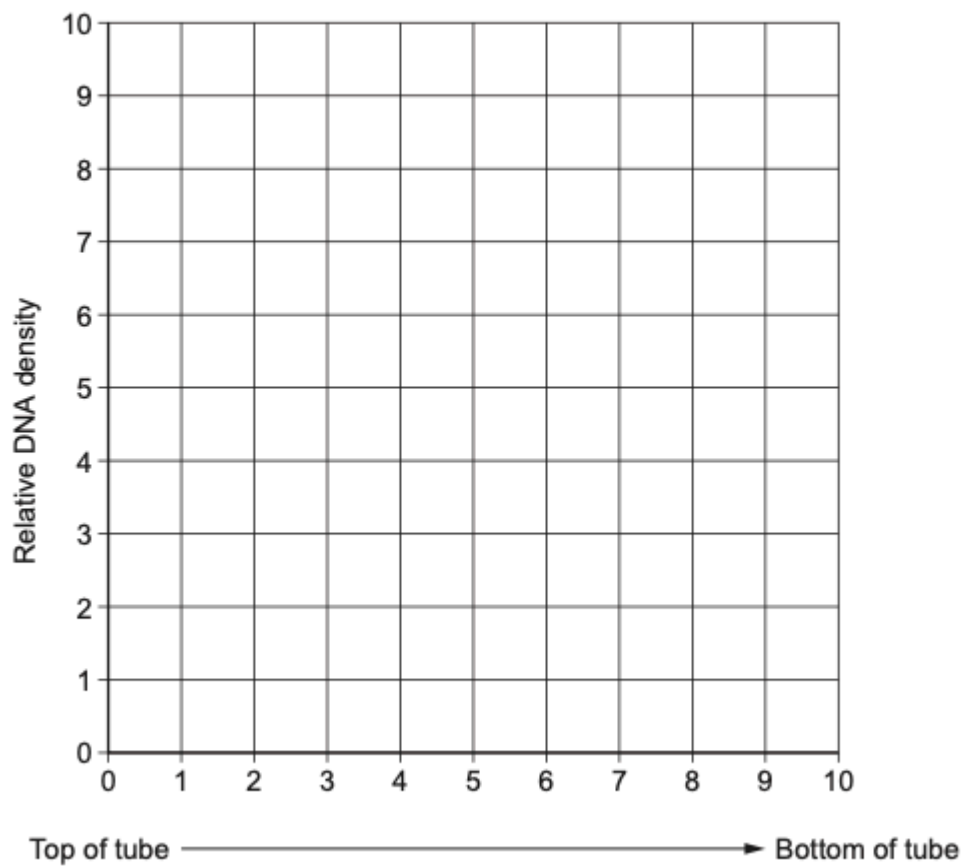
Top of tube → Bottom of tube



Top of tube → Bottom of tube

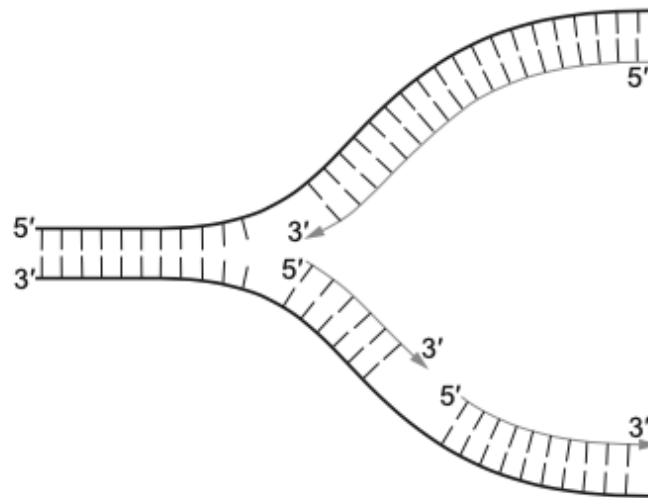
- (ii) On **Image 3.2** sketch the **graph** you would expect to see for Division 3. [3]

**Image 3.2**



- (b) **Image 3.3** shows a DNA replication fork.

**Image 3.3**



- (i) On **Image 3.3**, mark the position of **one** molecule of DNA polymerase with the letter **X**. [1]
- (ii) Describe the role of DNA helicase in this process. [1]
- .....
- .....
- (iii) Explain how **Image 3.3** shows that DNA is replicated by a semi-conservative mechanism. [2]
- .....
- .....
- .....
- .....
- .....
- .....

- (c) DNA polymerase has a proof-reading function, this means that it will check that it has replicated the DNA correctly and repair it if it is incorrectly synthesised.  
Use your knowledge of protein synthesis to suggest the importance of the proof-reading role to the organism. [3]

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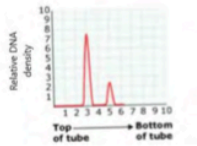
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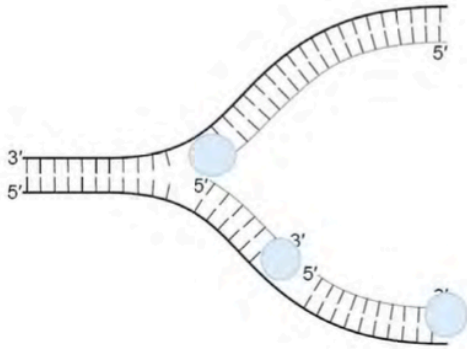
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|    |     |      |                                                                                                                                                                                  |  |   |  |   |   |  |
|----|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|--|---|---|--|
| 3. | (a) | (i)  | Start– 2 <sup>nd</sup> graph<br>Division 1 – 3 <sup>rd</sup> graph<br>Division 2 – 1 <sup>st</sup> graph<br><br>All correct (1)                                                  |  | 1 |  | 1 |   |  |
|    |     | (ii) | <br>2 peaks (1)<br>Peaks in correct place (3,5) (1)<br>Peaks relative heights correct 3:1 (1) |  | 3 |  | 3 | 2 |  |

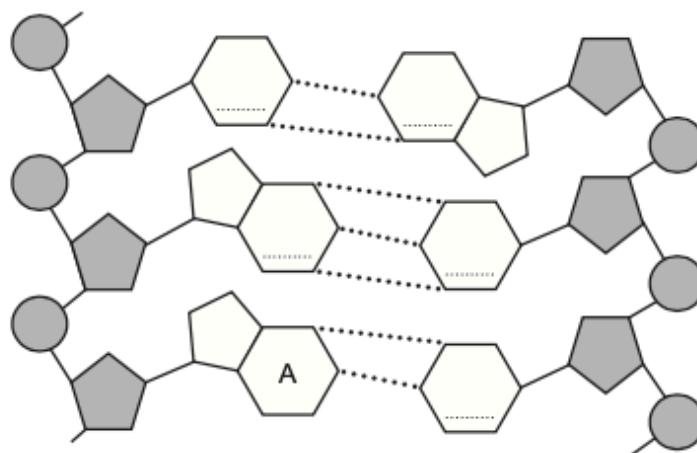
|     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |          |          |           |          |  |
|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|--|
| (b) | (i)   | <p>One X correct (1)<br/>cross should be centred on the newly synthesised strand.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                |          | 1        |          | 1         |          |  |
|     | (ii)  | <p>Unwind DNA/<br/>break <u>hydrogen</u> bonds between bases/<br/>{separates strands of / unzips} DNA</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1        |          |          | 1         |          |  |
|     | (iii) | <p>It shows that {each original strand of DNA/ the original DNA molecule} acts as a template {for a new strand}. (1)<br/>New {double stranded} DNA contains one old <b>and</b> one new strand (1)</p>                                                                                                                                                                                                                                                                                                                                                                                                 | 2        |          |          | 2         |          |  |
| (c) |       | <p><b>Any three (x1) from:</b><br/> A. The {nucleotide/base} sequence {in the DNA} is different / triplet code is different (1)<br/> B. This will lead to different mRNA being produced {during transcription}/ change the codons in the mRNA (1)<br/> C. And the different amino acid being incorporated into the protein {during translation} / different sequence of amino acids (1)<br/> D. The protein may {not function correctly / be non functional}/ different {polypeptide/ protein} formed (1)<br/> Reject wrong protein<br/> E. due to the changed {tertiary structure/ 3D shape} (1)</p> |          | 3        |          | 3         |          |  |
|     |       | <b>Question 3 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>3</b> | <b>8</b> | <b>0</b> | <b>11</b> | <b>2</b> |  |



# WJEC 2023

5. (a) **Image 5.1** represents part of a DNA molecule.

**Image 5.1**



- (i) Circle **one** nucleotide in **Image 5.1**. [1]
- (ii) Using the letters **A, C, T, G**, label **all** the organic bases in **Image 5.1**. One adenine is labelled already. [2]
- (iii) In the late 1940s, Erwin Chargaff published his research into the base composition of the DNA in different species. Some of his findings are shown in **Table 5.2**.

**Table 5.2**

| Organism   | Percentage of base |         |          |         |
|------------|--------------------|---------|----------|---------|
|            | Adenine            | Guanine | Cytosine | Thymine |
| maize      | 26.8               | 22.8    | 23.2     | 27.2    |
| chicken    | 28.0               | 22.0    | 21.6     | 28.4    |
| yeast      | 31.3               | 18.7    | 17.1     | 32.9    |
| sea urchin | 32.8               | .....   | .....    | .....   |

- I. Complete **Table 5.2** to suggest values for the sea urchin. [1]

II. Based on the results in **Table 5.2**, state the conclusion that can be made regarding the base composition of DNA. Explain your answer. [3]

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(b) Meselson and Stahl also carried out investigations on DNA in the 1950s. They proposed the theory of semi-conservative replication. State what is meant by semi-conservative replication. [2]

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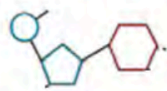
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(c) DNA contains only four different bases, but proteins can be made of up to 20 different amino acids. Explain how these four bases allow for the production of so many different proteins. [4]

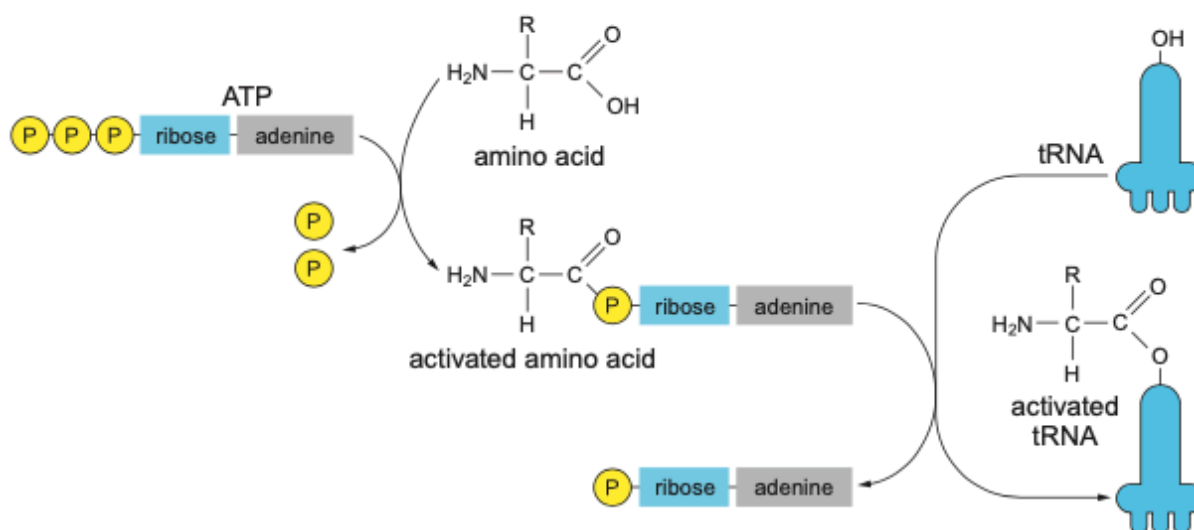
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|   |     |       |                                                                                                                                                                                                                                                    |   |   |   |   |   |   |
|---|-----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|
| 5 | (a) | (i)   | <br>or equivalent (1)                                                                                                                                           | 1 |   |   | 1 |   |   |
|   |     | (ii)  | T A<br>G C<br><b>A</b> T<br>One mark for both A-T<br>One mark for C-G                                                                                                                                                                              |   | 2 |   | 2 |   |   |
|   |     | (iii) | I Guanine 16.7 - 17.7%<br>Cytosine 16.7 - 17.7%<br>Thymine 32.3 - 33.3%                                                                                                                                                                            |   | 1 |   | 1 | 1 |   |
|   |     |       | II Complementary (base pairing) (1)<br>(For all organisms,) G is very close in {composition/percentage} to C <b>and</b> A is close in composition to T (1)<br>Variation from exact match due to {experimental error/ free nucleotides in cell} (1) |   | 2 | 1 | 3 |   | 1 |
|   | (b) |       | Each strand acts as a template (for a new strand) (1)<br>(Newly synthesised) DNA (molecule) contains one old and new strand (1)                                                                                                                    | 2 |   |   | 2 |   |   |

|     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |          |          |           |          |          |
|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
| (c) |  | <p><b>Any four (×1) from</b></p> <p>A. {Triplet code/ 3 bases code} for one amino acid (1)</p> <p>B. Explanation of why a minimum of three bases is needed (1)</p> <p>C. There are {64/ many} combinations of the 4 bases/ (genetic code is) degenerate/ or description of (1)</p> <p>D. Different {base / codon} sequences will give rise to different amino acid sequences (1)</p> <p>E. {DNA sequence/ genes} can be varying lengths (1)</p> <p>F. Different amino acid sequences lead to a wide variety of different {proteins/ polypeptides} / ref to exons or introns (1)</p> | 2        | 2        |          | 4         |          |          |
|     |  | <b>Question 5 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>5</b> | <b>7</b> | <b>1</b> | <b>13</b> | <b>1</b> | <b>1</b> |

7. DNA, mRNA, rRNA and tRNA are nucleic acids involved in protein synthesis. For tRNA to carry out its role in protein synthesis it needs to undergo the process shown in **Image 7**.

**Image 7**



Describe the function of each of the **four** types of nucleic acid involved in protein synthesis and state where in the cell each carries out its function. (A detailed description of protein synthesis is not required.)

Using the information given, explain the role of ATP in the process shown in **Image 7**. [9 QER]

7

Other nucleic acids involved in protein synthesis:

| Nucleic acid | role                                                                                                                            | Where it functions   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------|
| DNA          | provides the {genetic code/ template} for production of {all RNAs/ mRNA/ provides the genetic code for sequence of amino acids} | nucleus              |
| rRNA         | To form ribosomes/ the site of translation                                                                                      | cytoplasm/ rough ER  |
| mRNA         | Produced in transcription and used in translation/ transfers genetic code (from nucleus to ribosomes) / owtte                   | cytoplasm/ ribosomes |
| tRNA         | delivery of amino acids to the site of protein synthesis                                                                        | cytoplasm            |

**Section C:**

How tRNA activation works:

- ATP provides energy for the reaction
- Two phosphates are released
- Amino acid reacts with ATP to form activated amino acid. Amino acid is transferred to tRNA forming activated {tRNA/ amino acid}

6

3

9

Speciment

2. In 1961 a group of scientists demonstrated that the genetic code for a protein contained within DNA uses a triplet code of three bases for each amino acid.

(a) (i) How many different types of amino acid are used in protein synthesis? [1]

.....

(ii) Explain why there must be three bases in each code. [2]

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(b) The table shows some of the base sequences of mRNA together with their corresponding amino acids.

| <i>mRNA triplet</i> | <i>Amino acid</i> |
|---------------------|-------------------|
| UUU                 | phenyl alanine    |
| CUC                 | leucine           |
| AUG                 | methionine        |
| UAU                 | tyrosine          |
| CGU                 | arginine          |
| GAU                 | aspartic acid     |
| GUU                 | valine            |
| GAA                 | glutamic acid     |
| AGU                 | serine            |
| ACC                 | threonine         |
| ACA                 | serine            |
| GAC                 | aspartic acid     |
| UAG                 | stop code         |

A section of mRNA produced in the nucleus of a cell is shown below.

A U G G U U G A A G A U G U U G A C

- (i) Using the information given in the table write out the sequence of amino acids for which the section of mRNA codes. [2]

.....

.....

- (ii) Write the base sequence for the DNA template from which the strand of mRNA was produced. [2]

.....

- (iii) Describe the process occurring in the nucleus that gives rise to this strand of mRNA. [4]

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- (c) During DNA replication a mutation occurred in the sequence of bases from which the mRNA was produced. The mutated DNA resulted in an mRNA molecule with the following base sequence.

AUG GUA GAA GAC GUU GAC

- Explain why this mutation would have no effect on the protein that could be synthesised from this mRNA molecule. [2]

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| Question |     |       | Marking details                                                                                                                                                                                                                                   | Marks available |          |          |           |          |          |
|----------|-----|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |       |                                                                                                                                                                                                                                                   | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
| 2        | (a) | (i)   | Approximately 20 amino acids (1)<br>allow: 20 to 27                                                                                                                                                                                               | 1               |          |          | 1         |          |          |
|          |     | (ii)  | only 4 different bases so 1 or 2 bases could not code for all 20 amino acids/ OWTTE (1)<br>a three base system could code for 64 amino acids (1)                                                                                                  | 2               |          |          | 2         |          |          |
|          | (b) | (i)   | Methionine, valine, glutamic acid, aspartic acid, valine, aspartic acid<br>all correct = 3 marks<br>one incorrect = 2 marks<br>two incorrect = 1 mark<br>three incorrect = 0 mark                                                                 | 2               |          |          | 2         |          |          |
|          |     | (ii)  | TAC CAA CTT CTA CAA CTG<br>all correct = 2 marks<br>one incorrect triplet = 1 mark<br>two incorrect triplets = 0 marks                                                                                                                            | 2               |          |          | 2         |          |          |
|          |     | (iii) | Any <b>four</b> from:<br>DNA helicase uncoils DNA (1)<br>RNA polymerase separates/ breaks H bonds (1)<br>Exposed DNA bases acts as a template (1)<br>Complementary base pairing (1)<br>RNA polymerase attaches free nucleotides to mRNA strand(1) | 4               |          |          | 4         |          |          |
|          | (c) |       | The mutation from GAU to GAC codes for the same amino acid (1)<br>Therefore same primary structure of protein (1)                                                                                                                                 |                 |          | 2        | 2         |          |          |
|          |     |       | <b>Question 2 total</b>                                                                                                                                                                                                                           | <b>11</b>       | <b>0</b> | <b>2</b> | <b>13</b> | <b>0</b> | <b>0</b> |

8. It was once suggested that '*proteins are produced by ribosomes which translate a messenger RNA molecule produced by transcription of a single gene*'.

With reference to your knowledge and understanding of protein structure and synthesis, explain why the statement above does not **fully** describe how functional proteins are produced and secreted by eukaryotic cells.

(The quality of your extended response will be assessed in this question.) [9QER]

|   |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |   |  |   |  |  |
|---|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|--|---|--|--|
| 8 |  |  | <p><b>Indicative content</b></p> <p>Many proteins are made of several polypeptide chains that are bonded together to form a quaternary structure. Therefore, several genes must be involved to carry the information for a single protein because one gene would be needed for each polypeptide chain. This is called the one gene – one polypeptide hypothesis.</p> <p>All proteins are synthesised as a chain of amino acids which is called its primary structure. The primary structure needs to be folded and held in place by hydrogen bonds and interactions between variable groups. Therefore the polypeptide requires further processing to produce a functional protein.</p> <p>In addition, some proteins have additional functional groups added, eg, haem in haemoglobin or have carbohydrate groups attached eg, antigens involved in cell recognition.</p> <p>Ribosomes are only involved in translating the code carried by mRNA molecules. Endoplasmic reticulum is also needed to transport the polypeptide chains to Golgi bodies which then further process them to produce the final protein. Golgi bodies also package these proteins in liposomes or secretory vesicles so that they can be transported to their site of action inside or outside cells.</p> |  | 9 |  | 9 |  |  |
|---|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|--|---|--|--|

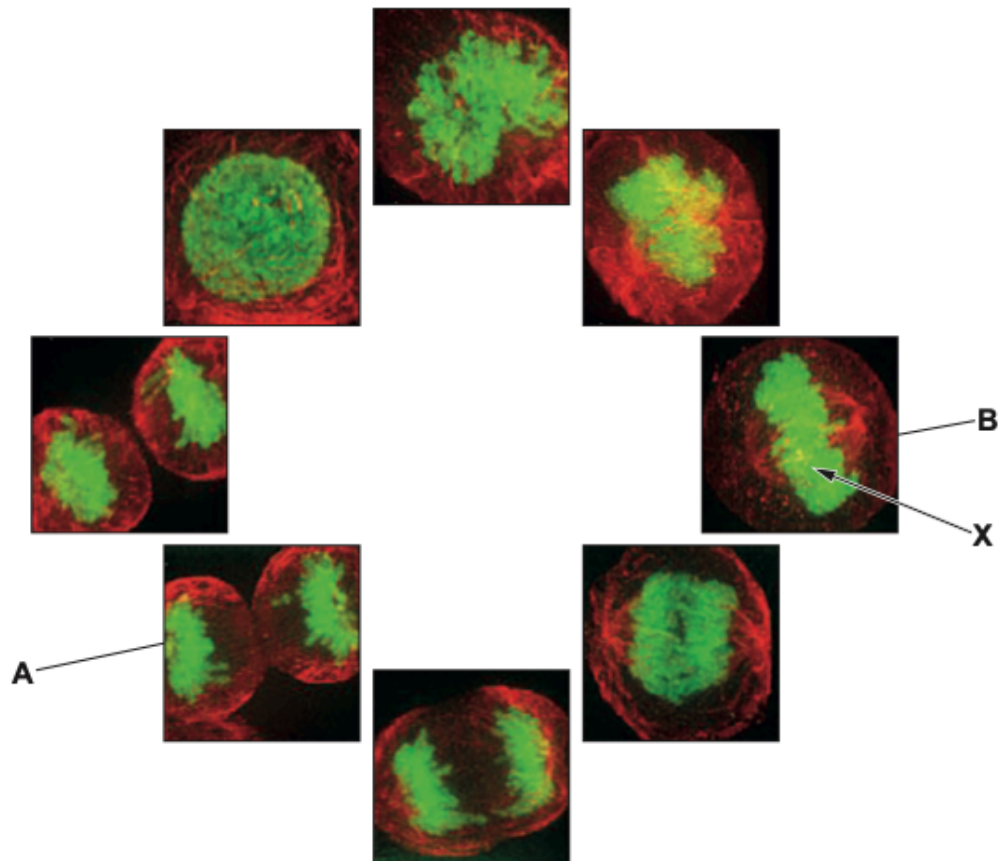


1.6

# WJEC 2016

1. In 1951, Henrietta Lacks died of cervical cancer. A research scientist called George Gey, grew a sample of her tumour. He found that these cells multiplied rapidly and could be grown indefinitely in culture. They became the first immortal human cell line, which he named HeLa. HeLa cells are now grown in research laboratories in many different countries.

(a) The photomicrographs below show images of HeLa cells during different stages of the cell cycle. The cells have been stained with a dye, which causes the DNA to be visible.



- (i) Name the stages shown in **A** and **B**.

[2]

Stage **A**: .....

Stage **B**: .....

(ii) Name the structures, labelled **X**, that can be seen in the photomicrograph. [1]

(iii) Explain why some stain would be seen in other parts of the cell. [2]

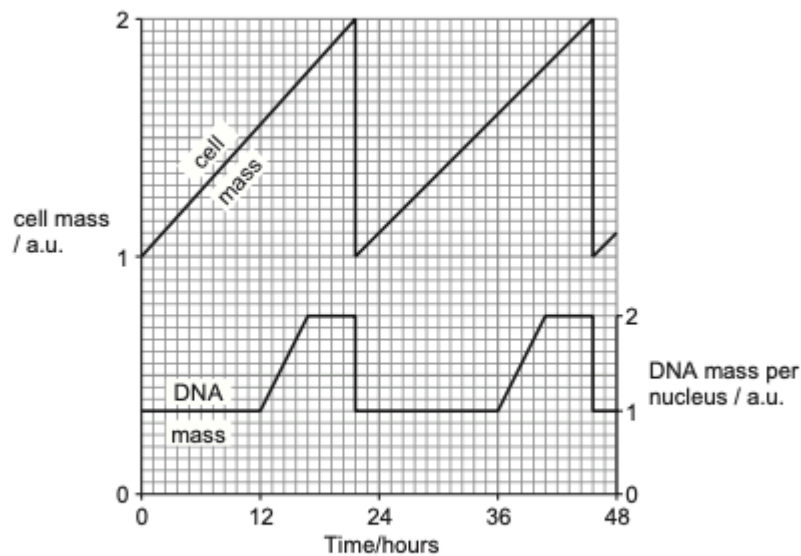
- (b) A biotechnology company that supplies HeLa cells to laboratories states that the number of HeLa cells double every 19 hours. They suggest that the starting culture of cells should have a density of 30 000 cells  $\text{cm}^{-3}$  and that the cells should be sub-cultured every 4 days.

Calculate the density of the cells that would be present after 4 days; give your answer in cells  $\text{cm}^{-3}$  in standard form. [3]

Cell density = .....  $\text{cm}^{-3}$

|   |     |       |                                                                                                                                                                                                                                                                                                                                                                                     |          |          |          |          |          |          |
|---|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|----------|----------|----------|
| 1 | (a) | (i)   | Telophase (1) Accept cytokinesis<br>Metaphase (1)<br>Any reference to meiosis – penalise 1 mark                                                                                                                                                                                                                                                                                     | 2        |          |          | 2        |          |          |
|   |     | (ii)  | Chromosomes<br>Accept chromatids<br>Reject chromatin/ spindle fibres                                                                                                                                                                                                                                                                                                                |          | 1        |          | 1        |          |          |
|   |     | (iii) | DNA is found( in other parts of the cell)(1)<br>in mitochondria (1)<br>Reject chloroplast                                                                                                                                                                                                                                                                                           |          | 2        |          | 2        |          |          |
|   | (b) |       | (4 days = 96 hours)<br>96/19 = 5 cycles (accept 5.05 cycles)<br>Evidence of 5 cycles e.g. 96/19 or 5 cycles/ 5.05 cycles/ 5.1<br>cycles = 1 mark<br><br>30 000 $\times 2^5$ (accept 30 000 $\times 2^{5.05/5.1}$ )<br>= 960 000/ 996 000/ 1 028 902 = 2 marks<br><br>= 9.6 $\times 10^5 \text{ cm}^{-3}$ (accept 9.96 $\times 10^5 \text{ cm}^{-3}$ / 1.0 $\times 10^6$ ) = 3 marks |          | 3        |          | 3        | 3        |          |
|   |     |       | <b>Question 1 total</b>                                                                                                                                                                                                                                                                                                                                                             | <b>2</b> | <b>6</b> | <b>0</b> | <b>8</b> | <b>3</b> | <b>0</b> |

7. Cells can be grown under laboratory conditions and used in the development of drugs to treat cancer. The graph below shows a model of the general changes in cell mass and changes in mass of DNA in two cell cycles.



Identify the type of cell division shown above and explain the changes in cell mass and mass of DNA per nucleus in one cell cycle. [9 QER]

7

Indicative content:

A: Cell division

- Mitosis
- DNA mass doubles, then halves/ returns to same content
- one cell cycle is within 24 hours

B: DNA mass

- DNA replication occurs during interphase
- Causing DNA to double
- After interphase DNA mass remains constant
- After interphase DNA present as chromosomes/ 2 chromatids
- Mass DNA halves at telophase/ cytokinesis
- when nuclear membrane reforms

C: Cell mass

- Interphase: mass of cell increases
- As does cell size
- Mass of cell continues to increase
- due to replication of DNA / organelles
- and possibly due to formation of spindle fibres
- cell undergoes cytokinesis and mass of cell halves

4

5

0

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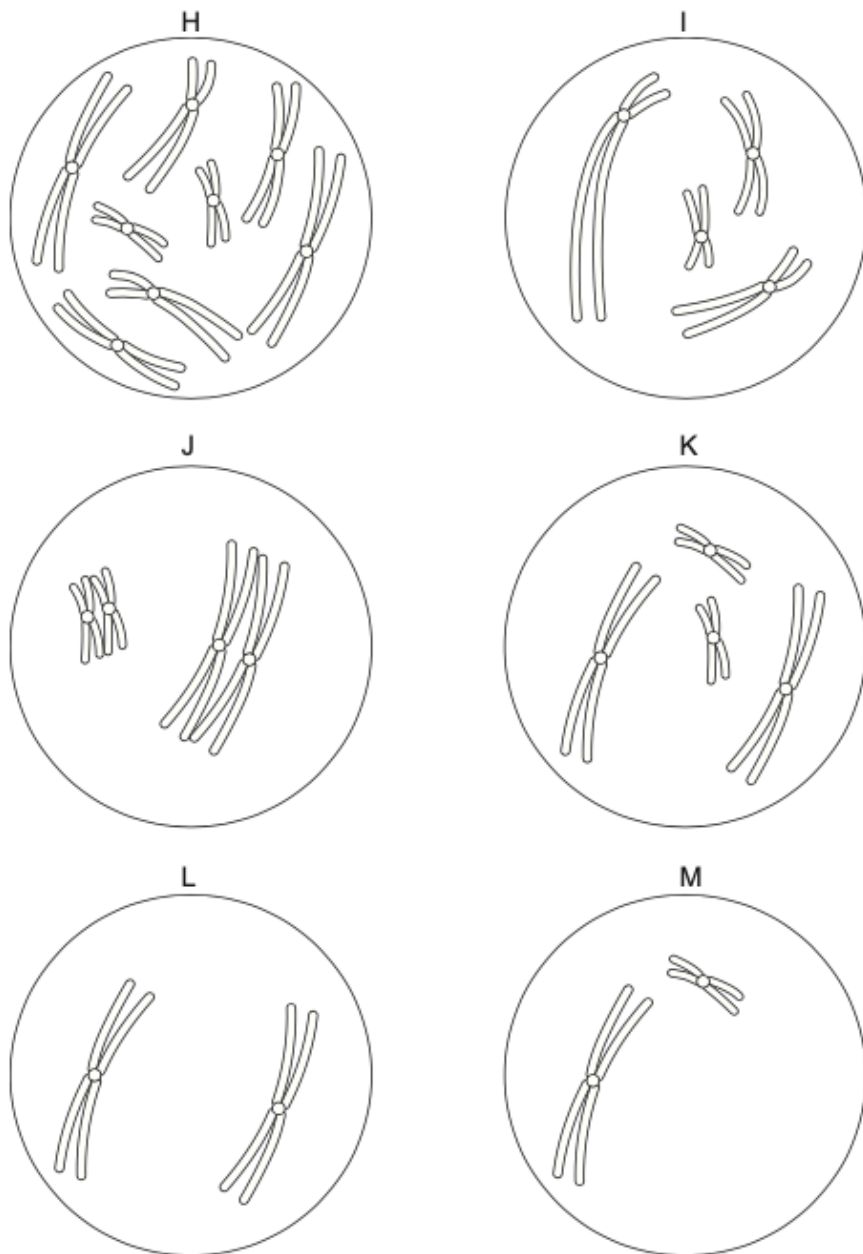
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## WJEC 2018

3. Spider mites are small invertebrates which are related to spiders and scorpions. One species, *Eutetranychus africanus*, which is native to the island of Mauritius, has very few chromosomes,  $2n = 4$ .

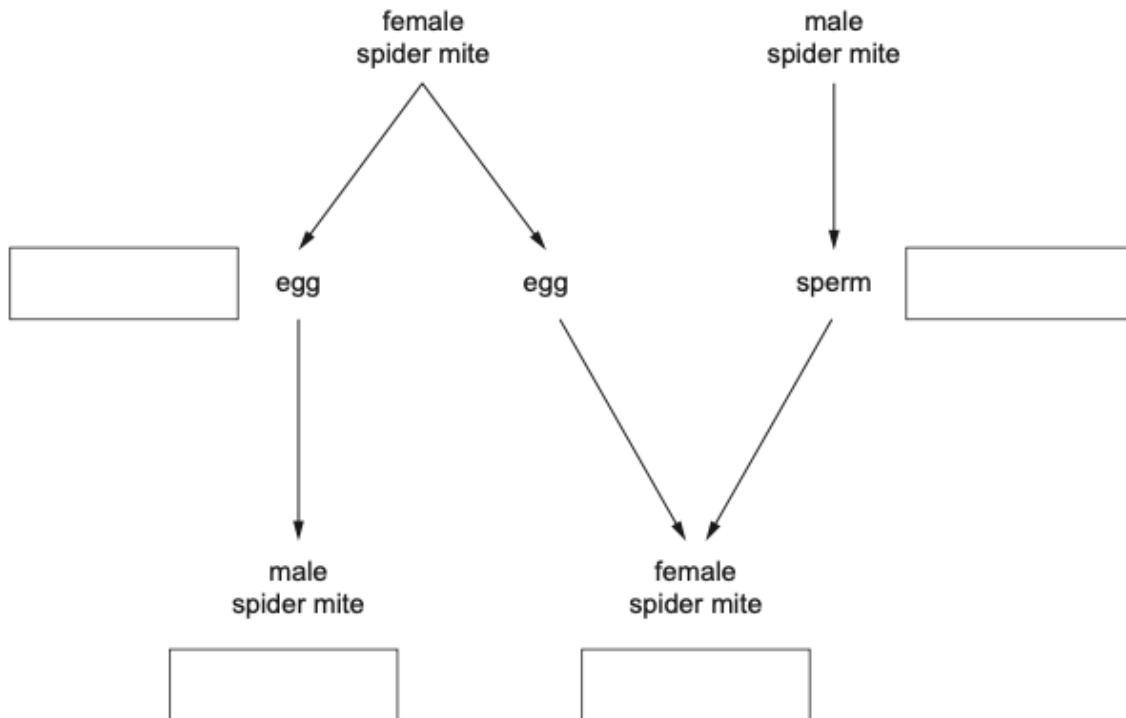
The diagrams below show the chromosomes in some cells undergoing mitosis or meiosis.



(a) Identify the cells that belong to this species of spider mite.

[2]

- (b) The diagram below shows the individuals that make up a colony of spider mites. Female spider mites are produced when a male sperm fertilises a female's egg. Male spider mites develop from an unfertilised female egg.



- (i) **Complete** the diagram to state which stages are **haploid** or **diploid**. [2]
- (ii) What conclusions can be made about the formation of male and female gametes in spider mites? [2]

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- (c) (i) A sample of tissue from the **leg muscle** of a young spider mite was analysed. The mass of DNA in some of the cells was 6.8 arbitrary units (au), whilst in other cells it was 3.4 au. Suggest explanations for this difference using your knowledge of the cell cycle. [3]

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- (ii) The table below shows the percentage of cells, with either 3.4 au or 6.8 au of DNA, in the leg muscle of a young spider mite and an older spider mite.

| Mass of DNA in cell /au | Percentage of cells |                   |
|-------------------------|---------------------|-------------------|
|                         | Young spider mite   | Older spider mite |
| 6.8                     | 20                  | 5                 |
| 3.4                     | 80                  | 95                |

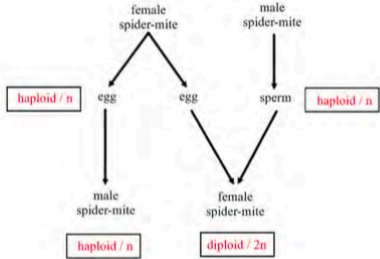
What conclusions can be drawn from the data regarding the significance of mitosis in these spider mites? [2]

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|   |     |      |                                                                                                                                                                                                                                                                                                                                                                      |          |          |          |           |          |          |
|---|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
| 3 | (a) |      | J, K & M = 2 marks<br>2 correct = 1 marks<br>0/1 correct = 0 marks<br>If use more than three letters then deduct one mark for each additional letter                                                                                                                                                                                                                 |          | 2        |          | 2         |          |          |
|   | (b) | (i)  |  <p>4 correct for 2 marks<br/>2/3 correct for 1 mark<br/>0/1 correct = 0</p>                                                                                                                                                                                                        |          | 2        |          | 2         |          |          |
|   |     | (ii) | Eggs/ female gametes/ ova are produced by <u>meiosis</u> (1)<br>Sperm/male gametes are produced by <u>mitosis</u> (1)                                                                                                                                                                                                                                                |          |          | 2        | 2         |          |          |
|   | (c) | (i)  | 1. 3.4 units DNA {before replication / in <u>early</u> interphase}/<br>{quantity of DNA halves/ returns to original value} {following<br>cytokinesis / (at the end of )telophase} (1)<br>2. {6.8 units of DNA/ DNA doubles} due to <u>DNA replication</u> (1)<br>3. (6.8 units of DNA will be present during) (Late) interphase/<br>prophase/ metaphase/ anaphase(1) |          | 3        |          | 3         |          |          |
|   |     | (ii) | {mitosis is faster/ <u>more</u> mitosis} in young spider mites than older<br>spider mites/ ORA (1)<br>Young spider mite - mitosis required for growth (and repair of<br>muscle tissue) / Older spider mite – mitosis required for repair (of<br>muscle tissue only) (1)                                                                                              |          |          | 2        | 2         |          |          |
|   |     |      | <b>Question 3 total</b>                                                                                                                                                                                                                                                                                                                                              | <b>0</b> | <b>7</b> | <b>4</b> | <b>11</b> | <b>0</b> | <b>0</b> |



## WJEC 2019

1. (a) The diagram below shows the chromosomes from a single cell of a Chinese daisy, *Aster shennongjiaensis*, during prophase of mitosis.



- (i) State the **haploid** number of chromosomes in this species. [1]

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- (ii) State the evidence from the diagram that the cell is undergoing mitosis and not meiosis. [1]

.....

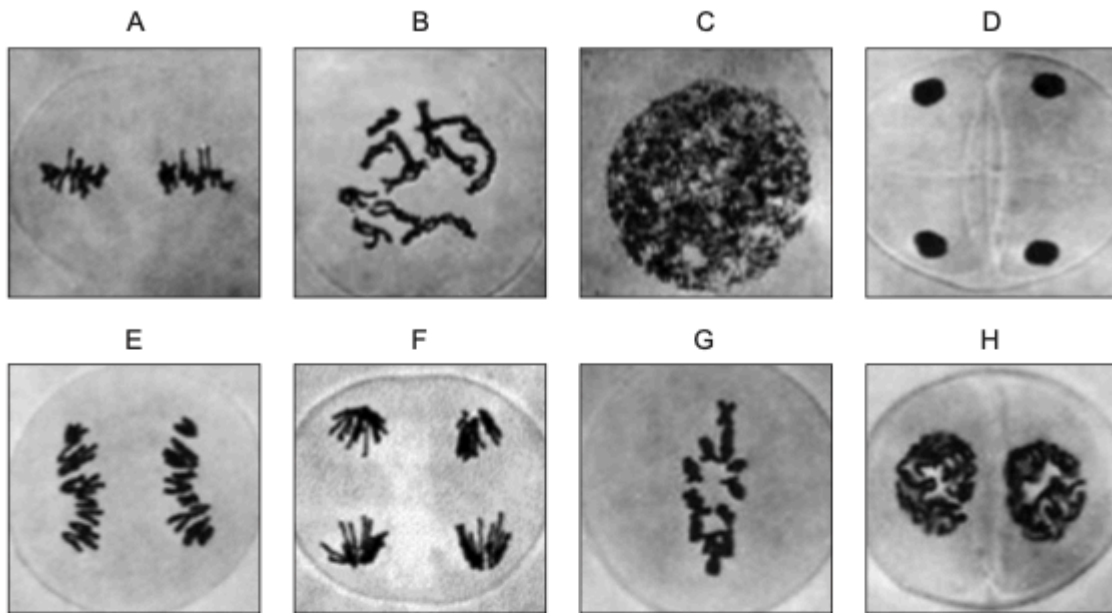
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- (iii) When the tissues from the plant were removed they were stained before being viewed under the microscope. Explain how this makes the chromosomes visible. [1]

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(b) The photomicrographs below show different stages of meiosis in a different plant species.



(i) State which organ of the plant could have been used to obtain the photomicrographs above. [1]

(ii) Place the photomicrographs in the correct order. Some letters have been done for you. [1]

|   |  |  |  |  |   |   |   |
|---|--|--|--|--|---|---|---|
| C |  |  |  |  | A | F | D |
|---|--|--|--|--|---|---|---|

(iii) Use your knowledge of cell division to describe **two** differences between the stages labelled E and F. [2]

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- (iv) With reference to photomicrographs **B** and **G**, state how meiosis contributes to genetic variation in the species and explain why this is important. [3]

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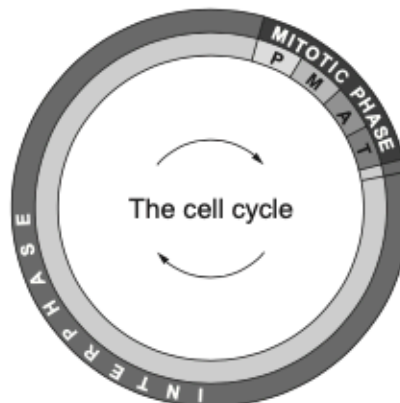
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|   |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |          |          |           |          |          |
|---|-----|-------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
| 1 | (a) | (i)   |  | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          | 1        |          | 1         |          |          |
|   |     | (ii)  |  | The chromosomes have not paired up /<br>Not (arranged) in homologous pairs/<br>Ignore homologous pairs unqualified<br>Absence of bivalents/<br>Synapsis does not occur<br>Ignore reference to chiasmata                                                                                                                                                                                                                                                                                                                          |          |          | 1        | 1         |          |          |
|   |     | (iii) |  | stains the {DNA/ nucleic acids/ histones}                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1        |          |          | 1         |          | 1        |
|   | (b) | (i)   |  | Anther/ ovary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1        |          |          | 1         |          | 1        |
|   |     | (ii)  |  | C BGEH AFD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          | 1        |          | 1         |          |          |
|   |     | (iii) |  | Any <b>two</b> (x1) from<br><ul style="list-style-type: none"> <li>In E {chromosomes/ homologous pairs} are {segregating / being pulled to opposite poles} whereas in F the chromatids are being pulled to opposite poles (1)</li> <li>In E the centromeres do not divide, in F they do divide (1)</li> <li>Correct reference to the spindle (1)</li> <li>E is diploid <b>and</b> F is haploid (1)</li> </ul> Ignore reference to the production of cells at the end of meiosis I and meiosis II                                 |          | 2        |          | 2         |          |          |
|   |     | (iv)  |  | <ul style="list-style-type: none"> <li>{Chiasmata / chiasma formation / crossing over/ or description of} is taking place in {photomicrograph B/ prophase 1} (1)<br/>Ignore reference to photomicrograph G</li> <li>{Independent assortment/random assortment/ or description of} is occurring in {photomicrograph G/ metaphase 1} (1)</li> <li>Genetic variation allows {the species to adapt / some of the offspring to survive} (in changeable environments.) (1) Reject reference to {them/ offspring} {adapting}</li> </ul> | 1        | 2        |          | 3         |          |          |
|   |     |       |  | <b>Question 1 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>3</b> | <b>6</b> | <b>1</b> | <b>10</b> | <b>0</b> | <b>2</b> |

# WJEC 2022

7. A eukaryotic cell cycle is represented in the **Image 7.1**.

**Image 7.1**



When looking at a group of cells under the microscope, cells undergoing mitosis can be seen as they will have visible chromosomes. The mitotic index, which represents the proportion of cells undergoing mitosis within a group of cells, can be calculated as follows:

$$\text{Mitotic index} = \frac{\text{cells in mitosis}}{\text{total number of cells in field of view}}$$

Cyclins are a family of regulatory proteins that control the progression of the cell cycle. The concentration of cyclin B peaks at the start of mitosis before dropping rapidly as the cells exit mitosis. Some mutations result in abnormally high concentrations of cyclin B.

Describe the events occurring during interphase that are necessary before mitosis can occur.

Using examples, explain why some tissues in humans naturally have a higher mitotic index than others.

Suggest and explain the possible effects of abnormally high concentrations of cyclin B.

[9 QER]

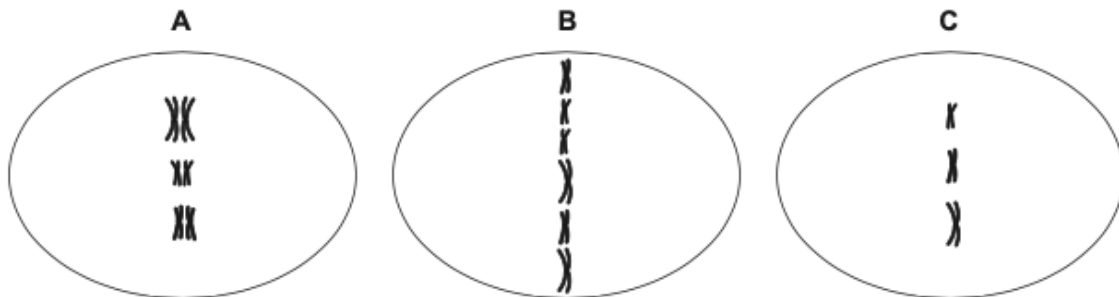
|    |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |   |   |  |  |  |
|----|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|--|--|--|
| 7. |  |  | <p><b>Indicative content:</b></p> <p>Events in interphase necessary for mitosis</p> <ul style="list-style-type: none"> <li>• DNA replication</li> <li>• {Replication/ Making new} {organelles/ named organelles}</li> <li>• Synthesis of ATP</li> <li>• Synthesis of proteins</li> </ul> <p>Mitotic index higher</p> <ul style="list-style-type: none"> <li>• Tissues where new cells are being made/ Growth of tissues</li> <li>• will have a higher proportion of cells undergoing mitosis</li> <li>• Example: skin (stem) cells, bone marrow (stem) cells/ gut lining cells / liver cells</li> <li>• Because cells are being {worn away rapidly/ degraded}/ for repair of {worn out/damaged} tissue/ replace cells</li> </ul> <p>Cancer cells and cyclin B</p> <ul style="list-style-type: none"> <li>• If cyclin B levels too high the cell will keep going into mitosis.</li> <li>• uncontrolled cell division</li> <li>• Cancer / Tumours</li> </ul> | 3 | 3 | 3 |  |  |  |
|----|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|--|--|--|

# WJEC 2023

2. Image 2.1 shows metaphase during cell division of three cells of the mosquito, *Culex pipiens*.

The mosquito has a diploid number of  $2n = 6$ .

Image 2.1



- (a) (i) Insert the correct letter (A, B or C) into the table to indicate which stage of cell division is represented by each cell. [2]

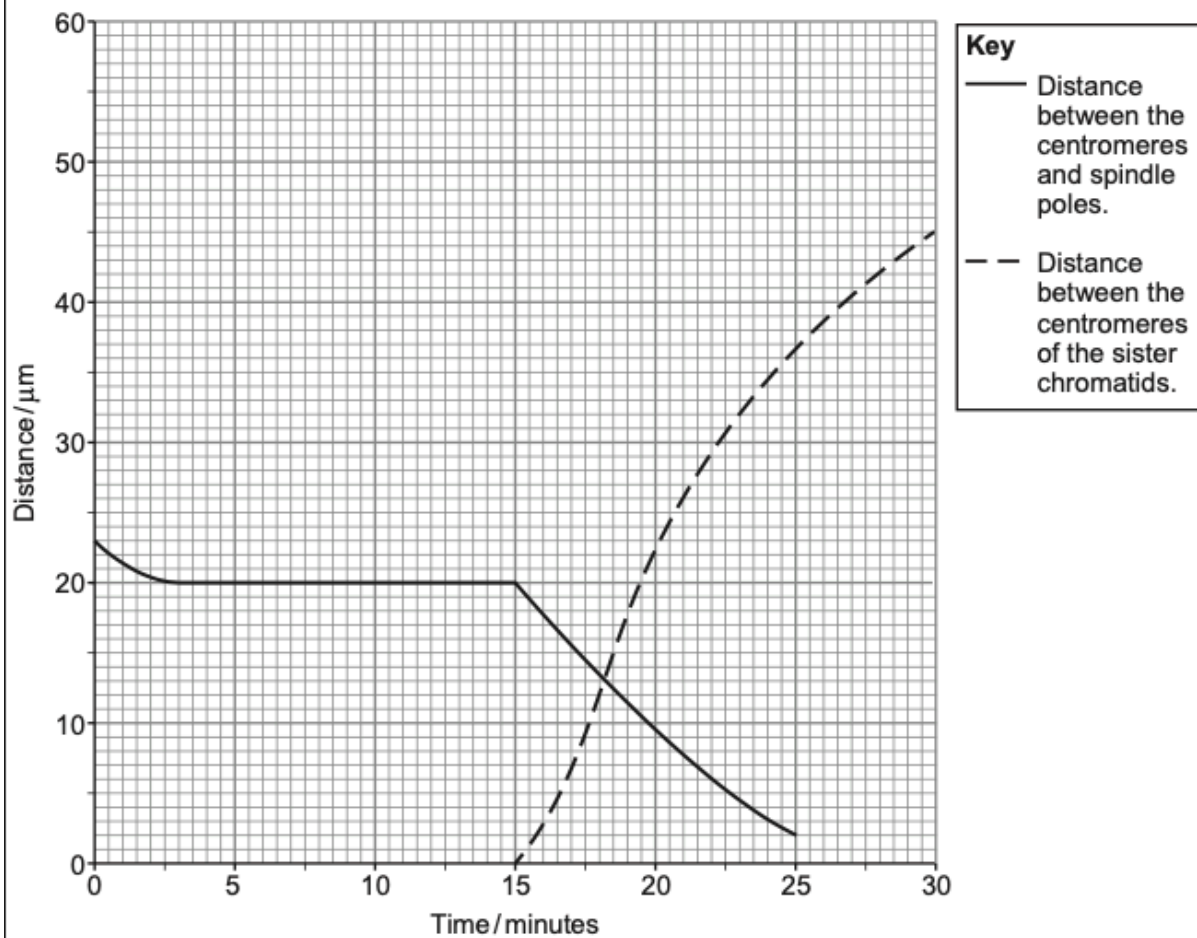
| Metaphase of: | Letter |
|---------------|--------|
| Mitosis       | .....  |
| Meiosis I     | .....  |
| Meiosis II    | .....  |

- (ii) State **two** differences in the daughter cells of *Culex pipiens* following mitosis and meiosis. [2]

| Difference | Mitosis        | Meiosis        |
|------------|----------------|----------------|
| 1.         | .....<br>..... | .....<br>..... |
| 2.         | .....<br>..... | .....<br>..... |

- (b) **Graph 2.2** shows some changes seen in a cell undergoing mitosis.

**Graph 2.2**



- (i) Calculate the rate of increase in distance between the centromeres of the sister chromatids between 15 and 30 minutes. [2]

Rate = .....  $\mu\text{m minute}^{-1}$

- (ii) Suggest how **Graph 2.2** provides evidence to show what happens to the chromosomes during anaphase of mitosis. [2]

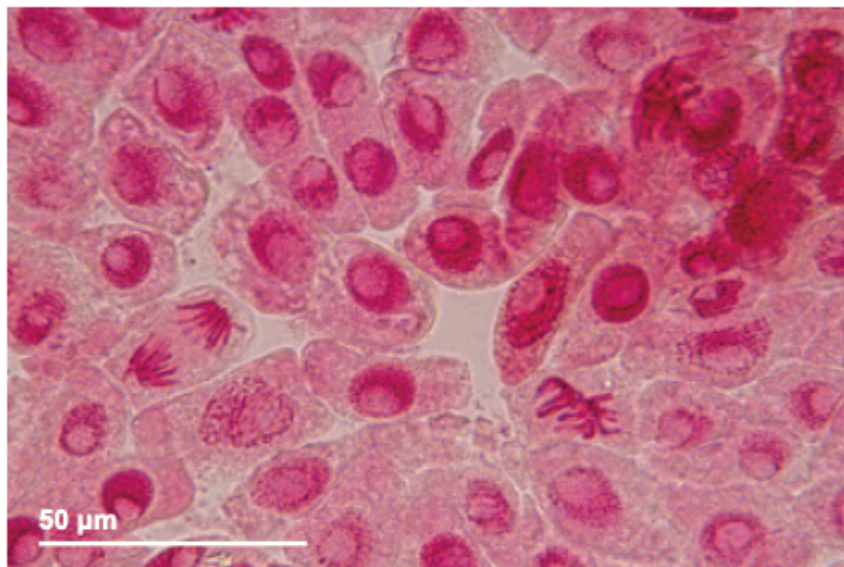
.....

.....

.....

- (c) The micrograph in **Image 2.3** shows onion plant cells undergoing mitosis.

**Image 2.3**



- (i) Use the scale bar to calculate the magnification of **Image 2.3**.

[2]

Magnification =  $\times$  .....

- (ii) On **Image 2.3** label:

[1]

**A** one cell in anaphase

**B** one cell in metaphase

- (iii) Suggest from which part of the onion plant the cells in **Image 2.3** were taken. Explain your answer.

[2]

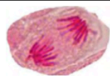
.....

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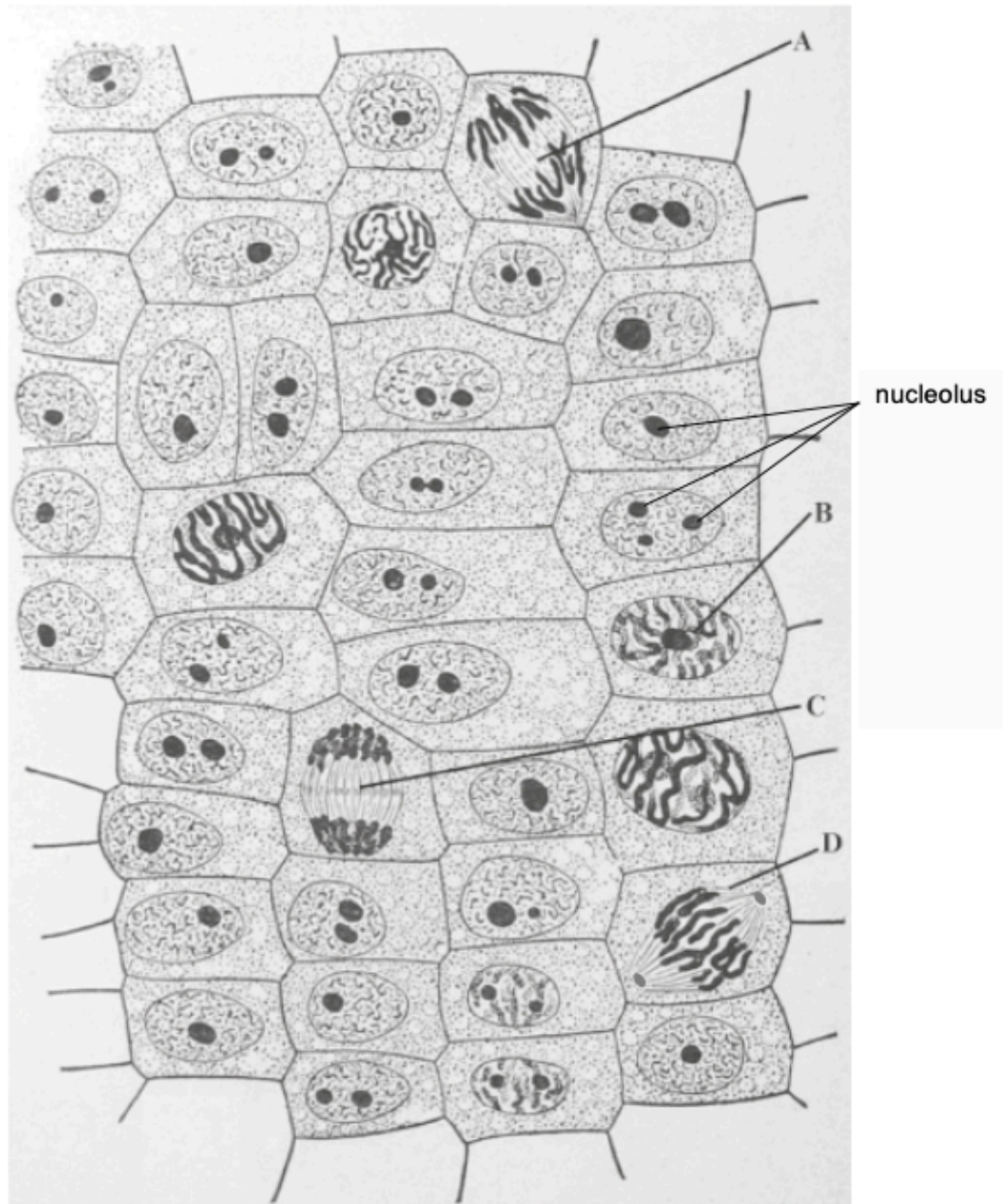
| 2          | (a)                                                               | (i)                                                          | B<br>A<br>C<br>All three correct = 2 marks<br>one/ two correct = 1 mark<br>0 correct = 0 marks                                                                                                                                                                                                                                         |            | 2       |            | 2 |                       |                           |   |                                                                   |                                                              |            |  |  |   |  |  |
|------------|-------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------|------------|---|-----------------------|---------------------------|---|-------------------------------------------------------------------|--------------------------------------------------------------|------------|--|--|---|--|--|
|            |                                                                   | (ii)                                                         | <table><tr><th>Difference</th><th>Mitosis</th><th>Meiosis</th></tr><tr><td>1</td><td>Genetically identical</td><td>genetically different (1)</td></tr><tr><td>2</td><td>6 chromosomes/<br/>chromosome number<br/>stays the same/<br/>diploid</td><td>3 chromosomes/<br/>chromosome number<br/>halved/<br/>haploid(1)</td></tr></table> | Difference | Mitosis | Meiosis    | 1 | Genetically identical | genetically different (1) | 2 | 6 chromosomes/<br>chromosome number<br>stays the same/<br>diploid | 3 chromosomes/<br>chromosome number<br>halved/<br>haploid(1) | 1<br><br>1 |  |  | 2 |  |  |
| Difference | Mitosis                                                           | Meiosis                                                      |                                                                                                                                                                                                                                                                                                                                        |            |         |            |   |                       |                           |   |                                                                   |                                                              |            |  |  |   |  |  |
| 1          | Genetically identical                                             | genetically different (1)                                    |                                                                                                                                                                                                                                                                                                                                        |            |         |            |   |                       |                           |   |                                                                   |                                                              |            |  |  |   |  |  |
| 2          | 6 chromosomes/<br>chromosome number<br>stays the same/<br>diploid | 3 chromosomes/<br>chromosome number<br>halved/<br>haploid(1) |                                                                                                                                                                                                                                                                                                                                        |            |         |            |   |                       |                           |   |                                                                   |                                                              |            |  |  |   |  |  |
|            | (b)                                                               | (i)                                                          | 3 $\mu\text{m minute}^{-1}$ (2)<br>1 mark for<br>45/15                                                                                                                                                                                                                                                                                 |            | 2       |            | 2 | 2                     |                           |   |                                                                   |                                                              |            |  |  |   |  |  |
|            |                                                                   | (ii)                                                         | (sister) Chromatids {separated/ pulled to poles} as distance<br>(between centromeres) increases (1)<br>{Chromosomes/ chromatids/ centromeres} pulled towards<br>(spindle) poles as distance between them and poles decreases<br>(1)                                                                                                    |            |         | 1<br><br>1 | 2 |                       |                           |   |                                                                   |                                                              |            |  |  |   |  |  |
|            | (c)                                                               | (i)                                                          | 760 = 2 marks<br>If incorrect award 1 mark for<br>$\frac{38 \times 1000}{50}$                                                                                                                                                                                                                                                          |            | 2       |            | 2 | 2                     | 2                         |   |                                                                   |                                                              |            |  |  |   |  |  |

|  |  |       |                                                                                                                                                                                                                               |          |          |          |           |          |          |
|--|--|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
|  |  | (ii)  |  anaphase<br><br> metaphase<br><br>Both correct for 1 mark | 1        |          |          | 1         |          | 1        |
|  |  | (iii) | Root tip/ Shoot tip/ meristem/ cambium (1)<br>regions of growth (1)                                                                                                                                                           | 2        |          |          | 2         |          | 2        |
|  |  |       | <b>Question 2 total</b>                                                                                                                                                                                                       | <b>5</b> | <b>6</b> | <b>2</b> | <b>13</b> | <b>4</b> | <b>5</b> |



Speciment

5. Growth of plants is largely limited to cell division in meristematic tissues located at the tips of roots and shoots and in the cambium of vascular bundles. The drawing below was made from a sample of meristematic plant tissue showing dividing cells.



- (a) Name the type of cell division taking place and identify from the diagram above the stages of cell division shown labelled **A** to **D**. [4]

**Type of cell division:** .....

**Stages:**

|          |       |
|----------|-------|
| <b>A</b> | ..... |
| <b>B</b> | ..... |
| <b>C</b> | ..... |
| <b>D</b> | ..... |

- (b) One stage of the cell cycle shown on the diagram is present in greater numbers than the others.

- (i) Name this stage and explain what this suggests about the relative length of this stage? [2]

.....

.....

- (ii) Describe how you could improve your confidence in your conclusion. [2]

.....

.....

|   |
|---|
|   |
| 8 |

| Question |     |      |  | Marking details                                                                                                                                                                                  | Marks available |          |          |          |          |          |
|----------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                  | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 5        | (a) |      |  | Mitosis (1)<br><br>A Anaphase<br>B Prophase<br>C Telophase<br>D Metaphase<br><br>all correct = 3 marks<br>one incorrect = 2 marks<br>two incorrect = 1 mark<br>three or four incorrect = 0 marks | 1<br><br>3      |          |          | 4        |          |          |
|          | (b) | (i)  |  | Interphase (1)<br>This is the longest stage (1)                                                                                                                                                  | 1               |          | 1        | 2        |          |          |
|          |     | (ii) |  | Count more cells in this specimen(1)<br>View more specimens (1)                                                                                                                                  |                 | 2        |          | 2        |          | 2        |
|          |     |      |  | <b>Question 5 total</b>                                                                                                                                                                          | <b>5</b>        | <b>2</b> | <b>1</b> | <b>8</b> | <b>0</b> | <b>2</b> |